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Supplementary appendix 2

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

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Supplementary Methods

Methods used in PK/PD and immunogenicity analysis:

PK analysis:

The endpoints for the PK assessment of cadonilimab included individual subject serum cadonilimab concentrations at different time points after cadonilimab administration.

Serum concentrations of cadonilimab were measured by a validated enzyme-linked immunosorbent assay with a lower limit of quantification of 0.0281 mg/L. Cadonilimab was administered as an intravenous infusion over 60 ±5 minutes. Serum samples were collected at Day 1 (prior to infusion, at the end of infusion and 3 hours after infusion), Day 2, 3, 8 and Day 15 (prior to the second dose infusion), which was defined as the first intensive collection for the Q2W treatment regimen. The second intensive collection was conducted between Day 57 and Day 71 when the fifth dose was administrated. The details of PK sample collections are shown in Tables S1 and S2. Pharmacokinetic parameters of cadonilimab were estimated using standard non-compartmental methods (WinNonlin version 8.3, Certara USA Inc; Princeton, NJ, USA) included area under the curve (AUC) from the time of dosing to the time of last observation (AUC0 – t), maximum observed concentration (Cmax), time of observed Cmax(Tmax), trough serum concentration (Cmin), total body clearance (CL), terminal elimination half-life (t1/2), volume of distribution (Vd), and the accumulation ratio.

Univariate binary logistic regression formula:

Univariate binary logistic regression was used to evaluate the correlation analysis of pharmacokinetics exposure of Cadonilimab and clinical safety indicator response. Where the formula of the model is as follows:

$$logit_i = \theta_1 + \theta_2\left(AUCss_i - Median(AUCss)\right) \qquad \text{Formula I}$$

$$P_i = \frac{e^{Logit_i}}{1+e^{Logit_i}} \qquad \text{Formula II}$$

where θ_1 is the typical value of the logit at the median AUCss. θ_2 is the typical value of the slope relating AUCss to the logit. $AUCss_i$ is the calculated AUCss in the ith patient (centered about the population median AUCss value). P_i is the predicted probability of experiencingrash in the ith patient.

The AUC_{ss} was the steady-state exposure estimated for subject based on a population pharmacokinetic model, and similarly, other PK parameters such as steady-state peak drug concentration $C_{max,ss}$, steady-state trough drug concentration $C_{min,ss}$ had been estimated based on a population pharmacokinetic model established for Cadonilimab. And the PK model as well as the estimation process were not presented here due to space limitations.

Pharmacodynamics:

Pharmacodynamic parameters including CTLA-4 suppression [intracellular proliferation marker Ki67], and target engagement biomarker, receptor occupancy (RO) on PD-1 and CTLA-4 of peripheral blood CD3+ T cells.

Circulating quantities of T cells expressing the Ki67 and RO were monitored using qualified flow

cytometry-based assays. Fresh peripheral blood were obtained in anti-coagulative tube on Cycle 1 day 1 pre-therapy, day 8 post-theray, Cycle 2 day 1 pre-therapy, Cycle 3 day 1 pre-therapy, Cycle 5 day 1 pre-therapy for Ki67 test, and on Cycle 1 day 1 pre-therapy, day 8 post-theray, Cycle 2 day 1 pre-therapy, Cycle 3 day 1 pre-therapy, Cycle 5 day 1 pre-therapy, Cycle 7 day 1 pre-therapy for RO test. Peripheral blood mononuclear cells were isolated and stained for CD3, CD4, and Ki67 in a certified clinical laboratory.

Proliferation marker Ki67 was monitored by the ratio of CD4+Ki67+ positive cells in total CD3+ T cell population following cadonilimab therapy to baseline, and target marker was monitored by RO on PD-1 and CTLA-4 of CD3+ T cell population by time point and dose cohort.

Immunogenicity test:

Samples for monitoring development of human antidrug-cadonilimab antibodies (ADA) and neutralizing antibody (NAb) were drawn before initial dosing and 3rd ,5th ,7th ,10th dosing, and every 3 cycles thereafter. Antidrug-cadonilimab antibodies were detected with an electrochemiluminescent immunoassay using Meso Scale Discovery technology (Rockville, Maryland, USA), validated according to current US FDA Medicines Agency guidelines for immunogenicity testing. The baseline and post-baseline ADA Results will be presented of cadonilimab by dose cohort. The percentage of overall treatment-emergentpositive, non-treatment-emergent-positive will be presented by dose cohort.

Supplementary Tables

Table S1. Summary of PK parameters after first dose of cadonilimab in the monotherapy part of AK104-201 phase Ib

Study Category/Number	Dose Level	AUC_{0-t} (day*μg/mL)	C_{max} (μg/mL)	T_{max} (h)[a]	$t_{1/2}$ (day)	V_d (L)	CL (L/Day)
AK104-201 (n=21)	6.0 mg/kg Q2W (n=9)	287 (271,378)	129 (116,142)	0.983 (0.967 , 3.95)	3.42 (2.41, 3.67)	5.32 (4.65 ,6.15)	1.11 (1.07 ,1.54)
	450 mg Q2W (n=9)	375 (312,466)	128 (124 ,150)	1.05 (0.967, 4.02)	4.00 (3.57, 4.81)	5.83 (5.56 ,6.91)	1.07 (0.824 ,1.38)
	10.0 mg/kg Q2W (n=3)	433 (385, 626)	174 (151 ,246)	1.10 (0.967, 3.73)	3.81 (3.33 ,8.16)	5.97 (5.44 ,12.4)	1.05 (0.989 ,1.27)

All parameter (except T_{max}) were reported as median (Q1, Q3). T_{max} was reported as median (Min, Max). C_{max}=the maximum concentration. T_{max}=time to maximum concentration. AUC_{0-t}=area under the serum concentration-time curve from 0h to the last measurable concentration. $t_{1/2}$=elimination half-life. V_d=Apparent volume of distribution. CL=Clearance.

Table S2. Summary of PK parameters after the fifth dose of cadonilimab in the monotherapy part of AK104-201 phase Ib

Study Category / Number	Dose Level	$AUC_{0\text{-}t,ss}$ (day*μg/mL)	$C_{min,ss}$ (μg/mL)	$T_{max,ss}$ (h)[a]	$C_{max,ss}$ (μg/mL)	$R_{ac,AUC}$	$R_{ac,Cmax}$	V_{ss} (L)	CL_{ss} (L/Day)
AK104-201 (n=9)	6.0 mg/kg Q2W (n=5)	389(268, 411)	3.98(3.93,6.52)	1.00 (0.983,1.15)	127(115,129)	1.03(0.945,1.04)	0.894(0.810,1.00)	6.24(3.84,6.44)	1.09(0.862, 1.20)
	450 mg Q2W (n=3)	408(359,532)	8.94(8.67,10.3)	1.05 (0.983,4.00)	104(103,142)	0.781(0.770,1.42)	0.839(0.736,1.34)	7.13(6.18,8.83)	1.10(0.883, 1.25)
	10.0 mg/kg Q2W (n=1)	470 (470,470)	8.42(8.42,8.42)	4.08 (4.08, 4.08)	168(168,168)	1.09(1.09,1.09)	0.966(0.966,0.966)	5.94(5.94,5.94)	1.15(1.15,1.15)

All parameter (except T_{max}) were reported as median (Q1, Q3). $T_{max,ss}$ was reported as Median(Min, Max). $C_{max,ss}$=the maximum concentration at steady-state. $C_{min,ss}$=the minimum concentration at steady-state. $T_{max,ss}$=time to maximum concentrationat steady-state. $AUC_{0\text{-}t,ss}$=area under the serum concentration-time curve during a dosing interval. $R_{ac,AUC}$ and $R_{ac,Cmax}$=the accumulation ratio of $AUC_{0\text{-}t,ss}$ to $AUC_{0\text{-}t,}$ and $C_{max,ss}$ to C_{max}, respectively. V_{ss}=Apparent volume of distribution at steady-state. CL_{ss}=Clearance at steady-state.

Table S3. Summary of anti-drug antibodies ratio of cadonilimab

	Phase Ib			Phase II			Total
	6 mg/kg Q2W (N=34)	10 mg/kg Q2W (N=3)	450 mg Q2W (N=32)	CC (N=92)	ESCC (N=19)	HCC (N=21)	(N=201)
Baseline							
ADA Negative	34 (100)	3 (100)	32 (100)	87 (94.6)	19 (100)	21 (100)	196 (97.5)
ADA Positive[1]	0	0	0	5 (5.4)	0	0	5 (2.5)
NAb Positive*	0	0	0	1 (1.1)	0	0	1 (0.5)
NAb Negative**	0	0	0	4 (4.3)	0	0	4 (2.0)
Post-baseline							
ADA Negative	30 (88.2)	3 (100)	30 (93.8)	83 (90.2)	19 (100)	16 (76.2)	181 (90.0)
ADA Positive	4 (11.8)	0	2 (6.3)	9 (9.8)	0	5 (23.8)	20 (10.0)
Non-treatment-emergent ADA Positive[2]	3 (8.8)	0	1 (3.1)	3 (3.3)	0	0	7 (3.5)
NAb Positive*	3 (8.8)	0	1 (3.1)	0	0	0	4 (2.0)
NAb-PD-1 Positive	3 (8.8)	0	0	0	0	0	3 (1.5)
NAb-PD-1 Negative	0	0	1 (3.1)	0	0	0	1 (0.5)
NAb-CTLA-4 Positive	0	0	1 (3.1)	0	0	0	1 (0.5)
NAb-CTLA-4 Negative	3 (8.8)	0	0	0	0	0	3 (1.5)
NAb Negative**	0	0	0	3 (3.3)	0	0	3 (1.5)
Treatment-emergent ADA Positive[3]	2 (5.9)	0	1 (3.1)	6 (6.5)	0	5 (23.8)	14 (7.0)
NAb Positive*	2 (5.9)	0	1 (3.1)	5 (5.4)	0	4 (19.0)	12 (6.0)
NAb-PD-1 Positive	2 (5.9)	0	1 (3.1)	4 (4.3)	0	4 (19.0)	11 (5.5)
NAb-PD-1 Negative	0	0	0	1 (1.1)	0	0	1 (0.5)
NAb-CTLA-4 Positive	1 (2.9)	0	1 (3.1)	4 (4.3)	0	4 (19.0)	10 (5.0)
NAb-CTLA-4 Negative	1 (2.9)	0	0	1 (1.1)	0	0	2 (1.0)
NAb Negative**	0	0	0	1 (1.1)	0	1 (4.8)	2 (1.0)

Baseline negative and post-baseline positive	2 (5.9)	0	1 (3.1)	6 (6.5)	0	5 (23.8)	14 (7.0)
NAb Positive*	2 (5.9)	0	1 (3.1)	5 (5.4)	0	4 (19.0)	12 (6.0)
NAb-PD-1 Positive	2 (5.9)	0	1 (3.1)	4 (4.3)	0	4 (19.0)	11 (5.5)
NAb-PD-1 Negative	0	0	0	1 (1.1)	0	0	1 (0.5)
NAb-CTLA-4 Positive	1 (2.9)	0	1 (3.1)	4 (4.3)	0	4 (19.0)	10 (5.0)
NAb-CTLA-4 Negative	1 (2.9)	0	0	1 (1.1)	0	0	2 (1.0)
NAb Negative**	0	0	0	1 (1.1)	0	1 (4.8)	2 (1.0)
Baseline positive and post-baseline titer >= 4 folder baseline titer	0	0	0	0	0	0	0
Baseline positive and missing baseline titer or missing post-baseline titer	0	0	0	0	0	0	0

Data are n(%). Q2W=every 2 weeks. CC=cervical cancer. ESCC=esophageal carcinoma. HCC=hepatocellular carcinoma. ADA=anti-drug antibody. PD-1=programmed death-1. CTLA-4=cytotoxic T lymphocyte antigen-4. [1] Baseline ADA Positive: Patients with positive ADA results and titer level>1:8 at baseline. [2] Non-treatment-emergent ADA Positive: Patients with no positive ADA result at baseline but with any positive response(titer level<=1:8) after receiving the treatment OR Patients with a baseline-positive ADA result in whom the assay signal was not enhanced (less than 4-fold over baseline titer level) at any time after initial drug administration. [3]Treatment-emergent ADA Positive: Patients with no positive ADA result at baseline but with any positive response (titer level>1:8) after receiving the treatment OR Patients with a baseline-positive ADA result in whom the assay signal was enhanced(greater than or equal 4-fold over baseline titer level) at any time after receiving the treatment OR when the baseline is positive without titer results. * NAB positive: Number of subjects positive for any neutralizing antibody (NAb-PD-1) or positive for any neutralizing antibody (NAb-CTLA-4) (%). **NAB negative: Number of subjects negative for neutralizing antibody (NAB-PD-1) and negative for neutralizing antibody (NAB-CTLA-4) (%).

Table S4. The enrolment date of first and last patients in 3 cohorts of phase II

	Cervical cancer (N=99)	Esophageal carcinoma (N=22)	Hepatocellular carcinoma (N=24)
First patient enrolment Date	Sep. 26, 2019	Dec.17, 32019	Oct. 31, 2019
Last patient enrolment Date	Jan.8,2021	Dec.2, 2020	Oct. 30, 2020

Table S5. The censoring rule for PFS

Situation	Outcome	Date of Progression or Censoring
No baseline or post-baseline tumour assessments	Censoring	Date of first dose of cadonilimab
No PD or deaths	Censoring	Date of last evaluable tumour assessment
PD	Event	Date of first documented PD
Death and no PD	Event	Date of death
Death or PD after ≥2 missed visits	Censoring	Date of last evaluable tumour assessment before missed visits
Death or PD after new anti-cancer treatment started	Censoring	Date of last evaluable tumour assessment before start of new treatment

In terms of OS, subjects who were alive by the date of cut-off would be censored on the last known alive date, if the last known alive date was later then the date of cut-off, the subject would be censored on the date of cut-off, and subjects with no survival follow-up after baseline would be censored on the date of the first dose of cadonilimab. PFS=progression-free survival. PD=progressive disease.

Table S6. The duration of follow up for PFS and OS in 3 cohorts of phase II

	Cervical cancer (N=99)		Esophageal carcinoma (N=22)		Hepatocellular carcinoma (N=24)	
	PFS	OS	PFS	OS	PFS	OS
Duration of follow-up (months)						
Median (95% CI)	13.1 (11.0, 15.6)	14.6 (13.7, 16.0)	23.9 (14.7, NE)	17.9 (10.8, 18.6)	24.3 (3.5, NE)	19.6 (15.7, 24.4)
min, max	0.0, 24.4	0.7, 27.4	0.0, 23.9	1.1, 24.5	0.0, 24.9	1.4, 28.1
Q1, Q3	1.8, 7.5	13.1, 17.5	1.8, 6.2	4.0, 15.1	1.7, 8.7	8.7, 19.8

PFS=progression-free survival. OS=overall survival. CI=confidence interval. NE=not evaluable.

Table S7. List of investigators, sites and enrolled patients

Principal investigator	Study site	Number of patients enrolled
Jiafu ji	Beijing Cancer Hospital	95
Nong xu	The First Affiliated Hospital, College of Medicine, Zhejiang University	21
Hanmei lou	Zhejiang Cancer Hospital (Gynecological Radiotherapy)	14
Li bai、enfeng zhao	Chinese PLA General Hospital	9
Xiaohua mu	Fudan University Shanghai Cancer Center	9
Yunxia li	General Hospital of Ningxia Medical University	7
Tianshu liu	Zhongshan Hospital	6
Yuzhi li	The First Affiliated Hospital of Bengbu Medical College	6
Qingxia fan	The First Affiliated Hospital of Zhengzhou University	6
Mei feng	Fujian Provincial Cancer Hospital (Gynecological Radiotherapy)	6
Haijun zhong	Zhejiang Cancer Hospital	5
Yi huang	Hubei Cancer Hospital	5
Ge lou	Harbin Medical University Cancer Hospital	5
Jing wang	Hunan Cancer Hospital	5
Xiaoyan lin	Fujian Medical University Union Hospital	4
Ye che	The First Affiliated Hospital of Henan Science & Technology University	4
Ruifang an	First Affiliated Hospital of Xi'an Jiaotong University	4
Changzheng li	Shandong Cancer Hospital	3
Qi qi	Chongqing University Cancer Hospital	3
Xin huang	SUN YAT-SEN University Cancer Center	3
Zengqing guo	Fujian Provincial Cancer Hospital	2
Shubin wang	Peking University Shenzhen Hospital	2
Guiling li	Union Hospital Tongji Medical College Huazhong University of Science and Technology	2
Junwei jia	The First Bethune Hospital of Jilin University	2
Jingli zhu	Nanjing Drum Tower Hospital	2
Hong zhu	Xiangya Hospital Central South University	2
Jimin li	Linyi Cancer Hospital	2
Fenghu li	Guizhou Cancer Hospital	2
Sihai liao	Affiliated Hospital of Guangdong Medical University	1
Qinghua min	The First Affiliated Hospital of Nanchang University	1

Table S8. Tumour types in patients enrolled in phase Ib

	Tumour type	Number
Phase Ib	Non–small-cell lung cancer	14
Phase Ib	Colorectal cancer	9
Phase Ib	Small-cell lung cancer	8
Phase Ib	Cervical cancer	8
Phase Ib	Ovarian cancer	8
Phase Ib	Duodenal cancer	1
Phase Ib	Gastric cancer	7
Phase Ib	Endometrial carcinoma	7
Phase Ib	renal cell carcinoma	5
Phase Ib	Esophageal carcinoma	2
Phase Ib	Hepatocellular carcinoma	2
Phase Ib	Breast cancer	2
Phase Ib	Mesothelioma	2
Phase Ib	Malignant tumour with unknown primary site	2
Phase Ib	Synovial sarcoma	1
Phase Ib	Neuroendocrine carcinoma	1
Phase Ib	Cholangiocellular carcinoma	1
Phase Ib	Thymic carcinoma	1
Phase Ib	Pancreatic cancer	1
Phase Ib	Bladder cancer	1

Table S9. Most common TRAEs and irAEs

	Total (N=240)	6 mg/kg Q2W (N=197)	10 mg/kg Q2W (N=3)	450 mg Q2W (N=40)	CC (N=111)	ESCC (N=22)	HCC (N=24)
TEAE	232 (96·7)	193 (98·0)	3 (100·0)	36 (90·0)	109 (98·2)	22 (100·0)	23 (95·8)
≥Grade 3 TEAE	103 (42·9)	91 (46·2)	1 (33·3)	11 (27·5)	49 (44·1)	14 (63·6)	9 (37·5)
TRAE	217 (90·4)	179 (90·9)	3 (100·0)	35 (87·5)	102 (91·9)	20 (90·9)	22 (91·7)
Aspartate aminotransferase increased	47 (19·6)	37 (18·8)	0	10 (25·0)	18 (16·2)	4 (18·2)	7 (29·2)
Alanine aminotransferase increased	45 (18·8)	39 (19·8)	0	6 (15·0)	20 (18·0)	2 (9·1)	7 (29·2)
White blood cell count decreased	37 (15·4)	37 (18·8)	0	0	17 (15·3)	5 (22·7)	9 (37·5)
Neutrophil count decreased	27 (11·3)	26 (13·2)	0	1 (2·5)	12 (10·8)	4 (18·2)	5 (20·8)
Platelet count decreased	14 (5·8)	13 (6·6)	0	1 (2·5)	5 (4·5)	0	5 (20·8)
Hypothyroidism	47 (19·6)	41 (20·8)	1 (33·3)	5 (12·5)	23 (20·7)	8 (36·4)	5 (20·8)
Anaemia	61 (25·4)	53 (26·9)	1 (33·3)	7 (17·5)	37 (33·3)	6 (27·3)	2 (8·3)
Rash	38 (15·8)	27 (13·7)	2 (66·7)	9 (22·5)	12 (10·8)	4 (18·2)	2 (8·3)
Infusion related reaction	9 (3·8)	9 (4·6)	0	0	2 (1·8)	0	6 (25·0)
Grade ≥3 TRAE	67 (27·9)	57 (28·9)	0	10 (25·0)	31 (27·9)	8 (36·4)	7 (29·2)
TRSAE	54 (22·5)	47 (23·9)	0	7 (17·5)	27 (24·3)	8 (36·4)	4 (16·7)
TRAE Leading to Permanently Discontinued	17 (7·1)	14 (7·1)	0	3 (7·5)	6 (5·4)	1 (4·5)	3 (12·5)
irAE	122 (50·8)	105 (53·3)	1 (33·3)	16 (40·0)	61 (55·0)	13 (59·1)	11 (45·8)
Hypothyroidism	45 (18·8)	39 (19·8)	1 (33·3)	5 (12·5)	21 (18·9)	8 (36·4)	5 (20·8)
Blood thyroid stimulating hormone increased	17 (7·1)	12 (6·1)	1 (33·3)	4 (10·0)	7 (6·3)	0	0
Hyperthyroidism	23 (9·6)	22 (11·2)	0	1 (2·5)	16 (14·4)	1 (4·5)	3 (12·5)
Blood thyroid stimulating hormone decreased	11 (4·6)	8 (4·1)	0	3 (7·5)	5 (4·5)	1 (4·5)	0
Hyperglycaemia	10 (4·2)	9 (4·6)	0	1 (2·5)	8 (7·2)	1 (4·5)	0
Rash	13 (5·4)	9 (4·6)	0	4 (10·0)	3 (2·7)	1 (4·5)	1 (4·2)
Grade ≥3 irAE	33 (13·8)	29 (14·7)	0	4 (10·0)	15 (13·5)	4 (18·2)	3 (12·5)

Data are n (%). TRAE=treatment-related adverse event. irAE=immune-related adverse event. TEAE=treatment-emergent adverse event. TRSAE=treatment-related serious adverse event.

Table S10. Grade 1-2 TEAE that occurred in at least 10% patients

	Total (N=240)	
	Grade 1	**Grade 2**
All TEAE	224 (93.3)	169 (70.4)
Investigations		
Aspartate aminotransferase increased	49 (20.4)	5 (2.1)
Alanine aminotransferase increased	45 (18.8)	4 (1.7)
White blood cell count decreased	38 (15.8)	21 (8.8)
Weight decreased	27 (11.3)	16 (6.7)
Neutrophil count decreased	25 (10.4)	15 (6.3)
Blood and lymphatic system disorders		
Anaemia	63 (26.3)	32 (13.3)
General disorders and administration site conditions		
Pyrexia	36 (15.0)	6 (2.5)
Endocrine disorders		
Hypothyroidism	21 (8.8)	31 (12.9)
Renal and urinary disorders		
Proteinuria	28 (11.7)	20 (8.3)

Data are n (%). TEAE=treatment-emergent adverse event.

Table S11. Grade 3-5 TEAE that occurred in all treated patients

	Total (N=240)		
	Grade 3	**Grade 4**	**Grade 5**
All TEAE	85 (35.4)	14 (5.8)	31 (12.9)
Investigations			
Weight decreased	5 (2.1)	0	0
Lipase increased	4 (1.7)	0	0
Blood creatine phosphokinase increased	3 (1.3)	0	0
Neutrophil count decreased	3 (1.3)	0	0
Platelet count decreased	3 (1.3)	0	0
Alanine aminotransferase increased	2 (0.8)	0	0
Aspartate aminotransferase increased	2 (0.8)	0	0
Blood creatine phosphokinase MB increased	2 (0.8)	0	0
Blood creatinine increased	2 (0.8)	0	0
Gamma-glutamyltransferase increased	2 (0.8)	0	0
Amylase increased	1 (0.4)	0	0
Bilirubin conjugated increased	0	1 (0.4)	0
Blood bilirubin increased	1 (0.4)	2 (0.8)	0
Blood cholesterol increased	1 (0.4)	0	0
Blood corticotrophin decreased	1 (0.4)	0	0
Blood pressure increased	1 (0.4)	0	0
C-reactive protein increased	1 (0.4)	0	0

	Total (N=240)		
	Grade 3	**Grade 4**	**Grade 5**
Cortisol decreased	1 (0.4)	0	0
Glucose urine present	1 (0.4)	0	0
Haemoglobin decreased	1 (0.4)	0	0
Lymphocyte count decreased	1 (0.4)	0	0
Troponin T increased	1 (0.4)	0	0
Urine bilirubin increased	1 (0.4)	0	0
Urobilinogen urine increased	1 (0.4)	0	0
White blood cell count decreased	1 (0.4)	0	0
White blood cells urine positive	1 (0.4)	0	0
General disorders and administration site conditions			
Death	0	0	14 (5.8)
Disease progression	0	0	10 (4.2)
Asthenia	1 (0.4)	0	0
Device related thrombosis	1 (0.4)	0	0
Multiple organ dysfunction syndrome	0	0	1 (0.4)
Sluggishness	1 (0.4)	0	0
Blood and lymphatic system disorders			
Anaemia	16 (6.7)	0	0
Immune thrombocytopenia	1 (0.4)	1 (0.4)	0
Metabolism and nutrition disorders			

	Total (N=240)		
	Grade 3	**Grade 4**	**Grade 5**
Decreased appetite	4 (1.7)	0	0
Hypokalaemia	3 (1.3)	2 (0.8)	0
Hypoalbuminaemia	2 (0.8)	0	0
Hyponatraemia	2 (0.8)	0	0
Diabetic ketoacidosis	0	1 (0.4)	0
Hypercalcaemia	0	1 (0.4)	0
Hyperkalaemia	1 (0.4)	0	0
Hypertriglyceridaemia	1 (0.4)	0	0
Hypoglycaemia	0	1 (0.4)	0
Infections and infestations			
Urinary tract infection	5 (2.1)	0	0
Pneumonia	2 (0.8)	1 (0.4)	1 (0.4)
Abdominal infection	0	0	1 (0.4)
Appendicitis	1 (0.4)	0	0
Bronchitis	1 (0.4)	0	0
Sepsis	1 (0.4)	0	0
Upper respiratory tract infection	1 (0.4)	0	0
Respiratory, thoracic and mediastinal disorders			
Respiratory failure	0	2 (0.8)	2 (0.8)
Apnoea	0	0	1 (0.4)

	Total (N=240)		
	Grade 3	**Grade 4**	**Grade 5**
Dyspnoea	1 (0.4)	1 (0.4)	0
Haemoptysis	1 (0.4)	0	0
Immune-mediated pneumonitis	0	1 (0.4)	0
Obstructive airways disorder	1 (0.4)	0	0
Pneumonia aspiration	0	1 (0.4)	0
Pneumonitis	1 (0.4)	0	0
Pulmonary embolism	1 (0.4)	0	1 (0.4)
Respiratory tract haemorrhage	1 (0.4)	0	0
Cardiac disorders			
Immune-mediated myocarditis	2 (0.8)	0	0
Acute myocardial infarction	1 (0.4)	0	1 (0.4)
Angina unstable	1 (0.4)	0	0
Myocarditis	1 (0.4)	1 (0.4)	0
Pericardial effusion	1 (0.4)	0	0
Ventricular arrhythmia	0	1 (0.4)	0
Gastrointestinal disorders			
Diarrhoea	2 (0.8)	0	0
Ileus	2 (0.8)	0	0
Upper gastrointestinal haemorrhage	2 (0.8)	0	0
Gastritis	1 (0.4)	0	0

	Total (N=240)		
	Grade 3	**Grade 4**	**Grade 5**
Haematochezia	1 (0.4)	0	0
Intestinal obstruction	1 (0.4)	0	0
Obstruction gastric	1 (0.4)	0	0
Renal and urinary disorders			
Haematuria	3 (1.3)	0	0
Nephrotic syndrome	1 (0.4)	0	0
Proteinuria	1 (0.4)	0	0
Renal failure	1 (0.4)	0	0
Renal hydrocele	1 (0.4)	0	0
Renal impairment	0	1 (0.4)	0
Musculoskeletal and connective tissue disorders			
Back pain	3 (1.3)	0	0
Arthralgia	2 (0.8)	0	0
Pain in extremity	2 (0.8)	0	0
Immune-mediated myositis	1 (0.4)	0	0
Endocrine disorders			
Adrenal insufficiency	1 (0.4)	0	0
Hyperthyroidism	0	1 (0.4)	0
Hypopituitarism	1 (0.4)	0	0
Immune-mediated hyperthyroidism	1 (0.4)	0	0

	Total (N=240)		
	Grade 3	**Grade 4**	**Grade 5**
Hepatobiliary disorders			
Hepatic function abnormal	2 (0.8)	0	0
Immune-mediated hepatitis	1 (0.4)	0	0
Reproductive system and breast disorders			
Vaginal haemorrhage	2 (0.8)	0	0
Female genital tract fistula	1 (0.4)	0	0
Immune system disorders			
Anaphylactic shock	0	1 (0.4)	0
Infusion related hypersensitivity reaction	1 (0.4)	0	0
Injury, poisoning and procedural complications			
Infusion related reaction	1 (0.4)	1 (0.4)	0
Skin and subcutaneous tissue disorders			
Rash	2 (0.8)	0	0
Congenital, familial and genetic disorders			
Tracheo-oesophageal fistula	1 (0.4)	0	0
Neoplasms benign, malignant and unspecified (incl cysts and polyps)			
Tumour pain	1 (0.4)	0	0
Vascular disorders			
Venous thrombosis	1 (0.4)	0	0

Data are n (%). TEAE=treatment-emergent adverse event.

Table S12. TRAE leading to permanent discontinuation

	Total (N=240)
TRAE leading to permanent discontinuation	17 (7.1)
Immune-mediated pneumonitis	2 (0.8)
Infusion related reaction	2 (0.8)
Obstructive airways disorder	1 (0.4)
Pneumonitis	1 (0.4)
Respiratory failure	1 (0.4)
Acute myocardial infarction	1 (0.4)
Myocarditis	1 (0.4)
Gastritis	1 (0.4)
Ileus	1 (0.4)
Anaphylactic shock	1 (0.4)
Infusion related hypersensitivity reaction	1 (0.4)
Diabetic ketoacidosis	1 (0.4)
Hypokalaemia	1 (0.4)
Immune thrombocytopenia	1 (0.4)
Venous thrombosis	1 (0.4)

Data are n (%). TRAE=treatment-related adverse event.

Table S13. Subsequent therapies

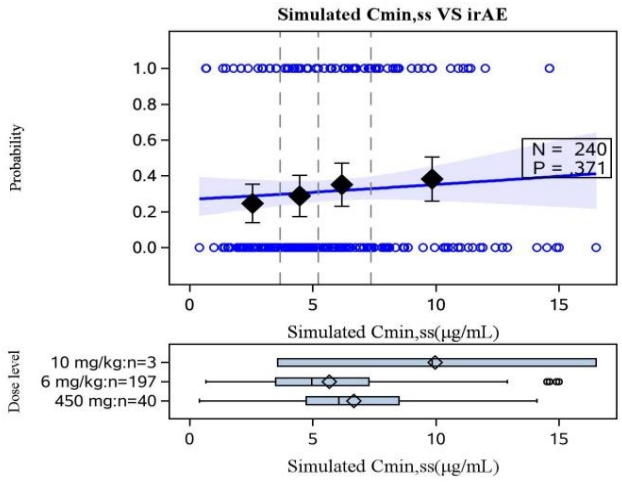
	Total (N=240)	Cervical cancer (N=111)	Esophageal carcinoma (N=22)	Hepatocellular carcinoma (N=24)	Phase Ib Total (N=83)
Any therapy	74 (30.8)	32 (28.8)	6 (27.3)	10 (41.7)	26 (31.3)
Chemotherapy	43 (17.9)	21 (18.9)	4 (18.2)	1 (4.2)	17 (20.5)
Immunotherapy	14 (5.8)	9 (8.1)	1 (4.5)	2 (8.3)	2 (2.4)
Targeted therapy	13 (5.4)	13 (11.7)	0	0	0
Combination	2 (0.8)	0	0	2 (8.3)	0
Hormone therapy	2 (0.8)	1 (0.9)	0	0	1 (1.2)
Radiotherapy	1 (0.4)	0	1 (4.5)	0	0
Surgery	1 (0.4)	0	0	1 (4.2)	0
Other	24 (10.0)	4 (3.6)	3 (13.6)	6 (25.0)	11 (13.3)

Data are n (%).

Supplementary Figures

Figure S1~S3. Logistic regression analysis of safety response and cadonilimab exposure. Logistic regression of safety response (irAE) and cadonilimab exposure ($C_{min,ss}$, AUC_{ss} and $C_{max,ss}$ of estimates for the Population PK model) in the monotherapy part of AK104-201 study. Exposure was presented on the X axis, probability of safety response was presented on the Y axis. The blue line was the mean probability of any irAE simulated with logistic regression models. The shaded region was the 95% confidence interval of probability. Blue points were the responses (with any irAE or no irAE) corresponding to an individual's concentration. The dotted line was the quartile line of the abscissa variable. Black diamonds and error bars showed the median and 95% CI of the observed probability of quantiles of concentrations. Sample size and P-value were recorded in the right box.

S1



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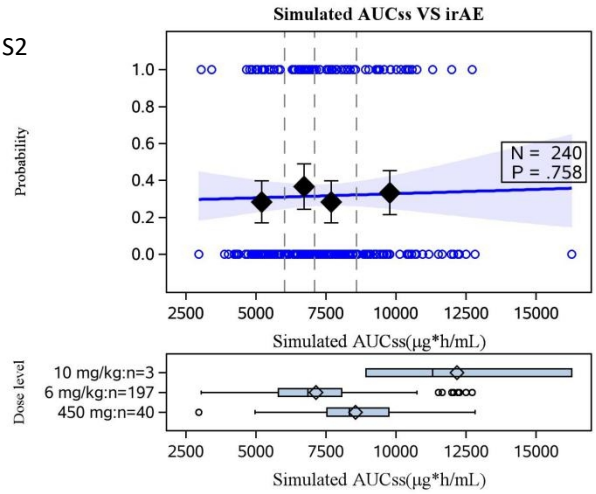


S3

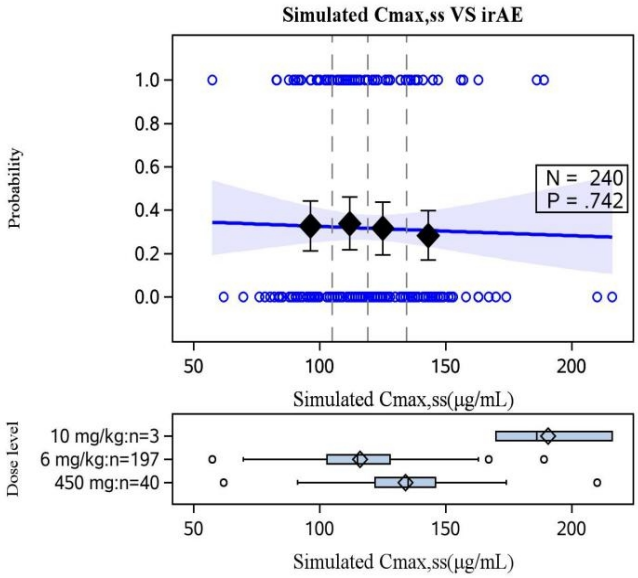


Figure S4. RO (mean+SD) - time plots following multiple dose of cadonilimab. C represent different Cycles, D represent different Days on each cycle.

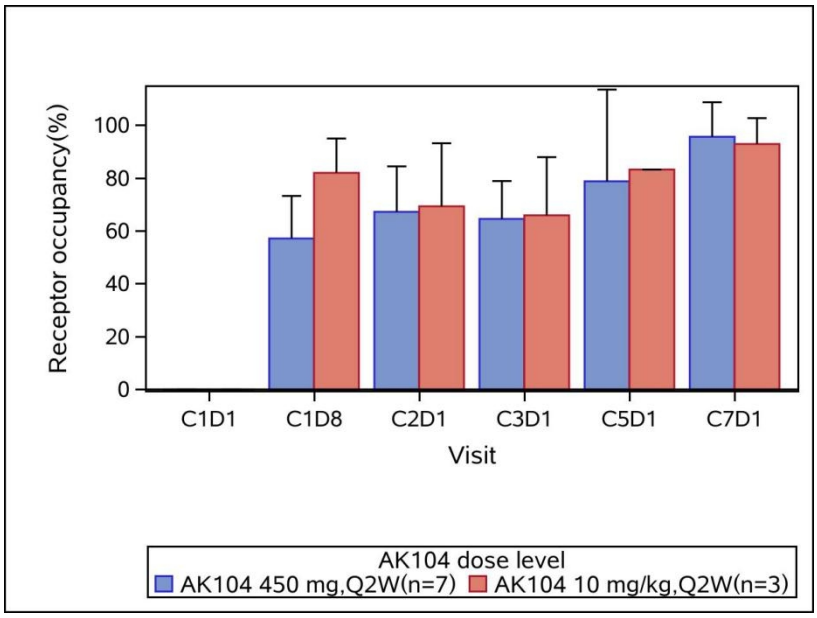
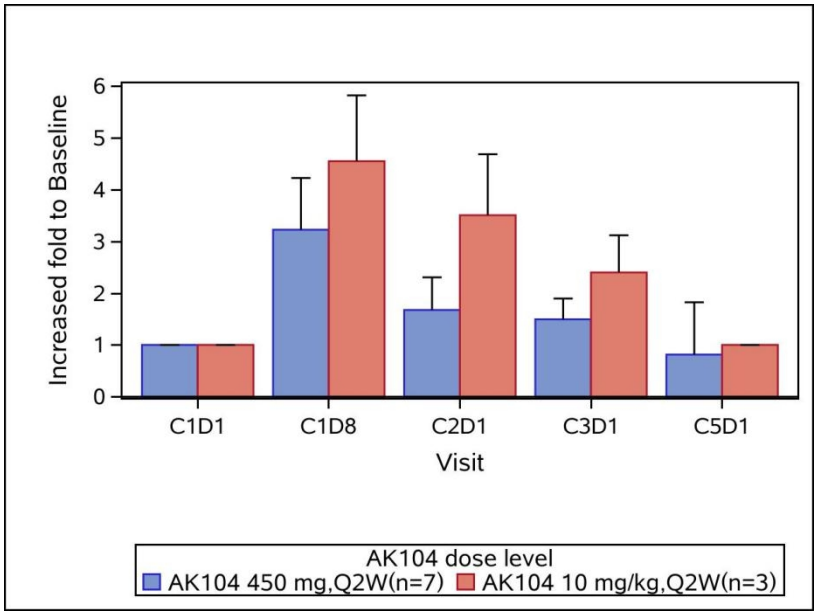


Figure S5. CD4+Ki67+ T cell parent ratio to baseline (mean+SD) - time plots following multiple dose of cadonilimab. C represent different Cycles, D represent different Days on each cycleC represent different Cycles, D represent different Days on each cycle. The postdose/baseline ratio of CD4+Ki67+ T cells reached its peak on Day 8 after the initial administration of cadonilimab in both the 450 mg Q2W and 10mg/kg Q2W groups. The average postdose/baseline ratio of Ki67 on Day 8 was approximately four times higher than that on Day 1 in the 10 mg/kg Q2W group.



Definition

The definition of relapsed or refractory cervical cancer included any of the following 2 items:

1. The recurrent/metastatic cervical cancer patients who were treated with platinum-based treatment, progressed during treatment or after receiving ≥4 treatment cycles, or were intolerant to toxicities caused by platinum-based treatment regimens.
2. The cervical patients progressed during or after receiving ≥4 platinum-based treatment cycles as neoadjuvant or adjuvant therapy, or progressed within 6 months after the end of the treatment.

Clinical Study Protocol

A Phase Ib/II, Open-label, Multi-center Trial to Evaluate AK104 (an Anti-PD-1 and Anti-CTLA-4 Bispecific Antibody) in the Treatment of Advanced Solid Tumors and as First-line Therapy with Combination Oxaliplatin and Capecitabine in Unresectable Advanced or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

AK104-201

Version: 5.1, June 06, 2021

Akeso Biopharma Co., Ltd.

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Protocol Synopsis

Study Title

A PhaseIb/II, Open-label, Multi-center Trial to Evaluate AK104 (an Anti-PD-1 and Anti-CTLA-4 Bispecific Antibody) in the Treatment of Advanced Solid Tumors and as First-line Therapy with Combination Oxaliplatin and Capecitabine in Unresectable Advanced or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

Study Objectives

Primary Objective:

- Phase Ib: To assess the safety and tolerability of AK104 as a monotherapy in subjects with advanced solid tumor and AK104 in combination with oxaliplatin and capecitabine (mXELOX/XELOX) as first-line therapy in subjects withadvanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase Ib: To determine the maximum tolerated dose (MTD) or the maximum administered dose (MAD) when the MTD is not reached, and the recommended Phase II dose (RP2D) of AK104 administered with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase II: To assess the objective response rate (ORR) of AK104 combined with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma as per RECIST v1.1.
- Phase II: To assess the objective response rate (ORR) of AK104 as a monotherapy in subjects with recurrent or metastatic cervical cancer as per RECIST v1.1.
- Phase II: To assess the objective response rate (ORR) of AK104 as a monotherapy in subjects with locally advanced or metastatic esophageal squamous cell carcinoma and advanced hepatocellular carcinoma as per RECIST v1.1.

Secondary Objectives:

- Phase Ib: To preliminarily evaluate the antitumor activity of AK104 as a monotherapy in subjects with advanced or metastatic solid tumors and AK104 combined with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase II: To assess the safety and tolerability of AK104 in combination with mXELOX/XELOX as first-line therapy in patients with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase II: To assess the efficacy of AK104 with mXELOX/XELOX as the first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma, including progression-free survival (PFS), duration of response (DoR), disease control rate (DCR), time to response (TTR) and overall survival (OS) based on RECIST v1.1 assessment.
- Phase II: To assess the efficacy of AK104 as a monotherapy in subjects with recurrent or metastatic cervical cancer, including PFS, DoR, DCR, TTR, and OS as assessed per RECIST v1.1.
- Phase II: To assess the efficacy of AK104 as a monotherapy in subjects with locally advanced or metastatic esophageal squamous cell carcinoma and advanced hepatocellular carcinoma, including PFS, DoR, DCR, TTR, and OS as assessed per RECIST v1.1.
- Phase II: To evaluate the safety and tolerability of AK104 as a monotherapy in subjects with recurrent or metastatic cervical cancer.
- Phase II: To evaluate the safety and tolerability of AK104 as a monotherapy in subjects with locally advanced or metastatic esophageal squamous cell carcinoma and advanced hepatocellular carcinoma.
- To assess the pharmacokinetic profile of AK104 as a monotherapy and in combination with mXELOX/XELOX.

- To assess the immunogenicity of AK104 as a monotherapy and in combination with mXELOX/XELOX.

Exploratory Objectives:

- To evaluate the correlation between PD-L1 expression and antitumor response in archived and/or fresh tumor tissue.
- To explore other AK104 potential biomarkers in blood and tumor tissue specimens, which may (1) be related to the biological activity (pharmacodynamics) of the drug; (2) prospectively determine the patients who most likely respond to AK104 treatment (predicted); and (3) have other correlations with study treatment response, drug resistance mechanism, and /or disease status.
- To preliminarily determine the antitumor activity of AK104 as a monotherapy in subjects with advanced or metastatic solid tumors and AK104 in combination with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma using the immune-related Response Evaluation Criteria in Solid Tumors (irRECIST).

Study Endpoints

Primary Endpoints:

- Phase Ib Safety Assessment: dose-limiting toxicity (DLT), incidence and severity of adverse events, clinically significant abnormal laboratory tests, vital signs, and electrocardiograms (ECGs) ; MTD, or MAD when the MTD is not reached, and recommended Phase II dose (RP2D).
- Phase II Efficacy Assessment:
  - ✓ ORR for squamous or adenosquamous cell carcinoma of the cervix will be assessed by Independent Radiology Review Committee (IRRC) as per RECIST v1.1.
  - ✓ ORR will be assessed by the investigator as per RECIST v1.1 for gastric adenocarcinoma or gastroesophageal junction adenocarcinoma, esophageal squamous cell carcinoma and hepatocellular carcinoma.

Secondary Endpoints:

- Phase Ib efficacy assessment: ORR, DCR, DoR, TTR, and PFS assessed as per RECIST v1.1.
- Phase II Safety Assessments: incidence and severity of adverse events, clinically significant abnormal laboratory tests, vital signs, and electrocardiogram (ECG) results.
- Phase II Efficacy Assessments:
  - ✓ DCR, DoR, TTR, and PFS for squamous or adenosquamous carcinoma of the cervix will be assessed by IRRC as per RECIST v1.1.
  - ✓ ORR, DCR, DoR, TTR, and PFS for squamous or adenosquamous carcinoma of the cervix will be assessed by the investigator as per RECIST v1.1.
  - ✓ DCR, DoR, TTR, and PFS will be assessed by the investigator as per RECIST v1.1 for gastric or gastroesophageal junction adenocarcinoma, esophageal squamous cell carcinoma and hepatocellular carcinoma.
- OS and 12-month overall survival rate (probability that the subject survives for 12 months after the first dose).
- Endpoints for AK104 PK assessment include serum drug concentration of each subject at different time points after AK104 administration. PK parameters will be measured with non-compartmental analysis, including observed maximum concentration (C_{max}), area under the plasma concentration-time curve (AUC), clearance (CL), and half-life ($t_{1/2}$).
- Endpoints for AK104 immunogenicity assessment include the number and percentage of subjects with detectable anti-drug antibody (ADA).

Exploratory Endpoints:

- To analyze the correlation between PD-L1 expression and the antitumor activity of AK104 [e.g., ORR

assessed by IRRC in subjects with PD-L1 positive (CPS ≥ 1) and PD-L1 negative (CPS < 1) cervical squamous/adenosquamous cell carcinoma], and other biomarkers that potentially predict the antitumor activity of AK104, including but not limited to the type and number of tumor infiltrating lymphaytes (TIL) and the immune-related gene mRNA expression characteristics in tumor tissues (tumor cells and tumor infiltrating immune cells);

- To obtain exploratory biomarkers in tumor tissues during the disease progress or response, including but not limited to the detection of immune-related gene characteristics and changes in the number of CD8+ effector T cells;
- Endpoints for assessment of potential pharmacodynamic biomarkers may include: (1) changes in CD4+ Ki67+ and CD8+ Ki67+ proliferating cells in blood from baseline over time, which will be determined using flow cytometry; and (2) PD-1 or CTLA-4 receptor occupancy in blood before and after treatment, which will be determined using flow cytometry.
- Anti-tumor activity endpoints assessed as per irRECIST criteria include ORR, DoR, DCR, TTR and PFS.

Study Design

This is a Phase Ib/II, open-label, multi-center clinical trial conducted in China to assess the safety, tolerability, PK profile, immunogenicity, pharmacodynamics, and antitumor activity of AK104 as a monotherapy in subjects with advanced solid tumors and AK104 in combination with oxaliplatin and capecitabine as a first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.

This study consists of 3 periods: a dose escalation period (Part 1 of Phase Ib), a dose expansion period (Part 2 of Phase Ib), and a dose confirmation period (Phase II). Subjects will be divided into different cohorts depending on the different tumor types and the AK104 treatment regimen in Phases Ib and II.

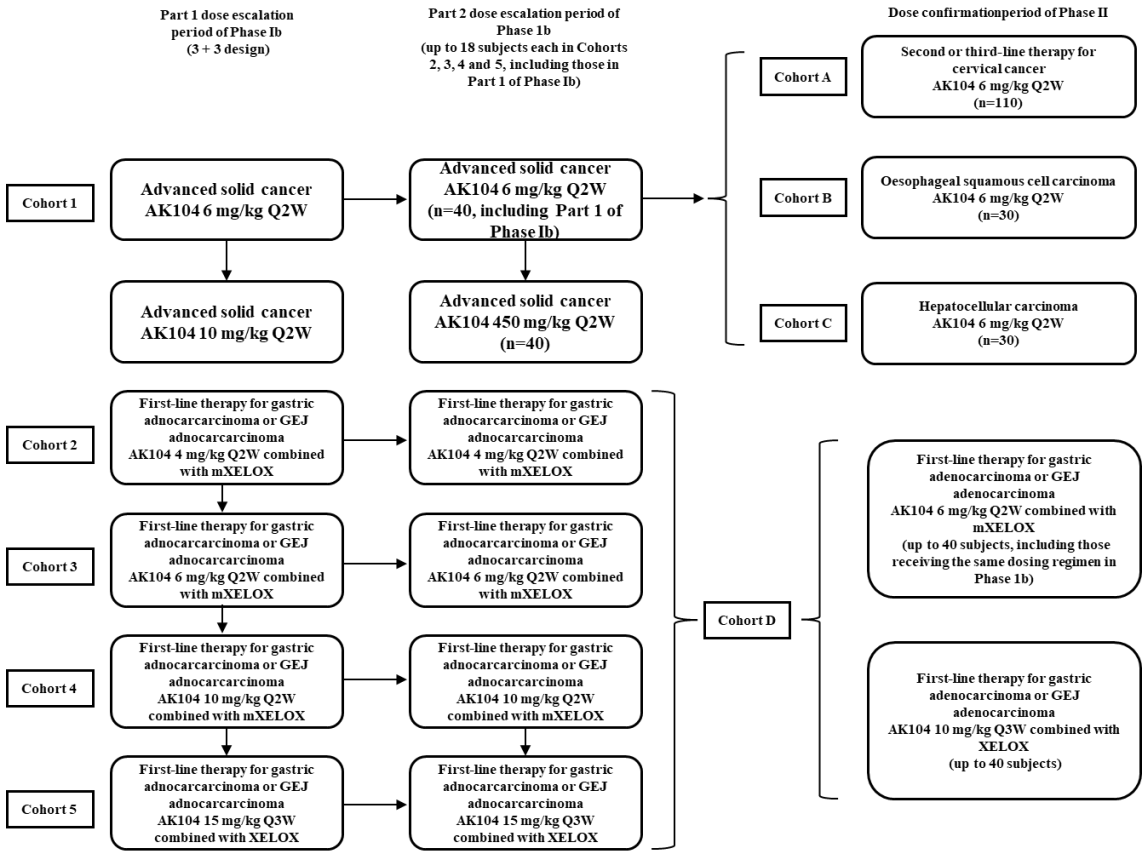
In the Phase Ib dose escalation period, AK104 as a monotherapy at doses of 6 mg/kg and 10 mg/kg every 2 weeks (Q2W) will be administered to treat the advanced solid tumors in Cohort 1, and AK104 at doses of 4mg/kg Q2W, 6mg/kg Q2W, 10 mg/kg Q2W, 15 mg/kg Q3W in combination with mXELOX/XELOX as first-line therapy will be administered to treat the advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma in the Cohort 2, 3, 4 and 5, respectively. Dose escalation and safety evaluation will be performed in accordance with the "3 + 3" principle.

In the Phase Ib dose expansion period, AK104 as a monotherapy at doses of 6 mg/kg and 450 mg will be administered to treat the advanced solid tumors in Cohort 1 (40 subjects for each dose group, including subjects in the dose escalation period), and for Cohort 2, 3, 4 and 5, if no DLT is observed or the DLTs occur in less than 2 out of 6 subjects in the dose escalation period, the enrollment will continue until there are 18 subjects in each corresponding cohort(including subjects in the dose escalation period).

In Phase II, subjects will be divided into four cohorts by tumor types, including Cohort A: cervical cancer (n = 110); Cohort B: esophageal squamous cell carcinoma (n = 30); Cohort C: hepatocellular carcinoma (n = 30); and Cohort D: advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma (2 dosing regimens including AK104 6mg/kg Q2W combined with mXELOX and AK104 10mg/kg Q3W combined with XELOX ,phase Ib with up to 40 subjects in each regimen, including those receiving the same dosing regimen in Phase Ib).

In this study, It is planned to enroll approximately 372 subjects (approximately 158in the dose escalation and expansion periods and approximately 214 in the dose confirmation period). Additional subjects may be enrolled if dose de-escalation cohorts or intermediate doses are tested, or additional dose escalation cohorts are required.

Figure Overall Study Design Diagram



Abbreviations: GEJ, gastroesophageal junction; Q2W, every 2 weeks; Q3W, every 3 weeks.

a. The dose escalation cohort includes the Cohort 1 of AK104 6 mg/kg Q2W and 10 mg/kg Q2W, the Cohort 2, Cohort 3and , Cohort 4 of AK104 (4 mg/kg Q2W, 6 mg/kg Q2W, 10 mg/kg Q2W) with mXELOX, and the Cohort 5 of AK104 (15mg/kg Q3W) with XELOX. Based on the safety data available, the sponsor in agreement with the Dose Escalation Committee (DEC) will decide whether to evaluate intermediate doses of AK104. If DLTs are observed in ≥ 2 out of 6 evaluable subjects in Cohort 2, the initial dose will be reduced to half of the previous dose (2 mg/kg Q2W). If the MTD is not reached in this study, the sponsor and DEC will discuss whether to continue the dose escalation in future protocol amendments. In each escalation cohort, the first subject will be treated first and then after a minimum of 24 hours, the next 2 or 3 subjects will be treated. For the subjects in each dose group, the dose level and dosing interval of AK104 remain unchanged.

b. In the dose escalation period, after tolerability is observed in 3~6 subjects, the dose group which has no DLT or has DLTs observed in fewer than 2 out of 6 subjects will be expanded after the discussion of the sponsor with DEC, based on preliminary safety and efficacy data. 6 mg/kg dose group in Cohort 1 can be expanded to a maximum ofa maximum of 40 subjects. Cohorts 2, 3, 4 and 5 can be expanded to include 18 subjects in each Cohort (including subjects enrolled in the dose escalation period). If DLT occurs in ≥ 2 out of 6 subjects, the dose group will not be expanded.

c. The AK104 450 mg group for treatment of solid tumors will enroll all solid tumor types except the tumor types selected in Cohorts A, B and C of Phase II. After the first 6 subjects receiving AK104 450mg Q2W complete the 28-day treatment, the investigator and sponsor will conduct a safety assessment analysis.

d. Cohort D including 2 dosing regimens of AK104 6mg/kg Q2W combined with mXELOX and AK104 10mg/kg Q3W combined with XELOX.

e. Dose escalation cohort: If 5 DLT-evaluable subjects out of 6 enrolled subjects at any dose have completed the safety assessment of 28 days/21 days after first dose of AK104 and have no DLT, the study can proceed to the next dose escalation cohort and/or to the dose expansion

enrollment.

The study consists of a screening period (no more than 28 days after the subject signs the informed consent up till the first AK104 dose), a treatment period (until confirmed disease progression, unacceptable toxicity, or withdrawal of the informed consent form, whichever comes first) and a follow-up period (including safety follow-up period and survival follow-up period). Subjects will be screened and evaluated 28 days before the first dose to determine if they meet the study eligibility criteria. Tumor samples from subjects should be collected during screening. Subjects who meet the study eligibility criteria will receive either AK104 as a monotherapy or in combination with oxaliplatin and capecitabine (mXELOX/XELOX) chemotherapy regimen based on their cohort.

All subjects will be evaluated periodically for tumor response and their clinical status will be categorized according to RECIST version 1.1 (see Section 7 for descriptions of tumor imaging assessments). In Phase Ib, tumor assessments will be performed every 6 weeks (± 7 days) within 54 weeks after enrollment, and then every 12 weeks (± 7 days). In Phase II, tumor assessments will be performed every 8 weeks (±7 days) within 56 weeks after enrollment, and then every 12 weeks (±7 days). For subjects receiving the treatment, the clinical decision will be made based on the response assessed by the investigator per RECIST version 1.1, including confirmation of progressive disease. A subject may continue receiving treatment between the initial assessment of progression and confirmation of progression if his/her performance status remains stable, the subject will benefit from ongoing treatment in the opinion of the investigator, and the subject meet other criteria listed in the protocol.

After completion of treatment, each subject will be followed-up for adverse event (AE) for at least 30 days and for serious adverse event(SAE) for at least 90 days or until the start of other anti-cancer treatments, whichever occurs first. Immune-related AEs (irAEs), regardless of serious or non-serious irAEs, will be followed until 90 days after the completion of treatment (whether other anti-cancer treatments have been initiated). A survival assessment will be conducted every 3 months after discontinuation of study drug to collect survival information and anti-cancer treatment information after discontinuation of study drug.

Dose escalation period (Phase Ib Part 1)

Cohort 1: AK104 monotherapy group, 6 mg/kg Q2W and 10 mg/kg Q2W, 3 subjects enrolled; Cohorts 2, 3, 4 and 5, AK104 with mXELOX/XELOX chemotherapy, starting dose of 4 mg/kg Q2W, 3 subjects enrolled. Based on the 3 + 3 dose escalation principle, the dose will be escalated in the order of 6 mg/kg Q2W, 10 mg/kg Q2W, and AK104 in combination with mXELOX/XELOX chemotherapy regimen, where the dose of AK104 is 4 mg/kg Q2W, 6 mg/kg Q2W, 10 mg/kg and 15 mg/kg Q3W in sequence.

Each cohort is expected to include at least 3 DLT-evaluable subjects and up to 6 evaluable subjects. In each dose escalation cohort, the first subject will be treated and then after a minimum of 24 hours, the next 2 or 3 subjects will be treated provided that there is no safety concern related to the treatment in the first subject. For subjects in each dose group, the dose level and dosing interval at which AK104 is administered remain unchanged.

After 3 subjects in a dose escalation cohort complete the DLT evaluation period (28 days/21 days after the first dose), dose escalation will continue to the next dose escalation cohort when all available safety data as well as safety data of the previous cohorts and any available PK / PD data obtained have been reviewed and unanimously agreed by the DEC. Dose escalation cohort: If 5 DLT-evaluable subjects out of 6 enrolled subjects at any dose have completed the safety assessment of 28 days/21 days after first dose of AK104 and have no DLT, the study can proceed to the next dose escalation cohort and/or to the dose expansion enrollment. If there are the delayed AEs that occur out of the DLT evaluation period and meet the defined DLT criteria, the DEC will consider these events during data reviews for dose escalation decisions.

Dose expansion period (Phase Ib Part 2)

After tolerability is observed in 3~6 subjects completing dose escalation, the dose group, in which no DLT is observed or DLTs are observed in fewer than 2 out of 6 subjects, will be expanded after discussion with the sponsor and DEC based on preliminary safety and efficacy data. The 6 mg/kg dose group in Cohort 1 can be expanded toa maximum of 40 subjects (including subjects enrolled in the dose escalation period), and the total number of subjects in the 450 mg dose group is 40.

A maximum of 18 subjects (including subjects enrolled in the dose escalation period) will be included in dose expansion group for Cohorts 2, 3, 4 and 5, respectively. If DLT occurs in ≥ 2out of 6 subjects, the dose

group will not be expanded. The sponsor may, at its discretion or after consultation with DEC, decide to discontinue enrollment in certain dose expansion cohorts after reviewing available clinical data and considering subjects' benefit/risk ratio. If the dose expansion Cohorts 2, 3, 4 and 5 are allowed to enroll subjects at the same time, the subjects will be randomly assigned to one of the dose expansion cohorts.

Dose confirmation period (Phase II)

Cohort A: Cervical cancer

This cohort is planned to include 110 subjects with recurrent or metastatic squamous or adenosquamous carcinoma of the cervix, who have progressed during or after treatment with platinum-containing doublet chemotherapy in combination with or without bevacizumab, and have undergone no more than 2 lines of therapy in the recurrent or metastatic phase. The treatment protocol adopts AK104 6 mg/kg once every 2 weeks.

Cohort B: Esophageal squamous cell carcinoma

30 subjects with unresectable locally advanced or metastatic esophageal squamous cell carcinoma who have failed, unable to tolerate or refused standard of care, and received a maximum of one line of treatment during the locally advanced or metastatic phase, will be enrolled into this cohort. The treatment protocol adopts AK104 6 mg/kg once every 2 weeks.

Cohort C: Hepatocellular carcinoma

30 patients with unresectable advanced hepatocellular carcinoma, who are diagnosed as stages C in accordance to Barcelona Clinic Liver Cancer (BCLC) system or stage Bare not suitable for surgery or local treatment, or have progressed after surgery and/or local treatment and are not suitable for radical treatment; who are categorized into Child-Pugh grade A and a better rating of B (≤7 points); and who failed, are unable to tolerate or refuse standard of care, and receive a maximum of one line of treatment during advanced stage, will be enrolled into the cohort. The treatment protocol adopts AK104 6 mg/kg once every 2 weeks.

Cohort D: Gastric adenocarcinoma and gastroesophageal junction adenocarcinoma

This cohort is planned to enroll two dose groups, each of which includes up to 40 subjects (including subjects receiving the same AK104 dosingregimen in Phase Ib) with unresectable advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma (GEJ) who have not previously received systemic chemotherapy duringadvanced or metastatic gastric adenocarcinoma or GEJ adenocarcinoma, will be enrolled into this cohort..

Enrollment into any cohort at the dose confirmation period may be discontinued at the discretion of the sponsor or after consultation with the investigators if available clinical data of each cohort are reviewed and subjects' risk/benefit ratio are considered, or recruitment rate of subjects from the cohorts is much lower than expectation.

Target subjects

Subjects to be included in this study are male or female aged ≥ 18 years and ≤ 75 years, with good organ function and a performance status score of 0 or 1 as assessed by the Eastern Cooperative Oncology Group (ECOG).

Cohort 1 in Phase Ib: subjects with advanced solid tumors confirmed by histological or cytological examination who failed to standard of care, or who have no standard of care, or who were unable to tolerate standard of care or refused to accept standard of care.

Cohort 2, 3, 4 and 5 in Phase Ib, and Cohort D in Phase II: Subjects with histologically or cytologically confirmed unresectable advanced or metastatic gastric adenocarcinoma or gastroesophageal junction (GEJ) adenocarcinoma (Subjects with squamous cells or undifferentiated or other histological types of stomach or GEJ cancer are not allowed to be enrolled). Subjects have not previously received systemic chemotherapy for advanced or metastatic gastric adenocarcinoma or GEJ adenocarcinoma. The subjects who have previously received adjuvant chemotherapy, neoadjuvant chemotherapy, or radical chemoradiotherapy with the aim of cure will be eligible, provided that the disease progression occurs > 6 months after the completion of the last treatment. Subjects of gastric or GEJ adenocarcinomas withHER2-positive are not allowed to be enrolled.

Cohort A in Phase II: subjects with histologically or cytologically confirmed recurrent or metastatic cervical cancer (pathological types include squamous or adenosquamous cell carcinoma); with radiographically confirmed disease progression during or after treatment with platinum-containing doublet chemotherapy with or without bevacizumab regimen; having received no more than 2 lines of therapies for recurrent or metastatic disease and with radiographically confirmed disease progression during or after the most recent treatment period.

Cohort B in Phase II: Subjects with histologically or cytologically confirmed unresectable locally advanced or metastatic esophageal squamous cell carcinoma; who failed to standard of care, or who were unable to tolerate standard of care or refused to accept standard of care, and have received a maximum of one line of therapy; if disease progression occurs during neoadjuvant/adjuvant chemotherapy or radical chemoradiotherapy or within 6 months after the completion of treatment, the subject is considered to have received first-line treatment.

Cohort C in Phase II: Subjects with histologically or cytologically confirmed unresectable advanced hepatocellular carcinoma; and subjects who are diagnosed as Stages C in accordance to Barcelona Clinic Liver Cancer (BCLC) system or Stages B and are not suitable for surgery or local treatment, or have progression after surgery and/or local treatment and are not suitable for radical treatment; who are categorized into Child-Pugh grade A and a better gradeof B (≤7 points); and who failed standard of care, who weree unable to tolerate or refuse to accept standard of care, and receive a maximum of one line of treatment during advanced stage.

Subjects have at least one measurable tumor lesion as per RECIST v1.1 criteria; subjects in Cohort A should have measurable disease confirmed by IRRC. A a measurable lesion that has been previously treated with local therapies such as radiotherapy can be considered a target lesion provided if the disease progression has been confirmed by radiographically .

Subjects must provide archived or freshly obtained tumor tissue samples after a definite diagnosis, and approximately 10 (at least 5 if approved by the medical monitor) unstained FFPE pathology slides (preferably freshly obtained tumor tissue samples).

Treatment Regimen

The AK104 intravenous infusion should be kept on the same day as the oxaliplatin intravenous infusion. AK104 will be administered intravenously at least 30 minutes before oxaliplatin intravenous injection and capecitabine administration. The administration time window is ± 3 days, but subjects must complete various examinations including vital signs, physical examination, laboratory examination, and physical status score to evaluate the safety and tolerability of continued treatment within 3 days before each administration.

<u>AK104</u>

During the dose escalation and dose expansion periods, subjects with solid tumors in Cohort 1 will be intravenously administered AK104 as a monotherapy at doses of 6 mg/kg Q2W, 10 mg/kg Q2W, 450 mg Q2W, respectively. Subjects in Cohorts 2, 3 and 4 will be administered with AK104 at doses of 4 mg/kg Q2W, 6 mg/kg Q2W, and 10 mg/kg Q2W in combination with mXELOX chemotherapy regimen, respectively. Subjects in Cohort 5 will be administered with AK104 at dose of 15 mg/kg Q3W in combination with XELOX chemotherapy regimen. In the dose confirmation period, subjects in Cohorts A, B and C will be treated with AK104 monotherapy at 6 mg/kg Q2W, and subjects in Cohort D will be treated with AK104 6mg/kg Q2W combined with mXELOX, AK104 10mg/kg Q3W combined with XELOX chemotherapy.

Subjects will receive AK104 combined with mXELOX/XELOX chemotherapy until the observation of unacceptable toxicity or evidence of confirmed disease progression, or withdrawal of informed consent. Subjects may continue to receive AK104 monotherapy between initial assessment of disease progression and confirmation of progression if the subject's performance status remains stable, and the subject can benefit from ongoing AK104 treatment in the opinion of the investigator, and other criteria listed in the protocol are met. Subjects will be permanently discontinued from AK104 monotherapy if their disease progressed again after continuing to receive AK104 monotherapy. In this study, the longest duration of AK104 treatment will be 24 months.

AK104 will be administered via infusion over 60 minutes (± 10 minutes). Ongoing monitoring of potential

infusion reactions will be performed and pre-medication of prophylaxis and/or adjustment of infusion rate will be allowed as per the protocol guidelines. For subjects who cannot tolerate a 60-minute infusion or may have infusion-related risks, the infusion time may be extended to a maximum of 120 minutes (±15 minutes). Dose modification is not allowed for AK104, but dose delay is allowed for a maximum of 12 weeks, calculated from the last planned administration time, and otherwise treatment should be discontinued.

mXELOX and XELOX Regimens

Cohorts 2-4, mXELOX regimen

- Oxaliplatin: 85 mg/m^2 intravenous infusion for 2-6 hours on Day 1 of each2-week treatment cycle;
- Capecitabine: 1000 mg/m^2 orally, twice daily on Days 1 to 10 of each 2-week treatment cycle.

During the first 10-12 cycles (every 2 weeks as one cycle), mXELOX chemotherapy can only be discontinued due to disease progression, unacceptable toxicity, or withdrawal of informed consent. After Cycles 10-12, if there is no confirmed disease progression or unacceptable toxicity, the subject and investigator can decide whether to continue the maintenance of capecitabine until disease progression, unacceptable toxicity, or subject`s request or at the discretion of investigator.

Cohort 5, XELOX regimen

- Oxaliplatin: 130 mg/m^2 intravenous infusion for 2-6 hours on Day 1 of each 3-week treatment cycle;
- Capecitabine: 1000 mg/m^2 orally, twice daily on Days 1 to 14 of each 3-week treatment cycle.

Chemotherapy (XELOX) can only be discontinued due to disease progression, unacceptable toxicity, or withdrawal of consent prior to the end of Cycle 6 (every 3 weeks as one cycle). After Cycle 6, if there is no confirmed disease progression or unacceptable toxicity, the subject and his/her investigator can decide whether to continue the maintenance of capecitabine until disease progression, unacceptable toxicity, or subject`s request or at the discretion of investigator.

Treatment of the toxicities associated with the listed drugs, including dose delay, dose reductions, and dose modifications, should be based on the instructions in oxaliplatin and capecitabine's prescription information and standard clinical practice. Under no circumstances should capecitabine be administered more than twice daily.

Statistical Methodology

Estimation of Sample Size

Approximately 372 subjects are planned to be enrolled to participate in the dose escalation and expansion period (about 158 subjects) and the dose confirmation period (about 214 subjects) of the study. More subjects may be required for the dose escalation and expansion periods if other dose escalation cohorts or dosing regimens are to be explored.

For the dose-escalation cohorts, the number of subjects to be enrolled will depend upon the toxicities observed as the study progresses. Dose escalation will follow the traditional 3 + 3 design, and at least 3 evaluable subjects and up to 6 evaluable subjects are expected to be included at each dose level. For the dose group with no DLT observed or DLT observed in fewer than 2 out of 6 subjects after tolerability observation for 3~6 subjects completing dose escalation, the group could be expanded to include additional subjects based on preliminary safety and efficacy data, after discussion with the sponsor and DEC. Up to 40 subjects (including subjects enrolled in the dose escalation period) will be included into the 6 mg/kg Q2W dose expansion group and 450 mg Q2W dose expansion group for Cohort 1, respectively, and up to 18 subjects (including subjects enrolled in the dose escalation period) will be included into the dose group for Cohort 2, 3, 4 and 5 respectively.

It is planned to enroll 110 subjects with cervical cancer in Cohort A (to ensure at least 100 subjects in the FAS set), 30 subjects with esophageal squamous cell carcinoma in Cohort B and 30 subjects with hepatocellular carcinoma in Cohort C during the dose confirmation period of this study. Up to 40 subjects (including subjects receiving the same dosing regimen in Phase Ib) will be planned to be enrolled in each of the two dose groups of cohort D for efficacy observation. In the ORR superiority test at a one-sided alpha

level of 0.05, when the sample size is 40 subjects, there will be a power of 83% to detect that the ORR evaluating AK104 combined with mXELOX/XELOX as the first-line therapy for gastric adenocarcinoma (GAC) or GEJ adenocarcinoma increases from 40% (assuming ORR of historical control mXELOX/XELOX first-line therapy for GAC or GEJ adenocarcinoma) to 60% (AK104 drug combination).

Statistical Analysis

The safety analysis set includes all enrolled subjects who have received any doseof the study drug and undergone at least one safety assessment. Analysis is based on the actual treatment received by the subject. The evaluation analysis of response-related efficacy endpoints is based on the full analysis set, which includes all subjects enrolled in the analysis set who have received any doseof AK104 and have baseline disease assessment data with measurable disease (defined as per RECIST v1.1 criteria).

The safety and efficacy data of the dose group cohorts in the dose escalation and expansion period (Phase Ib) of this study will be pooled and summarized according to the dose group and summary method. In addition, safety and efficacy data from the same dosing regimens in Phase Ib and II studies will be pooled and summarized.

Safety Analysis

Safety analysis will be performed using the safety analysis set according to the treatment actually received by the subjects. Safety analysis usually summarizes descriptive information appropriately in a table by dose group.

The summary statistical results of AE will be listed according to the seriousness (SAE), severity and causality with the test drug, and AEs and AESIs that caused the discontinuation will also be summarized. AEs will be graded according to version 5.0 of the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE), and will be described with the preferred term in the Medical Dictionary for Regulatory Activities (MedDRA). Laboratory abnormalities and toxicity levels will be derived and summarized according to NCI CTCAE version 5.0. A two-way frequency table of baseline and worst post-treatment values based on CTCAE grade will be shown in the transitional table of clinical laboratory tests.

Efficacy Analysis

The primary endpoint of this study's dose confirmation period (phase II) is the ORR assessed per RECIST v1.1. The analysis of the primary endpoint will be performed after all subjects who meet the statistical and radiographic RECIST v1.1 assessment requirements are withdrawn from the study or complete at least 24 weeks of follow-up (if the subject have achieved PD before Week 24 , he/she will needn't to wait for a follow-up assessment at week 24). The analysis for ORR and DCR will be based on the full analysis set (FAS). For Cohort A in Phase II, its 95% exact 2-sided confidence interval can be estimated using the Clopper Pearson method and the 80% exact 2-sided confidence interval for the other cohorts can be estimated using the Clopper Pearson method. Time to event endpoints (DoR, TTR, PFS and OS) will be analyzed using the Kaplan-Meier method. PFS and OS analysis will be performed based on the safety analysis set. The graphical analysis will include a spider plotsof the tumor burden percentage of the target lesions changing from baseline over time, and a waterfall plots of the bestpercentage of the target lesions tumor burden changing from baseline.

Pharmacokinetic Analysis

Individual AK104 concentrations will be tabulated by dose cohort along with descriptive statistics. Noncompartmental PK data analysis and planned PK sample collection will be performed for each dose cohort where data is available. The relevant descriptive statistics of the PK parameters of the non-compartmental model may include AUC, C_{max}, CL, and $t_{1/2}$. Linearity of PK parameters (AUC and C_{max}) will be explored. A population PK model will be established in this trial. The impact of physiologically relevant subject characteristics (covariates) and disease characteristics on PK will be evaluated. The relationship between exposure and response or safety will be evaluated. Such analysis can be described in PK/PD analysis plans for different populations, and the result report of this exploratory analysis can be separated from the main study report.

Immunogenicity Analysis

The immunogenic potential of AK104 will be assessed by summarizing the number and percentage of subjects who develop detectable ADA. The impact of ADA on PK will be assessed if data is available. Samples will be collected for evaluating the neutralizing capacity of ADAs in the future. Only subjects who have received at least one dose of AK104 and provided a baseline sample and at least one post-treatment sample will be evaluated.

Pharmacodynamics/Biomarker Analysis

The frequency change of CD4+ Ki67+ (as primary pharmacological biomarker) proliferative cell from baseline over time is analyzed with dose cohort and scheduled evaluation time. Other pharmacodynamic markers such as PD-1 or CTLA-4 receptor occupancy and CD8+ Ki67+ will be analyzed similarly. The relationship between exposure and pharmacodynamic biomarkers will be explored.

The relationship between PD-L1 expression and efficacy as a predictive biomarker in the archived tumor tissue samples or baseline fresh biopsy samples will be analyzed. PD-L1 expression will be assessed by the central laboratory for the determination of IHC of PD-L1. For Cohort A in Phase II, the ORR of subjects with PD-L1 positive (CPS ≥ 1) and PD-L1 negative (CPS < 1) squamous/adenosquamous carcinoma of the cervix will be assessed by IRRC based on RECIST 1.1.

If additional tumor tissue samples and data are available, other possible predictive biomarkers in the tumor tissue samples will be explored and analyzed, including but not limited to the type and number of tumor infiltrating lymphaytes (TIL) and correlation analysis between immune-related gene mRNA expression characteristics with AK104 antitumor activity.

Interim Analysis

For Cohort A in Phase II, an interim analysis may be planned after at least 70 subjects have been enrolled and these subjects have completed at least 16 weeks of follow-up (interval between the data cutoff date and the first dose date ≥ 16 weeks), and the results of the interim analysis will likely be used to communicate with regulatory authorities. The Sponsor dose not plan to prematurely discontinue the study based on these efficacy or safety findings.

Dose Escalation Committee (DEC) and Safety Review

During the dose escalation and expansion periods (Phase Ib), the Dose Escalation Committee (DEC) will conduct ongoing safety oversight over the study and carry out planned reviews on a regular basis to safety and other relevant data. The DEC includes, at minimum, the study chair, the medical monitor, the Sponsor's representative and other Principal Investigators or their representatives (from each site with subjects enrolled in this study). The DEC will be responsible for making decisions on dose escalation and expansion.

After initiation of enrollment of subjects in the dose confirmation period (Phase II) of this study and the formation of a safety review team, the team will replace the DEC for ongoing safety oversight over the study and carry out planned reviews on a regular basis to safety and other relevant data. The safety review team includes, at minimum, the study chair, medical monitors, sponsor representatives, and other medical oncologists.

Following safety review and discussion, the sponsor will make all final decisions regarding any change in study conduct.

List of Abbreviations

Abbreviations or Specific Terms	Definition
ADA	Anti-drug antibody
ADL	Activities of daily living
AE	Adverse event
AESI	Adverse event of special interest
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
AST	Aspartate aminotransferase
AUC	Area under concentration-time curve
BsAb	Bispecific antibody
BID	Bis in die (twice a day)
BOR	Best overall response
CI	Confidence interval
CL	Clearance
C_{max}	Maximum concentration
CNS	Central nervous system
CR	Complete response
CRO	Contract research organization
CrCl	Creatinine clearance
CT	Computed tomography
ctDNA	circulating tumor DNA
CTLA-4	Cytotoxic T lymphocyte associated antigen 4
DCR	Disease control rate
DEC	Dose Escalation Committee
DLT	Dose limiting toxicity
DLCO	Diffusion capacity for carbon monoxide
DoR	Duration of response
DPD	Dihydropyrimidine dehydrogenase
EBV	Epstein-Barr virus
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	electronic Case Report Form
EDC	Electronic data capture
FBC	Full blood count
Fc	Fragment crystallizable
FDA	Food and Drug Administration
FFPE	Formalin fixed paraffin embedded

FoxP3	Fork head box P3
FP	5-Fluorouracil + Cisplatin
GCP	Good Clinical Practice
GEJ	Gastroesophageal junction
GLP	Good Laboratory Practices
HBV	Hepatitis B virus
HCC	Hepatocellular carcinoma
HCV	Hepatitis C virus
HIV	Human immunodeficiency virus
HNSTD	Highest non-severely toxic dose
HR	Hazard ratio
ICF	Informed consent form
ICOS	Inducible co-stimulator
ICH	International Conference on Harmonization
IEC	Independent Ethics Committee
IgG1	Immunglobulin G1
IHC	Immunohistochemistry
irAE	Immune-related adverse event
INR	International normalized ratio, an anticoagulation parameter
IRB	Institutional Review Board
ISH	in situ hybridization
irCR	Immune-related complete response
irPD	Immune-related progressive disease
irPR	Immune-related partial response
IRRC	Independent Radiology Review Committee
irRECIST	immune-related Response Evaluation Criteria in Solid Tumors
irSD	Immune-related stable disease
IV	Intravenous(ly)
IVIG	Intravenous immuneglobulin
LLN	Lower limit of normal
LFTs	Liver function tests
MAD	Maximum administered dose
MedDRA	Medical Dictionary for Regulatory Activities
MRA	MRI angiography
MRI	Magnetic resonance imaging
mRNA	messenger RNA
MSI	Microsatellite Instability
MTD	Maximum tolerated dose
dMMR	deficient Mismatch Repair
mXELOX	modified Oxaliplatin + Capecitabine

NCI CTCAE	National Cancer Institute-Common Terminology Criteria for Adverse Events
NSAID	Non-steroidal anti-inflammatory drugs
ORR	Objective response rate
OS	Overall survival
PBMC	Peripheral blood mononuclear cell
PCR	Polymerase chain reaction
PD	Progressive disease
PD-1	Programmed cell death-1
PD-L1	Programmed cell death ligand-1
PD-L2	Programmed cell death ligand-2
PET	Positron emission tomography
PFS	Progression-free survival
PK	Pharmacokinetics
PR	Partial response
Q2W	Every two weeks
QTcB	QT interval corrected for heart rate using Bazett's formula
QTcF	QT interval corrected for heart rate using Fridericia's formula
RCC	Renal cell carcinoma
RECIST	Response Evaluation Criteria in Solid Tumors
RO	Receptor occupancy
RP2D	Recommended Phase II Dose
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SCr	Serum Creatinine
SD	Stable disease
SID	Subject Identification
$t_{1/2}$	Half life
TBL	Total bilirubin level
TRAE	Treatment-related adverse events
TCGA	The Cancer Genome Atlas
TFT	Thyroid function test
TIL	Tumor infiltrating lymphocytes
TSH	Thyroid stimulating hormone
ULN	Upper limit of normal
UEC	Urea-electrolyte-creatinine
WHO-DD	World Health Organization Drug Dictionary
XELOX	Oxaliplatin + Capecitabine
XP	Capecitabine + Cisplatin

1. Study Background

1.1. Investigational Drug

1.1.1. Name

AK104

1.1.2. Description

AK104 is a humanized immunoglobulin G1 (IgG1) bispecific antibody (BsAb) with fragment crystallizable (Fc) mutations to eliminate Fc receptor- and complement-mediated cytotoxic effects. AK104 binds to both programmed cell death receptor-1 (PD-1) and cytotoxic T lymphocyte associated antigen-4 (CTLA-4) and blocks PD-1/programmed cell death ligand-1 (PD-L1), PD-1/PD-L2, CTLA-4/B7.1 and CTLA-4/B7.2 interactions.

AK104 is a tetravalent IgG-ScFv bispecific antibody. AK104 contains a point mutation in its constant region that prevents binding of complement C1q to Fc gamma receptor involved in cytotoxic effects. AK104 is expressed in a Chinese hamster ovary cell line with a total molecular weight (including that of oligosaccharides) of approximately 200 kDa.

1.1.3. Intended Use Under Investigation

The target indications of AK104 are treatment of recurrent or metastatic cervical cancer, unresectable locally advanced or metastatic esophageal squamous cell carcinoma, and unresectable advanced hepatocellular carcinoma as monotherapy, and treatment of unresectable advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma in combination with oxaliplatin and capecitabine as first-line therapy.

1.1.4. Summary of Preclinical Studies

For complete details on the results of preclinical studies, please refer to the Investigator's Brochure.

1.1.4.1. Summary of Preclinical Pharmacokinetics

The pharmacokinetic properties of AK104 were investigated in cynomolgus monkeys. Following single intravenous administration of AK104 injection at 1, 4, and 16 mg/kg to cynomolgus monkeys, the clearance (Cl) observed was 2.2, 1.6, and 1.9 ml/h/kg, respectively, and the apparent volume of distribution (Vss_obs) observed were 91, 92, and 113 mL/kg, respectively, indicating that AK104 was significantly distributed in tissues. The half-life ($t_{1/2}$) after administration of 4 mg/kg AK104 was 47.9 hours, approximately 1/3 to 1/2 of that of typical antibodies. It should be noted that the volume of distribution of AK104 in monkeys was significantly greater than the blood volume, indicating significant tissue distribution. Rapid clearance from blood can reduce nonspecific (off-target) cytotoxicity. The preferential distribution of AK104 in tumor tissues may lead to better safety than the conventional anti-PD-1 and anti-CTLA-4 antibodies described above.

A nearly dose-proportional increase between the exposure (C_{max} and AUC) and dose was

observed in the dose range from 1 to 16 mg kg. The increase in exposure (AUC_{last}) to AK104 increased dose proportionally with a power function model beta of 1.05, suggesting a linear pharmacokinetic profile of AK104.

A multiple dose PK study of AK104 was conducted in cynomolgus monkeys. After 4 weeks of weekly intravenous injection of 4 mg/kg AK104, the ratios of AUC 0-168h and C_{max} after the last and first dose were 0.554 and 0.230, respectively, suggesting that anti-drug antibodies may interfere with the detection of analytes or accelerate the clearance of AK104, which resulted in a decrease in AK104 exposure after the last dose compared to that after the first dose.

1.1.4.2. Summary of Preclinical Pharmacology

PD-1 is a member of the immunoglobulin superfamily (IgSF) that regulates the activation of T cells. PD-1 can be expressed in T cells, B cells, macrophages, NK cells, dendritic cells and mast cells. PD-1 inhibits activation of T cells by down regulating cytokine production, cell proliferation, cell survival, and transcription factors associated with the functions of effector T cells.

CTLA-4 is a cell surface receptor mainly expressed on activated T cells. CTLA-4 binds to its ligands B7.1 and B7.2 to produce inhibitory signals, thus inhibiting T cell activity and cytokine synthesis, and limiting cell proliferation. After T cell activation, CTLA-4 expression is upregulated and plays a role in inhibiting the immune response, regulating and ultimately shutting down T cell activation. Binding of CD80/86 to CTLA-4 provides a negative regulatory signal for T cells.

AK104 exhibits the following activities as a bispecific antibody (BsAb):

- AK104 can bind to PD-1 and CTLA-4 simultaneously with high affinity.
- AK104 can effectively block the PD-1/PD-L1, PD-1/PD-L2, CTLA-4/B7.1 and CTLA-4/B7.2 interactions.
- AK104 can significantly increase the secretion of IL-2 and IFN-γ by human primary T cells cocultured with cells expressing B7 (e.g., Raji cells) or cells expressing PD-L1 (e.g., dendritic cells).
- AK104 can inhibit tumor growth in several xenograft models, in which human peripheral blood mononuclear cells (PBMCs) were injected into immunocompromised mice to determine the inhibition on inoculated tumor cells, including hepatocellular carcinoma (HCC) 827 cells and Raji tumor cell lines which have stable PD-L1 transduction .
- AK104 can inhibit tumor growth in an murine colorectal (MC) 38 allograft tumor model developed in the human PD-1 knock-in mice.

1.1.4.3. Summary of Preclinical Toxicology

The cynomolgus monkey is considered the only pharmacologically relevant animal species in

the preclinical safety study of AK104. The highest dose of AK104 was 60 mg/kg/week in a 4-week repeated-dose GLP toxicity study of AK104 injection administered intravenously to cynomolgus monkeys followed by a 4-week recovery period.

AK104 injection was well tolerated at intravenous doses up to 60 mg/kg. Elevated ALT and/or AST and a decrease in the percentage of CD16+ cells were observed at the end of the dosing period, and at the end of the recovery period, the recovery of ALT and AST parameters was observed, but a trend towards a decrease in CD16+ was still observed. Histopathological changes related to AK104 were mainly characterized by slight to mild lymphocytic infiltration in the sciatic nerve, interstitium of lacrimal glands, interstitium of salivary glands (submandibular gland) and renal interstitium. At the end of the recovery period, slight to mild lymphocytic infiltration was still observed in the interstitium of the salivary glands (submandibular glands) as well as in the renal interstitium. Therefore, the highest non-severely toxic dose (HNSTD) was considered to be 60 mg/kg, corresponding to an AUC (mean exposure in both male and female animals after repeated dosing) of 19,514 μg*hr/mL.

1.1.5. Potential differences between AK104 and anti-PD-1/L1 in combination with anti-CTLA-4 antibody

The bispecific antibody AK104, which targets both PD-1 and CTLA-4, has a quadrivalent structure, with a shorter half-life ($t_{1/2}$) and lower toxicity in monkey toxicity studies compared to anti-PD-1/L1 in combination with anti-CTLA-4 antibody. The use of AK104 in cancer patients may have better efficacy and safety than that achieved by anti-PD-1/L1 in combination with anti-CTLA-4 antibody.

1.1.5.1. Higher PD-1 and CTLA-4 coexpression in tumor tissue than that in normal tissue and its possible effect on enhanced efficacy

The level of coexpressed PD-1 and CTLA-4 in tumor infiltrating lymphocytes (TILs) is higher than that in normal tissues and peripheral blood cells (**Ahmadzadeh M, 2009**). A search of the public TCGA database revealed that PD-1 and CTLA-4 messenger RNA (mRNA) are coexpressed in a large number of solid tumors (**Hedvat M, 2016**). It is well known that bispecific antibodies tend to have high affinity and long-term target retention due to their multivalent nature (**Holliger P, 2005**). The higher coexpression of PD-1 and CTLA-4 in tumor tissue compared to that in normal tissue has this high affinity and high retention effect more enriched in tumor tissue. In contrast, anti-PD-1 in combination with anti-CTLA-4 antibody does not have this advantage. In addition, CTLA-4 blocking antibodies have been shown to upregulate PD-1, whereas PD-1 blocking antibodies upregulate CTLA-4 in some studies (**Curran, MA, 2010**). Prolongation of tumor retention by tetravalent PD-1/CTLA-4 bispecific antibodies such as AK104 can be enhanced by cross-upregulation of PD-1 and CTLA-4. Thus, the tetravalent bispecific antibodies targeting PD-1 and CTLA-4 can provide a therapeutic advantage superior to that provided by the combination of anti-PD1/L1 and anti-CTLA-4.

1.1.5.2. AK104 showed lower toxicity than anti-PD-1 and anti-CTLA-4 antibodies in monkey toxicity studies

The results of the 4-week repeated-dose toxicity study of AK104 in monkeys were compared to those of the 4-week repeated-dose toxicity study of Nivolumab combined with Ipilimumab in monkeys. The comparison of results across studies may be confusing, but the safety results are compelling when compared indirectly with other studies in this case. Less toxicities were observed with AK104 compared to the combination of Nivolumab and Ipilimumab.

Of note, the monkey toxity study in which Ipilimumab 3 mg/kg was administered weekly in combination with nivolumb 10 mg/kg once weekly (BLA # 125377 Study No. SUV00106 [FDA, 2014]) showed severe toxicity, including treatment-related diarrhea, fluid swelling and darkening of the large intestine, infiltration of mononuclear cells of multiple organs into the liver and kidney, inflammatory changes in the large intestine including infiltration of lymphocytes, eosinophils and neutrophils, and mucosal epithelial degeneration.

Additional toxicities including watery stools, decreased food consumption, body weight loss, and pancreatic degranulation of acinar cells were noted in the high dose group of Ipilimumab 10 mg/kg in combination with nivolumb 50 mg/kg in this study. One animal in this group died due to acute gastric dilatation.

The antibody dose of 60 mg/kg used in the AK104 repeated dose toxicity study in monkeys was similar in molar concentration (equivalent to 45 mg/kg of conventional antibody) to 50 mg/kg of Nivolumab (an anti-PD-1) in the high dose group of the SUV00106 study, and was well above the dose of Ipilimumab (an anti-CTLA-4) in the high dose group of the SUV00106 study. The dosing regimen (4 IV doses per week) in the SUV00106 study was the same as that in the AK104 study.

1.1.6. Summary of Clinical Studies

AK104 is a bispecific antibody (BsAb) capable of binding both PD-1 and CTLA-4 with high affinity. AK104 is a novel tumor immunotherapeutic agent with a quadrivalent structure and a short half-life. It was shown to be less toxic than the combination of anti-PD-1 and anti-CTLA-4 antibody in monkey toxicity studies. These characteristic features may enhance the efficacy of the combination of anti-PD-1/L1 and anti-CTLA-4 antibody and reduce treatment-related toxicities. For details on safety findings from clinical studies of anti-PD-1/L1 in combination of anti-CTLA-4 antibody, please refer to the Investigator's Brochure. For a detailed summary of potential differences between AK104 and anti-PD-1/L1 in combination with anti-CTLA-4 antibody, please see Section 1.1.5.

At present, no clinical studies of any BsAb with similar mechanisms to AK104 have been published. For potential treatment risks associated with AK104, please refer to 1.3.1.

Multiple Phase I and II clinical studies of AK104 have been conducted at home and abroad, involving the treatment regimens of AK104 monotherapy and AK104 in combination with chemotherapy. As of March, 2021, four studies of AK104 have involved 446 subjects with

advanced solid tumors treated with AK104 as monotherapy and 79 subjects with first-line gastric cancer treated with AK104 in combination with chemotherapy. Given that source data verification (SDV) has not been 100% completed currently, the preliminary analysis is as follows:

1.1.6.1. Preliminary Safety

In the dose escalation study of AK104 alone, the escalation from 0.2 mg/kg Q2W to 10 mg/kg Q2W, 15 mg/kg Q3W, and 25 mg/kg Q3W has completed. Only 1 subject in 1.0 mg/kg group had a DLT event: grade 3 AST increased. The MTD was not reached at any dose level.

Table 1-1 AK104 Monotherapy-Treatment-related Adverse Events (TRAEs) with Incidence ≥ 5%

	Total (N=446)	**6 mg/kg Q2W (N=325)**	**450 mg Q2W (N=58)**	**10 mg/kg Q2W (N=21)**	**15 mg/kg Q3W (N=12)**	**25 mg/kg Q3W (N=14)**
At least one TRAE	80.0%	80.3%	81.0%	85.7%	75.0%	78.6%
Hypothyroidism	15.2%	16.6%	10.3%	14.3%	16.7%	14.3%
Rash	14.6%	12.9%	20.7%	23.8%	16.7%	14.3%
Anaemia	11.7%	13.5%	12.1%	0.0%	8.3%	0.0%
ALT increased	11.7%	12.9%	10.3%	4.8%	0.0%	7.1%
AST increased	11.0%	11.7%	15.5%	0.0%	0.0%	0.0%
Pruritus	9.9%	8.0%	13.8%	28.6%	25.0%	0.0%
Hyperthyroidism	9.0%	8.9%	3.4%	9.5%	25.0%	14.3%
Pyrexia	8.5%	8.3%	8.6%	19.0%	8.3%	0.0%
Proteinuria	8.1%	8.6%	13.8%	0.0%	0.0%	0.0%
Leukocyte count decreased	7.8%	10.8%	0.0%	0.0%	0.0%	0.0%
Infusion-related reactions	7.2%	6.5%	6.9%	9.5%	16.7%	0.0%
Nausea	7.0%	6.2%	8.6%	9.5%	8.3%	7.1%
Diarrhoea	5.8%	5.2%	5.2%	4.8%	25.0%	0.0%
Neutrophil count decreased	5.2%	7.1%	0.0%	0.0%	0.0%	0.0%

In the escalation study of AK104 in combination with chemotherapy, the escalation to 4 mg/kg, 6 mg/kg and 10 mg/kg Q2W in combination with chemotherapy has completed, and no DLTs were observed in any dose group. The MTD was not reached at any dose level.

Table 1-2 AK104 Combination therapy- Safety overall table

	Total (N=79)	**4 mg/kg** combination with chemotherapy **(N=18)**	**6 mg/kg** combination with chemotherapy **(N=40)**	**10 mg/kg** combination with chemotherapy **(N=18)**	**10 mg/kg** combination with chemotherapy **(N=18)**

	Total (N=79)	**4 mg/kg** combination with chemotherapy **(N=18)**	**6 mg/kg** combination with chemotherapy **(N=40)**	**10 mg/kg** combination with chemotherapy **(N=18)**	**10 mg/kg** combination with chemotherapy **(N=18)**
TRAE	75 (94.9)	18 (100.0)	38 (95.0)	18 (100.0)	1 (33.3)
TRAE associated with AK104	66 (83.5)	16 (88.9)	33 (82.5)	16 (88.9)	1 (33.3)
≥ Grade 3 TRAE	41 (51.9)	12 (66.7)	21 (52.5)	8 (44.4)	0 (0.0)
≥ Grade 3 TRAE associated with AK104	27 (34.2)	8 (44.4)	12 (30.0)	7 (38.9)	0 (0.0)
irAE	41 (51.9)	11 (61.1)	20 (50.0)	9 (50.0)	1 (33.3)
≥ Grade 3 irAE	14 (17.7)	3 (16.7)	7 (17.5)	4 (22.2)	0 (0.0)
TRSAE	29 (36.7)	7 (38.9)	14 (35.0)	8 (44.4)	0 (0.0)
TRAE leading to drug discontinuation	3 (3.8)	1 (5.6)	1 (2.5)	1 (5.6)	0 (0.0)

Table 1-3 AK104 Combination Therapy-TRAEs with Incidence ≥ 15%

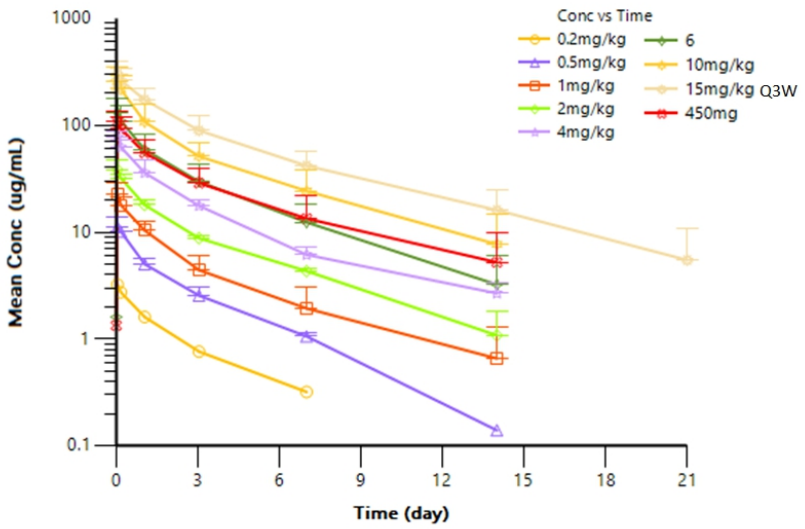
	Total (N=79)		**4 mg/kg** combination with chemotherapy **(N=18)**		**6 mg/kg** combination with chemotherapy **(N=40)**		**10 mg/kg** combination with chemotherapy **(N=18)**		**15 mg/kg** combination with chemotherapy **(N=3)**	
	Any grade n (%)	**Grade 3/4 n (%)**	**Any grade n (%)**	**Grade 3/4 n (%)**	**Any grade n (%)**	**Grade 3/4 n (%)**	**Any grade n (%)**	**Grade 3/4 n (%)**	**Any grade n (%)**	**Grade 3/4 n**
At least one TRAE	75 (94.9)	41 (51.9)	18 (100.0)	12 (66.7)	38 (95.0)	21 (52.5)	18 (100.0)	8 (44.4)	1 (33.3)	0 (0.0)
Leukocyte count decreased	42 (53.2)	5 (6.3)	12 (66.7)	3 (16.7)	20 (50.0)	1 (2.5)	10 (55.6)	1 (5.6)	0 (0.0)	0 (0.0)
Neutrophil count decreased	41 (51.9)	10 (12.7)	12 (66.7)	6 (33.3)	22 (55.0)	4 (10.0)	7 (38.9)	0 (0.0)	0 (0.0)	0 (0.0)

Platelet count decreased	37 (46.8)	7 (8.9)	10 (55.6)	1 (5.6)	20 (50.0)	5 (12.5)	7 (38.9)	1 (5.6)	0 (0.0)	0 (0.0)
Anaemia	32 (40.5)	5 (6.3)	6 (33.3)	0 (0.0)	21 (52.5)	5 (12.5)	5 (27.8)	0 (0.0)	0 (0.0)	0 (0.0)
Vomiting	23 (29.1)	1 (1.3)	8 (44.4)	0 (0.0)	10 (25.0)	1 (2.5)	4 (22.2)	0 (0.0)	1 (33.3)	0 (0.0)
Pyrexia	21 (26.6)	1 (1.3)	3 (16.7)	0 (0.0)	13 (32.5)	1 (2.5)	4 (22.2)	0 (0.0)	1 (33.3)	0 (0.0)
AST increased	19 (24.1)	1 (1.3)	3 (16.7)	1 (5.6)	9 (22.5)	0 (0.0)	7 (38.9)	0 (0.0)	0 (0.0)	0 (0.0)
Nausea	18 (22.8)	1 (1.3)	8 (44.4)	0 (0.0)	6 (15.0)	1 (2.5)	4 (22.2)	0 (0.0)	0 (0.0)	0 (0.0)
Asthenia	14 (17.7)	3 (3.8)	3 (16.7)	1 (5.6)	6 (15.0)	0 (0.0)	5 (27.8)	2 (11.1)	0 (0.0)	0 (0.0)
ALT increased	14 (17.7)	1 (1.3)	3 (16.7)	1 (5.6)	6 (15.0)	0 (0.0)	5 (27.8)	0 (0.0)	0 (0.0)	0 (0.0)
nfusion-related reactions	14 (17.7)	0 (0.0)	6 (33.3)	0 (0.0)	5 (12.5)	0 (0.0)	3 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)
Hypothyroidism	12 (15.2)	0 (0.0)	3 (16.7)	0 (0.0)	4 (10.0)	0 (0.0)	5 (27.8)	0 (0.0)	0 (0.0)	0 (0.0)
Hypoesthesia	12 (15.2)	0 (0.0)	1 (5.6)	0 (0.0)	6 (15.0)	0 (0.0)	5 (27.8)	0 (0.0)	0 (0.0)	0 (0.0)

1.1.6.2. Preliminary Pharmacokinetic Profile

Preliminary pharmacokinetic results showed that the exposure of AK104 increased with increasing dose, showing a linear pharmacokinetic profile (Figure 1-1). No significant differences in pharmacokinetics were found between Caucasians and Chinese. The median trough concentration after reaching steady state at 6 mg/kg Q2W was 5 ug/ml.

Figure 1-1 Pharmacokinetic profiles of different doses of AK104 after single dose administration



1.1.6.3. Preliminary Efficacy

AK104 monotherapy and combination therapy can not only significantly reduce immunotoxicity, but also show better clinical efficacy than the combination therapy of anti-PD-1 antibody and anti-CTLA-4 antibody.

As of May 2021, preliminary results of AK104 in combination with mXELOX/XELOX as the first-line therapy for advanced gastric cancer or gastroesophageal junction adenocarcinoma showed an ORR of 65.8% (48/73) in 73 subjects with first-line gastric cancer who could be evaluated for efficacy; DCR was 93.2% (68/73), with 2 CR, and 46 PRs; there was no correlation between tumor response and PD-L1 expression.

The staged efficacy and safety data of AK104 for recurrent or metastatic squamous carcinoma of the cervix after standard of care showed an ORR of 47.6%, a DCR of 66.7%, the median DoR was not reached (0.03 + to 10.35 months), and the maximum duration of treatment had exceeded 1 year. One subject had stable disease for nearly 1 year with a maximum reduction in target lesions of -25.9%. Compared with the results of PD-1 monotherapy or PD-1 in combination with CTLA-4 in the treatment of cervical cancer, AK104 can show a more potential and sustainable tumor response in subjects who have failed platinum-based therapy. In August 2020, the U.S. Food and Drug Administration (FDA) granted AK104 a fast track qualification for the treatment of advanced squamous carcinoma of the cervix; in October 2020, the Center for Drug Evaluation (CDE) of the National Medical Products Administration also included AK104 in the list of breakthrough therapies for recurrent or metastatic squamous carcinoma of the cervix.

The results of AK104 in the treatment of malignant mesothelioma were also of concern. At the 2020 ESMO meeting, Australian investigators orally reported the results of the study with AK104 monotherapy in 18 subjects with malignant mesothelioma. Of the 15 efficacy evaluable subjects, ORR was 20.0% and DCR was 80.0%. The longest DoR exceeded 12.9 months. One subject who had received anti-PD-1 therapy was treated with 6 mg/kg AK104, and the duration of stable disease was 7.2 months. For subjects at doses above 1 mg/kg, the 9-month progression-free survival rate was 54.5%.

MSI-H/dMMR phenotype, as a biomarker of tumor immunotherapy, has predictive value for efficacy. Preliminary clinical data of AK104 in the treatment of third-line or higher advanced MSI-H solid tumors showed 7 PRs in 7 efficacy evaluable subjects (6 of whom were colorectal cancer), of which 3 subjects with colorectal cancer achieved CR (43%). At present, the sponsor has conducted a neoadjuvant study of AK104 in the treatment of MSI-H gastric cancer and colorectal cancer to preliminarily assess and explore the safety and efficacy of AK104 in perioperative treatment.

1.2. Study Rationale

1.2.1 Disease Treatment Status

Gastric cancer is the fifth most common malignant tumor and the second leading cause of cancer deaths in the world, which is associated with a variety of environmental factors and oncogene mutations. Gastric cancer is one of the most common malignant tumors in China, with an incidence rate of about 36 per 100,000 persons and a mortality rate of 26 per 100,000 persons. Adenocarcinoma is the major histologic subtype, representing approximately 90% of gastric cancer. About two-thirds are true gastric cancer (non-cardia), and the remainder are gastroesophageal junction cancers (cardiac). The most common anatomical subsites differ in world regions due to differences in risk factors.

Even after radical resection, the risk of local recurrence and distant metastasis remains very high. For advanced gastric cancer patients with extensive metastasis, the possibility of cure may be minimal, and the goal of treatment is to control disease and prolong survival. One of the most critical factors affecting the survival in patients with extensive metastatic gastric cancer remains the efficacy of first-line therapy. Asian patients with newly diagnosed inoperable locally advanced or metastatic disease generally receive doublet chemotherapy regimens containing a platinum and a fluorouracil. The current first-line standard of care remains chemotherapy-based, and commonly used doublet regimens include: 5-FU/LV + cisplatin (FP), capecitabine + cisplatin, tegafur gimeracil oteracil potassium + cisplatin, capecitabine + oxaliplatin (XELOX), FOLFOX, capecitabine + paclitaxel, and FOLFIRI. The triplet regimen may provide additional clinical benefit. However, due to increased toxicity, up to 60% of grade 3/4 myelotoxicity limits its clinical use, and no efficacy and safety data of triplet regimen from a large sample in the Asian population are available to date. Response rates in patients receiving first-line chemotherapy regimens are around 30% to 50% with median progression-free survival (PFS) ranging from 5 to 7 months. The median overall survival (OS) is less than 12 months, and less than 10% of patients remain alive in 5 years. Approximately 12% to 20% of patients with gastric adenocarcinomas are HER2-positive. These patients may receive a unique therapy, that is, trastuzumab, an anti-HER2 antibody in combination with platinum- and fluoropyrimidine-based first-line chemotherapy. The traditional three-week XELOX regimen has poor compliance due to toxicity. Only 67% of patients completed the planned cycles, chemotherapy dose changed due to adverse events in 90% of patients, the dose of capecitabine was reduced in 33% of patients, treatment cycle was interrupted in 30% of patients, treatment cycle was delayed in 74% of patients, and oxaliplatin dose was reduced in 33% of patients (**Boutros C, 2016**). The two-week mXELOX regimen is superior to the three-week XELOX regimen in both operability and efficacy (**Yung-Chia Kuo1, 2014**).

The duration of first-line chemotherapy usually does not exceed 6 months due to progressive disease (PD) or the cumulative toxicity of chemotherapy. Therefore, further well-tolerated therapies are encouraged for patients who have achieved maximum tumor shrinkage in first-line chemotherapy. However, there are currently no approved drugs for maintenance treatment after first-line therapy. In second-line therapies, there are several available single-agent chemotherapy regimens including irinotecan, docetaxel, or paclitaxel, which have been shown to improve OS compared with best supportive care. Ramucirumab, an anti-VEGFR-2 antibody, is another option for second-line therapy and has been shown to

provide survival benefits comparable to that achieved in chemotherapy when used alone and to provide survival benefits superior to that achieved with paclitaxel alone when combined with paclitaxel.

1.2.2 Advance in research of immunotherapy for gastric cancer

Immune checkpoints are a class of inhibitory molecules in the immune system. Physiologically, they play an important role in avoiding damage to normal tissues and organs by the immune response. However, in tumors, immune checkpoints can mediate immune escape of tumors, so immune checkpoints have been developed as targets of anti-tumor therapies in recent years. The rationale of immune checkpoint targeted therapy is to mobilize the activity of T cells to kill tumor cells. In recent years, anti-CTLA-4 and anti-PD-1/PD-L1 drugs have gained some indications in the treatment of a variety of malignant tumors. Anti-PD-1/PD-L1 therapy has great potential in the treatment of gastric cancer. The treatment modalities explored range from monotherapy to various forms of combination therapies, and the study population moves from last-line treatment to second-line, first-line, maintenance and even adjuvant and neoadjuvant therapy. At present, the two PD1 monoclonal antibodies approved in the United States or Japan are Nivolumab and Pembrolizumab. Pembrolizumab is approved for the second-line and subsequent treatment of MSI-high or dMMR advanced gastric cancer, as well as for the third-line and subsequent treatment of PD-L1-positive advanced gastric cancer. Nivolumab is approved for the treatment of advanced gastric cancer that has failed to respond to at least 2 other conventional therapies.

The KEYNOTE-059 study is a Phase II study to investigate the safety and efficacy of Pembrolizumab as monotherapy or in combination with chemotherapy in the treatment of advanced G/GEJ cancer (**Fuchs CS, 2018**). Cohort 1 consisted of 259 patients who received Pembrolizumab monotherapy at a dose of 200 mg every 3 weeks for advanced gastric cancer that had failed to respond to second-line therapy, and the response rate was higher in PD-L1-positive patients than that in negative patients (15.5% vs. 6.4%); the monotherapy group showed an ORR of 13.3% (95% CI, 8.2 to 20.2), an mPFS of 2.0 months, an mOS of 5.5 months, and a better efficacy in the PD-L1-positive population (including 148 PD-L1-positive patients, with a PD-L1 positive rate of 57%). Based on the results of this study, Pembrolizumab was approved by the FDA for the back-line treatment of PD-L1-positive recurrent or advanced gastric cancer.

ATTRACTION2 is a multicenter, randomized, double-blind, placebo-controlled Phase III clinical trial conducted in Asia to investigate Nivolumab in the treatment of progressed G/GEJ cancer in progressive stage after second- and multiple-line chemotherapies (**Kang YK, 2017**). In this study, a total of 493 patients who failed chemotherapy were enrolled. The mOS was 5.32 months in the Nivolumab group and 4.14 months in the placebo group (HR 0.63, 95% CI 0.50 to 0.78, $P < 0.0001$); and the 6- and 12-month OS rates were 46.4% vs. 34.7%, and 26.6% vs. 10.9% for the Nivolumab and placebo groups, respectively. The ORR was 11.2% for the Nivolumab group and 0% for the placebo group. Based on the results of this study, Nivolumab was approved for the treatment of advanced gastric cancer that had failed to respond to at least 2 other conventional therapies.

PD-1/PD-L1 is not the only immunosuppressive molecule in the immune microenvironment, and other immunosuppressive molecules, including CTLA-4, may also cooperate with PD-1/PD-L1 to inhibit T cell killing activity. Therefore, the combination of targeted drug therapies with different immunosuppressive molecules is a potential approach to potentially increase efficacy. At present, in addition to anti-PD-1/PD-L1 targeted drugs, anti-CTLA-4 targeted drugs have also been developed well, becoming the first immune-targeted drug to be used in combination with anti-PD-1 in clinical studies. PD-1 and CTLA-4, of which the mechanisms of action are to prevent T-cell failure and promote T-cell activation, respectively, have complementary and nonredundant roles in modulating immune responses. CTLA-4-mediated T cell activation and proliferation inhibition occurr predominantly at sites of antigen presentation in the lymphatic chamber, whereas PD-1 pathway-mediated inhibition occurs predominantly at tumor sites, although CTLA-4 distribution is specific, whereas PD-1 is not specific enough. Targeting both pathways may have additive or synergistic activity (**Pardoll DM, 2011**), which indicates that combination therapy can produce superior antitumor activity compared to monotherapy. For example, concomitant treatment with Nivolumab and Ipilimumab would result in higher ORR than monotherapy in patients with advanced melanoma. Importantly, the treatment response appears to be deep and durable (**Wolchok JD, 2013**). The combination of Nivolumab and Ipilimumab has been approved for the treatment of unresectable or metastatic melanoma. In early clinical trials, the combination of Nivolumab and Ipilimumab has also shown encouraging antitumor activity and manageable toxicity in a variety of tumor species.

The CheckMate 032 study explored Nivolumab alone or in combination with Ipilimumab in a Western population with advanced gastric or gastroesophageal junction cancer that have progressed after at least one line of chemotherapy (**Janjigian YY, 2016**). A total of 160 patients (of whom 79% were resistant to second-line and above therapies were enrolled into 3 groups, including 59 patients in the Nivolumab 3 mg/kg monotherapy group, 49 subjects in the Nivolumab 1 mg/kg + Ipilimumab 3 mg/kg group and 52 patients in the Nivolumab 3 mg/kg + Ipilimumab 1 mg/kg group. The ORR in the nivo 1 mg/kg + ipi 3 mg/kg group was 26% (95% CI, 14-41%),The ORR was 10% (95% CI, 3-22%) in the nivo 3 mg/kg + ipi 1 mg/kg group and 14% (95% CI, 6-25%) in the nivo alone 3 mg/kg Q2W group. In a subgroup analysis based on PD-L1 status, the ORR for subjects with tumor PD-L1 expression ≥ 1% was 44% (4/9) in the nivo 1 mg/kg + ipi 3 mg/kg group, 27% (3/11) in the nivo 3 mg/kg + ipi 1 mg/kg group, and 27% (4/15) in the nivo alone 3 mg/kg Q2W group; the incidence of Grade 3 to 4 treatment-related AEs was 45% in the nivo 1 mg/kg + ipi 3 mg/kg group, 27% in the nivo 3 mg/kg + ipi 1 mg/kg group and 17% in the nivo alone 3 mg/kg Q2W group; and the incidence of treatment-related AEs leading to discontinuation was 22% in the nivo 1 mg/kg + ipi 3 mg/kg group, 12% in the nivo 3 mg/kg + ipi 1 mg/kg group and 5% in the nivo alone 3 mg/kg Q2W group. Combination therapy with Ipilimumab significantly increased the incidence of toxic and side effects. The major grade 3 to 4 treatment-related adverse reactions in all three groups were diarrhea and elevated liver enzymes. These preliminary findings suggest that although the nivo 1 mg/kg + ipi 3 mg/kg group had the highest ORR, its side effects cannot be ignored.

1.2.3 Justification for the development of immunotherapy combined with chemotherapy for gastric cancer

For locally advanced or metastatic gastric cancer, platinum-based chemotherapy remains the standard first-line chemotherapy option. The survival benefit from cytotoxic chemotherapy has reached a plateau. Exposure of the immune system to high levels of tumor antigens can reasonably anticipate tumor cell killing effect of cytotoxic chemotherapy and by inhibiting the PD-L1/PD-1 signaling pathway, restoration of tumor-specific T-cell immunity in this setting may produce deeper and durable responses than standard chemotherapy alone. The use of chemotherapeutic drugs can activate the tumor interferon pathway, increase the infiltration of immune cells, and provide a suitable microenvironment for anti-PD-1 therapy. In a clinical study comparing Pembrolizumab with chemotherapy or chemotherapy alone as first-line treatment for non-squamous non-small cell lung cancer (KEYNOTE-021, NCT02039674), the combination regimen significantly increased the ORR compared with chemotherapy alone (55% vs. 29%, p = 0002). In May 2018, FDA accelerated approval of Pembrolizumab in combination with chemotherapy as first-line treatment for non-squamous non-small cell lung cancer, which is not limited by PD-L1 expression.

In gastric cancer, preliminary data are also available for anti-PD-1 therapy in combination with chemotherapy. In the KEYNOTE-059 study (Cohort 2), Pembrolizumab was used in combination with a first-line FP (5-fluorouracil + cisplatin)/XP (capecitabine + cisplatin) regimen for treatment-naïve advanced HER2-non-overexpressing advanced gastric or gastroesophageal junction adenocarcinoma. Twenty-five patients were enrolled. The ORR and DCR were 60% and 80%, respectively, and the median PFS and OS were 6.6 months and 13.8 months, respectively. The ORR and DCR were 69% and 81% in PD-L1-positive patients and 38% and 75% in PD-L1-negative patients, respectively. In terms of adverse reactions, grade 3 to 4 treatment-related adverse reactions occurred in 76% of patients and the most common was neutropenia (64%),and the others included stomatitis (20%), anemia (8%), decreased appetite (8%), fatigue (8%), hand-and-foot syndrome (8%) and thrombocytopenia (8%). Of these, immune-related Grade 3 or higher adverse reactions occurred in 12% of patients, including rash (8%) and nephritis (4%). This small sample study suggests that anti-PD-1 therapy in combination with chemotherapy has great potential in gastric or gastroesophageal junction adenocarcinomas.

Further, several clinical studies will explore the efficacy of anti-PD-1 therapy in combination with chemotherapy in palliative or adjuvant treatment of gastric or gastroesophageal junction adenocarcinoma. In the already ongoing Phase III study KEYNOTE-062, investigators will compare the efficacy of Pembrolizumab and chemotherapy (FP/xeloda + DDP regimen) alone or in combination in the first-line treatment of HER-2-non- overexpressing and PD-L1 positive advanced gastric or gastroesophageal junction adenocarcinoma. The primary endpoints of the study are PFS and OS, and 750 patients are planned to be enrolled. While the Phase III study CheckMate-649 (NCT02872116) included 3 groups: Nivo + Ipi group (Nivo 1 mg/kg + Ipi 3 mg/kg*4 cycles → Nivo maintenance designed based on CheckMate-032 data) vs. Nivo + XELOX/FOLFOX vs.XELOX/FOLFOX. The primary endpoint of the study was

OS of Nivo 1 mg/kg + Ipi 3 mg/kg versus XELOX/FOLFOX in tumor PD-L1 positive patients.

AK104 is a bispecific antibody that binds both PD-1 and CTLA-4 with high affinity and efficiently blocks the PD-1/PD-L1, PD-1/PD-L2, CTLA-4/B7.1, and CTLA-4/B7.2 interactions. AK104 is a novel tumor immunotherapy agent with a quadrivalent structure, short half-life and minimal toxicity observed in monkey toxicity studies. These characteristic features may enhance the efficacy of the combination of anti-PD-1/L1 and anti-CTLA-4 antibody and reduce treatment-related toxicities. These hypotheses need to be validated in prospective clinical trials of AK104. For a detailed summary of potential differences between AK104 and anti-PD-1/L1 in combination with anti-CTLA-4 antibody, please see Section 1.1.5.

Therefore, this study will explore the preliminary antitumor activity, safety, pharmacokinetics, and immunogenic profile of anti-PD-1 and CTLA-4 bispecific antibody AK104 in combination with XELOX as the first-line treatment for unresectable advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma, to provide sufficient evidence for further clinical study development of AK104 in combination with standard chemotherapy. The justification for the selection of the starting dose to be evaluated in the dose escalation period of this study is provided in Section 3.2.1. Following the determination of the maximum tolerated dose (MTD)/maximum administered dose (MAD) or the recommended Phase II dose (RP2D), the dose confirmation period will be conducted for efficacy observation.

1.2.4 Justification for the development of immunotherapy in solid tumors

Cervical cancer

Cervical cancer is the fourth most common malignant tumor in women around the world, while the treatment of advanced and recurrent cervical cancer is limited, with a 5-year survival rate of less than 20%. Squamous cell carcinoma is the most common cancer of the cervix, accounting for about 70%, followed by adenocarcinoma, accounting for 25%. Persistent HPV infection results in almost all cervical cancers. In view of the unmet clinical needs, the efficacy, safety, predictive markers and treatment regimens of immunotherapy are also becoming the focus of the gynecological field.

In the KEYNOTE-028 study, 475 patients with 20 species of PD-L1-positive advanced solid tumors were enrolled to receive Pembrolizumab 10 mg/kg Q2W. Of the 24 PD-L1-positive cervical cancer patients with a median age of 42 years (**Frenel JS, 2017**), 92% (n = 22) had received prior radiotherapy and 63% (n = 15) had received at least 2 prior lines of chemotherapy. As of the median follow-up duration of 11 months, 4 patients achieved PR with an ORR of 17% and a median DOR of 5.4 months. The median PFS was 2 months, the PFS rate was 21% at 6 months and 4% at 12 months for all patients. Drug-related adverse reactions were mainly rash, fever, asthenia, etc. There were no serious adverse reactions and treatment-related deaths.

Based on the KEYNOTE-028 study, KEYNOTE-158 further validated the antitumor activity in 11 species of treatment-naïve or previously treated advanced solid tumors (**Chung HC, 2019**). Of 98 patients with cervical cancer, positive PD-L1 expression was observed in 82 patients (83.7%). Patients were treated with 200 mg Pembrolizumab every 3 weeks. As of January 2018, the median duration of follow-up reached 10.2 months; and the results showed 3 CRs, 9 PRs, and an ORR of 12.2%, of which the ORR was 17.6% (6/34) in patients who had ever received ≤ 1 line of prior systemic therapy and 11.7% (4/34) in patients who had received 2 lines of prior systemic therapies; and the proportion of patients who received ≤ 2 prior therapies was 83.3% (10/12) in all patients with a tumor assessment of PR or CR (n = 12). The ORR was 14.6% in PD-L1 positive patients, and was 14.3% in patients who had received ≥ 1 prior line of therapy. A total of 18 patients achieved SD, 15 of whom were PD-L1 positive, with a disease control rate of 30.6% and 32.9% for the total population and PD-L1-positive population, respectively. The results of the KEYNOTE-158 study were consistent with those of the KEYNOTE-028 study, in both of which Pembrolizumab demonstrated durable antitumor activity and manageable safety in patients with previously treated advanced cervical cancer. In June 2018, the US FDA accelerated approval of it for the treatment of patients with recurrent or metastatic cervical cancer (PD-L1 CPS ≥ 1) with disease progression during or after chemotherapy.

In a Phase II clinical study (JapicCTI-163212) conducted in Japan (**Kosei Hasegawa, 2018**), 20 patients with advanced recurrent cervical cancer, including 14 squamous cell carcinomas, 5 adenocarcinomas, and 1 adenosquamous carcinoma by histological subtype, were treated with Nivolumab (Nivo 240 mg every 2 weeks). The results showed the ORR was 25%; the median PFS was 5.6 months; the 12-month survival (OS%) was 73%; the ORR in patients with PD-L1 ≥ 1% (33%; 5/15) was higher than that in patients with PD-L1 < 1% (0%; 0/5); and the HPV status did not affect ORR (33% in both HPV+ [3/9] and HPV- [1/3] patients). An additional clinical trial to evaluate the efficacy and safety of Ipilimumab (**Lheureux S, 2018**) included 42 patients with metastatic or recurrent cervical cancer, who were treated with Ipilimumab intravenously every 3 weeks at a dose of 10 mg/kg, for a total of 4 cycles followed by a course of maintenance therapy every 12 weeks thereafter. Of the 34 evaluable patients, 1 achieved PR and 10 achieved SD. The median PFS was 2.5 months and the median overall survival was 8.5 months. The CheckMate 358 study (**Hollebecque A, 2017**) was designed to explore the combination of Nivolumab and Ipilimumab in multiple tumors, including cervical cancer. ASCO 2017 showed an objective response rate of 26.3% (95% CI 9.1-51.2) in 19 patients with advanced cervical cancer treated with Nivolumab, who did not reach the median duration after 6 months of follow-up. The combination of double-target immunotherapy in cervical cancer is worthy of further expectation.

As of August 01, 2019, a total of 5 patients with recurrent or metastatic cervical cancer, all treated with AK104 6 mg/kg monotherapy, had been included in the Phase Ib portion of the study. Results showed that AK104 was safe and in 3 patients with evaluable efficacy, all of the tumor lesions were reduced somewhat, with the maximum reduction up to 28%, tumor assessment of all of them was stable disease, and the longest time of treatment was more than 5 months. The preliminary results suggest that AK104 has a potential therapeutic effect on

cervical cancer. It also provides the basis for further exploring the treatment of advanced cervical cancer.

Esophageal cancer

At present, the incidence of esophageal cancer in China is the highest in the world. Compared with adenocarcinoma-predominant pathologic types in Europe and America, squamous cell carcinoma is predominant in China, accounting for more than 90%. Early diagnosis of esophageal squamous cell carcinoma (ESCC) is difficult and ESCC invades and metastasizes easily. Chemotherapy is still the main treatment for advanced esophageal cancer. Platinum-based chemotherapy doublet is the backbone of first-line treatment, which is usually platinum + paclitaxel or platinum + 5-fluorouracil, and the second-line therapy is mainly paclitaxel or irinotecan mono-chemotherapy, but the prognosis is poor, with a 5-year survival rate of only about 30%. Both Pembrolizumab and Nivolumab have been prospectively studied in esophageal squamous cell carcinoma. Results of these studies indicate that Nivolumab is effective in approximately 15% of the non-screened and independently assessed population, and Pembrolizumab is effective in approximately 20% of PD-L1-positive patients.

Based on the prospect of Pembrolizumab in the treatment of esophageal cancer shown in studies KEYNOTE-028 and KEYNOTE-180, the investigator further conducted the Phase III KEYNOTE-181 study (**Kojima T, 2019**). Pembrolizumab was compared head-to-head with investigator-selected chemotherapy in patients with advanced or metastatic esophageal squamous cell carcinoma or adenocarcinoma/Siewert type I adenocarcinoma of the esophagogastric junction. A total of 628 patients were enrolled in the study and randomized in a 1:1 ratio to receive Pembrolizumab 200 mg every 3 weeks for 2 years, or investigator-selected chemotherapy (including paclitaxel, docetaxel or irinotecan). Of these patients, 401 were with squamous cell carcinomas and 222 were with PD-L1 positive cancer (CPS ≥ 10). In patients with esophageal squamous cell carcinoma, the ORR in the Pembrolizumab and chemotherapy groups was 16.7% and 7.4%, respectively (P = 0.0022); OS of the Pembrolizumab group also improved clinically, reaching 8.2 months vs. 7.1 months in the chemotherapy group (HR 0.78; 95% CI 0.63 to 0.96; P = 0.0095); and the 18-month OS rate in the two groups was 23% and 12%, respectively. In patients that were PD-L1-positive, the ORR in the two groups was 21.5% and 6.1%, respectively; and the DOR in the Pembrolizumab group was significantly longer than that in the chemotherapy group (9.3 months vs. 7.7 months). The study reached the primary OS study endpoint and was the first to demonstrate a survival benefit in immunotherapy for esophageal cancer. From the results of KEYNOTE-181 study, patients with esophageal squamous cell carcinoma responded better to PD-1 immunotherapy. Secondly, PD-L1 expression could be an important Biomarker in esophageal squamous cell carcinoma patients, and OS and ORR were significantly improved in PD-L1-positive patients. On 31 July 2019, FDA approved pembrolizumab as monotherapy in the treatment of recurrent locally advanced or metastatic ESCC. These patients had PD-L1 expression on tumor (combined positive score [CPS] ≥ 10) and still had disease progression after one or more prior systemic therapies. Phase III clinical studies (Checkmate 648 study) to evaluate Nivolumab in combination with Ipilimumab or Nivolumab in combination with

chemotherapy as first-line treatment in patients with advanced, recurrent, or metastatic esophageal squamous cell carcinoma are also ongoing.

Hepatocellular carcinoma

Hepatocellular carcinoma is characterised by a high degree of malignancy, rapid intrahepatic and extrahepatic metastasis and poor prognosis. Following sorafenib and lenvatinib are used as first-line treatment for advanced hepatic cancer, and regorafenib, cabozantinib, ramucriumab are recommended as second-line treatment for advanced hepatic cancer, immune checkpoint inhibitors are drugs that can significantly improve the prognosis of advanced hepatic cancer after molecular targeted therapy.

CheckMate 040 study has undoubtedly played a crucial role in promoting the development of immunotherapy in the field of hepatic cancer. The objective of this study was to evaluate the efficacy of Nivolumab or a Nivolumab-based combination regimen in patients with advanced HCC, including patients with or without chronic viral hepatitis and patients with intolerance or disease progression after previous or no sorafenib treatment. In 2017, the FDA approved Nivolumab as a monotherapy for second-line treatment of advanced liver cancer treated with sorafenib based on data from Checkmate-040. The study cohort showed that Nivolumab monotherapy in patients with advanced hepatocellular carcinoma of different disease causes, different stages, and different regions showed significant improvement in the efficacy of treatment with the characteristics of translation of the efficacy into survival benefits, comparable efficacy between East and Western populations, and comparable efficacy between hepatitis B and C patients.

In the Checkmate-040 study, three different dosing regimens for the combination of Nivolumab and Ipilimuma in a nivolumab-based combination regimen were evaluated: Group A: Nivolumab 1 mg/kg in combination with Ipilimumab 3 mg/kg every 3 weeks for 4 consecutive cycles, and then followed by Nivolumab 240 mg every 2 weeks; Group B: Nivolumab 3 mg/kg in combination with Ipilimumab 1 mg/kg every 3 weeks for 4 consecutive cycles, and then followed by Nivolumab 240 mg every 2 weeks; Group C: Nivolumab 3 mg/kg every 2 weeks in combination with Ipilimumab 1 mg/kg every 6 weeks. The 2019 ASCO results showed an ORR of 31%, a DCR of 49% and a mDoR of 17.5 months as assessed by the blinded independent central review committee per RECIST V1.1 criteria after a minimum of 28 months of follow-up. DCR was 54%, 43%, and 49% and mDoR was 17.5, 22.2, and 16.6 months in Groups A, B, and C, respectively. Patients in Group A had the longest median OS, reaching 22.8 months and their 30-month overall survival rate achieved 44%. Regardless of the PD-L1 expression level in tumor tissues, significant efficacy was observed. ORR and mOS benefit were observed in both patients with PD-L1 < 1% and PD-L1 ≥ 1% at baseline. The study further clarified the value of anti-PD-1 and anti-CTLA-4 (double-target immune agents) in the treatment of hepatocellular carcinoma.

1.3. Study Risk Assessment

1.3.1 Potential Risks Associated with AK104

Potential risks based on the mechanism of action of AK104 and related molecules include immune-mediated responses. Potential immune-related adverse events (irAEs) may be similar to those caused by anti-PD-1/L1 and/or anti-CTLA-4 drugs, which may include, but are not limited to the following types of toxicity: infusion-related reactions (pyrexia, rash, pruritus, hypotension, dyspnea, chest discomfort, wheezing, tachycardia, stiffness), skin toxicity (rash, pruritus, vitiligo), endocrine toxicity(hyperglycemia, hypothyroidism, hyperthyroidism, primary hypoadrenalism, hypophysitis), hepatotoxicity (hepatitis, aspartate aminotransferase increased, alanine aminotransferase increased, bilirubin increased), gastrointestinal toxicity (diarrhoea, colitis), pulmonary toxicity (pneumonia, pulmonary sarcoidosis), rheumatoid/boneiliac toxicity (rheumatoid arthritis, myositis, myalgia), neurotoxicity (myasthenia gravis, aseptic meningitis, encephalitis, transverse myelitis, Guillain-Barre syndrome), hematotoxicity (autoimmune hemolytic anemia, aplastic anemia, immune thrombocytopenia,acquired hemophilia), nephrotoxicity (nephritis, renal insufficiency), cardiotoxicity (myocarditis, pericarditis, cardiovascular abnormalities), ocular toxicity (uveitis, scleritis). At present, the safety events found in the AK104 clinical study are consistent with the characteristics of tumor immunotherapy targeting PD-1 and/or CTLA-4 without new adverse reactions, as detailed in 1.1.6.

According to the available clinical data of anti-PD-1 monoclonal antibody drugs, although the incidence of adverse reactions is high, these antibodies are well tolerated, only a small proportion of subjects may discontinue the drug due to adverse reactions, and most adverse reactions can be resolved with treatment. The serum half-life t1/2 of AK104 is approximately 1/3 to 1/2 of typical antibodies. Rapid clearance from blood can reduce nonspecific (off-target) cytotoxicity. As described in Section 1.1.5, the preferential distribution of AK104 in tumor tissues may lead to better safety. Please refer to the Investigator's Brochure for safety results from clinical studies of anti-PD-1/L1 in combination with anti-CTLA-4.

As with the administration of some therapeutic antibodies, other potential risks associated with AK104 are infusion or allergic reactions. Subjects may develop neutralizing ADAs, and this may be associated with allergic or allergic toxicity, or may induce and enhance AK104 toxicity. Standard clinical assessments and interventions for allergy or allergic reactions are recommended. In this study, subjects should be monitored for these events and the formation of anti-drug antibodies.

The recommended exclusion criteria, safety monitoring, starting dose, dose escalation plans, discontinuation criteria, and toxicity management guidelines (Section 3.1.4) are to minimize the potential risks to patients participating in this study. At present, no clinical studies of any BsAb with similar mechanisms to AK104 have been published. Clinical data on AK104, a bispecific antibody targeting both anti-PD-1 and CTLA-4, are very limited. Exclusion criteria for this study are to ensure that subjects who might be at risk (i.e., those with previous autoimmune diseases, inflammatory bowel disease, unresolved toxicity from prior therapy, or

blood disease or organ dysfunction) are excluded. Toxicity management guidelines are based on other anti-PD-1/L1 and/or anti-CTLA-4 agents, and focus on frequent monitoring and early detection of potential immune-mediated events. Principles for managing toxicities include discontinuation and/or corticosteroids or more effective immunosuppressive therapy as needed. The information obtained in this study will provide a reference for AK104 treatment decisions and provide potential combination therapy strategies for later clinical studies.

1.3.2 Potential Benefits

Phase Ia clinical data of AK104 have preliminarily shown that AK104 is safe and well tolerated in patients with advanced tumors, and its antitumor activity and definite pharmacological activity are preliminarily observed. Given that the recently published data have suggested significant efficacy of PD-1 inhibitors as monotherapy, anti-PD-1 in combination with CTLA-4, and anti-PD-1 in combination with standard chemotherapy for advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma, it is concluded that AK104, a bispecific antibody targeting both anti-PD-1 and CTLA-4, may also achieve good efficacy in patients with the same disease condition, and relieve disease and prolong progression-free survival after combining with standard chemotherapy.

The level of coexpressed PD-1 and CTLA-4 receptors in tumor infiltrating lymphocytes is higher than that in lymphocytes found in T cells from normal tissues and peripheral blood. AK104 is a bispecific antibody that targets both PD-1 and CTLA-4, with a quadrivalent structure and a short half-life. It is less toxic compared to anti-PD-1/L1 in combination with anti-CTLA-4 (observed in monkey studies). It is superior to anti-PD-1/L1 antibody alone or anti-PD-1/L1 in combination with anti-CTLA-4 in terms of efficacy and/or safety. These hypotheses need to be validated in prospective clinical trials.

2. Study Objectives and Endpoints

2.1. Study Objectives

2.1.1. Primary Objectives

- Phase Ib: To assess the safety and tolerability of AK104 as a monotherapy in subjects with advanced solid tumor and AK104 in combination with oxaliplatin and capecitabine (mXELOX/XELOX) as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase Ib: To determine the maximum tolerated dose (MTD) or the maximum administered dose (MAD) when the MTD is not reached, and the recommended Phase II dose (RP2D) of AK104 administered with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase II: To assess the objective response rate (ORR) of AK104 combined with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma as per RECIST v1.1.
- Phase II: To assess the objective response rate (ORR) of AK104 as a monotherapy in subjects with recurrent or metastatic cervical cancer as per RECIST v1.1.
- Phase II: To assess the objective response rate (ORR) of AK104 as a monotherapy in subjects with locally advanced or metastatic esophageal squamous cell carcinoma and advanced hepatocellular carcinoma as per RECIST v1.1.

2.1.2. Secondary Objectives

- Phase Ib: To preliminarily evaluate the antitumor activity of AK104 as a monotherapy in subjects with advanced or metastatic solid tumors and AK104 combined with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase II: To assess the safety and tolerability of AK104 in combination with mXELOX/XELOX as first-line therapy in patients with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.
- Phase II: To assess the efficacy of AK104 with mXELOX/XELOX as the first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma, including progression-free survival (PFS), duration of response (DoR), disease control rate (DCR), time to response (TTR) and overall survival (OS) based on RECIST v1.1 assessment.
- Phase II: To assess the efficacy of AK104 as a monotherapy in subjects with recurrent or metastatic cervical cancer, including PFS, DoR, DCR, TTR, and OS as assessed per RECIST v1.1.

- Phase II: To assess the efficacy of AK104 as a monotherapy in subjects with locally advanced or metastatic esophageal squamous cell carcinoma and advanced hepatocellular carcinoma, including PFS, DoR, DCR, TTR, and OS as assessed per RECIST v1.1.
- Phase II: To evaluate the safety and tolerability of AK104 as a monotherapy in subjects with recurrent or metastatic cervical cancer.
- Phase II: To evaluate the safety and tolerability of AK104 as a monotherapy in subjects with locally advanced or metastatic esophageal squamous cell carcinoma and advanced hepatocellular carcinoma.
- To assess the pharmacokinetic profile of AK104 as a monotherapy and in combination with mXELOX/XELOX.
- To assess the immunogenicity of AK104 as a monotherapy and in combination with mXELOX/XELOX.

2.1.3. Exploratory Objectives

- To evaluate the correlation between PD-L1 expression and antitumor response in archived and/or fresh tumor tissue.
- To explore other AK104 potential biomarkers in blood and tumor tissue specimens, which may (1) be related to the biological activity of the drug (pharmacodynamics); (2) prospectively determine the patients who most likely respond to AK104 treatment (predicted); and (3) have other correlations with study treatment response, drug resistance mechanism, and / or disease status.
- To preliminarily determine the antitumor activity of AK104 as a monotherapy in subjects with advanced or metastatic solid tumors and AK104 in combination with mXELOX/XELOX as first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma using the immune-related Response Evaluation Criteria in Solid Tumors (irRECIST).

2.2. Study Endpoints

2.2.1. Primary Endpoints

- Phase Ib Safety Assessments: dose-limiting toxicity (DLT), incidence and severity of adverse events, clinically significant abnormal laboratory tests, vital signs, and electrocardiograms (ECGs) ; MTD, or MAD when the MTD is not reached, and recommended Phase II dose (RP2D).
- Phase II Efficacy Assessment:
  - ✓ ORR for squamous or adenosquamous cell carcinoma of the cervix will be assessed by Independent Radiology Review Committee (IRRC) as per RECIST v1.1.

- ✓ ORR will be assessed by the investigator as per RECIST v1.1 for gastric adenocarcinoma or gastroesophageal junction adenocarcinoma, esophageal squamous cell carcinoma and hepatocellular carcinoma.

2.2.2. Secondary Endpoints

- Phase Ib Efficacy Assessment: ORR, DCR, DoR, TTR, and PFS assessed as per RECIST v1.1.
- Phase II Safety Assessments: incidence and severity of adverse events, clinically significant abnormal laboratory tests, vital signs, and electrocardiogram (ECG) results.
- Phase II Efficacy Assessments:
  - ✓ DCR, DoR, TTR, and PFS for squamous or adenosquamous carcinoma of the cervix will be assessed by IRRC as per RECIST v1.1.
  - ✓ ORR, DCR, DoR, TTR, and PFS for squamous or adenosquamous carcinoma of the cervix will be assessed by the investigator as per RECIST v1.1.
  - ✓ DCR, DoR, TTR, and PFS will be assessed by the investigator as per RECIST v1.1 for gastric or gastroesophageal junction adenocarcinoma, esophageal squamous cell carcinoma and hepatocellular carcinoma.
- Overall survival (OS) and 12-month overall survival rate (probability that the subject survives for 12 months after the first dose).
- Endpoints for AK104 PK assessment include serum drug concentration of each subject at different time points after AK104 administration. PK parameters will be measured with non-compartmental analysis, including observed maximum concentration (C_{max}), area under the plasma concentration-time curve (AUC), clearance (CL), and half-life ($t_{1/2}$).
- Endpoints for AK104 immunogenicity assessment include the number and percentage of subjects with detectable anti-drug antibody (ADA).

2.2.3. Exploratory Endpoints

- To analyze the correlation between PD-L1 expression and the antitumor activity of AK104 [e.g., ORR assessed by IRRC in subjects with PD-L1 positive (CPS ≥ 1) and PD-L1 negative (CPS < 1) cervical squamous/adenosquamous cell carcinoma], and other biomarkers that potentially predict the antitumor activity of AK104, including but not limited to the type and number of tumor infiltrating lymphaytes (TIL) and the immune-related gene mRNA expression characteristics in tumor tissues (tumor cells and tumor infiltrating immune cells);
- To obtain exploratory biomarkers in tumor tissues during the disease progress or response, including but not limited to the detection of immune-related gene

characteristics and changes in the number of CD8+ effector T cells;

- Endpoints for assessment of potential pharmacodynamic biomarkers may include: (1) the detection of changes in CD4+ Ki67+ and CD8+ Ki67+ proliferating cells in blood from baseline over time using flow cytometry; and (2) the detection of PD-1 or CTLA-4 receptor occupancy in blood before and after treatment using flow cytometry.
- Anti-tumor activity endpoints assessed as per irRECIST criteria include ORR, DoR, DCR, TTR and PFS.

3. Study Design

3.1. Study Description

3.1.1. Overview

This is a Phase Ib/II, open-label, multi-center clinical trial conducted in China to assess the safety, tolerability, PK profile, immunogenicity, pharmacodynamics, and antitumor activity of AK104 as a monotherapy in subjects with advanced solid tumors and AK104 in combination with oxaliplatin and capecitabine (mXELOX/XELOX) as a first-line therapy in subjects with advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.

This study consists of 3 periods: a dose escalation period (Part 1 of Phase Ib), a dose expansion period (Part 2 of Phase Ib), and a dose confirmation period (Phase II). Subjects will be divided into different cohorts depending on the different tumor types and treatment regimen in Phases Ib and 2.

In the Phase Ib dose escalation period, AK104 as a monotherapy at doses of 6 mg/kg and 10 mg/kg every 2 weeks (Q2W) will be administered to treat the advanced solid tumors in Cohort 1, and AK104 at doses of 4mg/kg Q2W, 6mg/kg Q2W, 10 mg/kg Q2W, 15 mg/kg Q3W in combination with mXELOX/XELOX as first-line therapy will be administered to treat the advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma in the Cohort 2, 3, 4 and 5, respectively. Dose escalation and safety evaluation will be performed in accordance with the "3 + 3" principle.

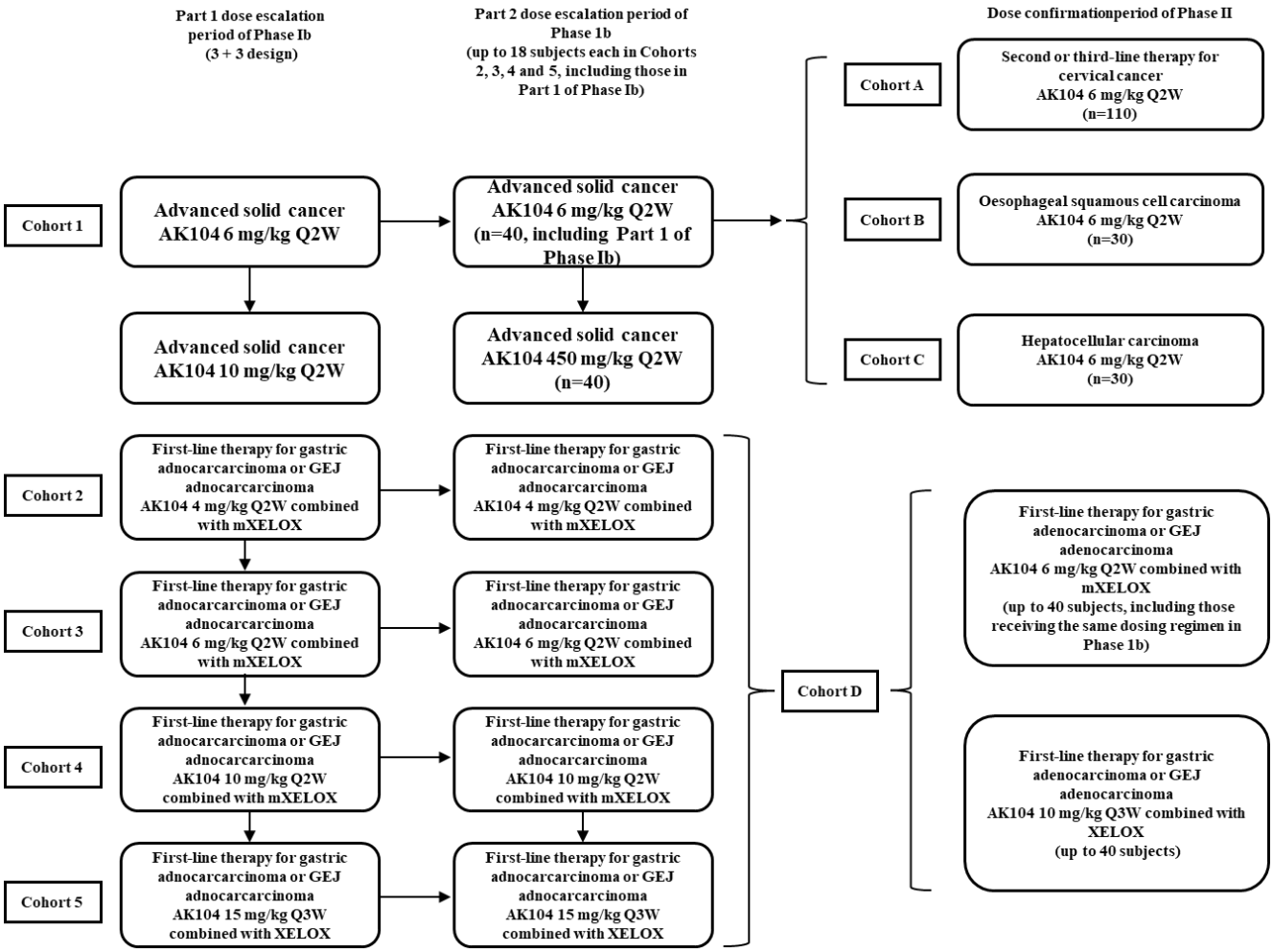
In the Phase Ib dose expansion period, AK104 as a monotherapy at doses of 6 mg/kg and 450 mg will be administered to treat the advanced solid tumors in Cohort 1 (40 subjects for each dose group, including subjects in the dose escalation period), and for Cohort 2, 3, 4 and 5, if no DLT is observed or the DLTs occur in less than 2 out of 6 subjects in the dose escalation period, the enrollment will continue until there are 18 subjects in each corresponding cohort(including subjects in the dose escalation period).

In Phase II, subjects will be divided into four cohorts by tumor types, including Cohort A: cervical cancer (n = 110); Cohort B: esophageal squamous cell carcinoma (n = 30); Cohort C: hepatocellular carcinoma (n = 30); and Cohort D: advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma (2 dosing regimens including AK104 6mg/kg Q2W combined with mXELOX and AK104 10mg/kg Q3W combined with

XELOX, with up to 40 subjects in each regimen, including those receiving the same dosing regimen in phase Ib).

In this study, It is planned to enroll approximately 372 subjects (approximately 158 in the dose escalation and expansion periods and approximately 214 in the dose confirmation period). Additional subjects may be enrolled if dose de-escalation cohorts or intermediate doses are tested, or additional dose escalation cohorts are required.

Figure 3-1 Overall Study Design Diagram



Abbreviations: GEJ, gastroesophageal junction; Q2W, every 2 weeks; Q3W, every 3 weeks.

a. The dose escalation cohort includes Cohort of AK104 6 mg/kg Q2W and 10 mg/kg Q2W, the Cohort 2, Cohort 3, Cohort 4 of AK104 (4 mg/kg Q2W, 6 mg/kg Q2W, 10 mg/kg Q2W) with mXELOX, and the Cohort 5 of AK104 (15mg/kg Q3W) with XELOX. The intermediate dose of AK104 may be evaluated at the discretion of the sponsor in agreement with the DEC based on the available safety data. If DLTs are observed in ≥ 2 out of 6 evaluable subjects in Cohort 2, the initial dose will be reduced to half of the previous dose (2 mg/kg Q2W). If the MTD is not reached in this study, the sponsor and DEC will discuss whether to continue the dose escalation in future protocol amendments. In each escalation

cohort, the first subject will be treated and then after a minimum of 24 hours, the next 2 or 3 subjects will be treated. For the subjects in each dose group, the dose level and dosing interval of AK104 remain unchanged.

b. In the dose escalation period, after tolerability is observed in 3~6 subjects, the dose group which has no DLT or has DLTs observed in fewer than 2 out of 6 subjects will be expanded after the discussion of the sponsor with DEC, based on preliminary safety and efficacy data. 6 mg/kg dose group in Cohort 1 can be expanded to a maximum ofa maximum of 40 subjects. Cohorts 2, 3, 4 and 5 can be expanded to include 18 subjects in each Cohort (including subjects enrolled in the dose escalation period). If DLT occurs in ≥ 2 out of 6 subjects, the dose group will not be expanded.

c. The Phase Ib dose expansion cohort of AK104 450 mg Q2W group will enroll all solid tumor types except the selected tumor types in the Cohorts A, B, and C from Phase II. After the first 6 subjects receiving AK104 450mg Q2W complete the 28-day treatment, the investigator and sponsor will conduct a safety assessment analysis.

d. Cohort D ncluding 2 dosing regimens of AK104 6mg/kg Q2W combined with mXELOX and AK104 10mg/kg Q3W combined with XELOX.

e. Dose escalation cohort: If 5 DLT-evaluable subjects out of 6 enrolled subjects at any dose have completed the safety assessment of 28 days/21 days after first dose of AK104 and have no DLT, the study can proceed to the next dose escalation cohort and/or to the dose expansion enrollment.

The study consists of a screening period (no more than 28 days after the subject signs the informed consent form up till the first AK104 dose, see Section 6.1), a treatment period (until confirmed disease progression, unacceptable toxicity, or withdrawal of informed consent, whichever occurs first, see Section 6.2), and a follow-up period(including safety follow-up period and survival follow-up period, see Section 6.3). Subjects will be screened and evaluated 28 days before the first dose to determine if they meet the study eligibility critieria. During the screening period, tumor biopsy specimens must be collected (for Cohorts 2, 3, 4 and 5 in Phase Ib and for Cohort D in Phase II, archived tumor tissue samples collected within the previous 6 months or freshly obtained tumor tissue samples are required). Subjects who meet the study eligibility criteria will receive either AK104 as a monotherapy or in combination with oxaliplatin and capecitabine (mXELOX/XELOX) chemotherapy regimen based on their cohort.

All subjects will be evaluated periodically for tumor response and their clinical status will be categorized according to RECIST version 1.1 (**Eisenhauer EA, 2009**) (see Section 7 for descriptions of tumor imaging assessments). In Phase Ib, tumor assessments will be performed every 6 weeks (± 7 days) within 54 weeks after enrollment, and then every 12 weeks (± 7 days). In Phase II, tumor assessments will be performed every 8 weeks (±7 days) within 56 weeks after enrollment, and then every 12 weeks (±7 days).

For subjects receiving the treatment, the clinical decision will be made based on the response assessed by the investigator per RECIST version 1.1, including confirmation of progressive disease. A subject may continue receiving treatment between the initial assessment of progression and confirmation of progression if his/her performance status remains stable, and the subject will benefit from ongoing treatment in the opinion of the investigator, and the

subject meet other criteria listed in the protocol (see Section 3.1.2.3).

After completion of treatment, each subject will be followed-up for AE for at least 30 days and for SAE for at least 90 days or until the start of other anti-cancer treatments, whichever occurs first. Immune-related AEs (irAEs), regardless of serious or non-serious irAEs, will be followed until 90 days after the completion of treatment (whether other anti-cancer treatments have been initiated). A survival assessment will be conducted every 3 months after discontinuation of study drug to collect survival information and anti-cancer treatment information after discontinuation of study drug.

3.1.1.1. Dose escalation period (Part 1 of Phase Ib)

Cohort 1: AK104 monotherapy group, 6 mg/kg Q2W and 10 mg/kg Q2W, 3 subjects enrolled; Cohorts 2, 3, 4 and 5, AK104 with mXELOX/XELOX chemotherapy, starting dose of 4 mg/kg Q2W, 3 subjects enrolled. Based on the 3 + 3 dose escalation principle as described in Table 3-1 3 + 3 Dose Escalation Procedure, the dose will be escalated in the order of 6 mg/kg Q2W, 10 mg/kg Q2W, and AK104 in combination with mXELOX/XELOX chemotherapy regimen, where the dose of AK104 is 4 mg/kg Q2W, 6 mg/kg Q2W, 10 mg/kg Q2W, 15mg/kg Q3W in sequence. Each cohort is expected to include at least 3 DLT-evaluable subjects and a maximum of 6 evaluable subjects. DLTs and DLT-evaluable subjects are defined in Section 3.1.3. For Cohorts 1, 2, 3 and 4, the DLT evaluation period is 28 days after the first dose; for Cohort 5, the DLT evaluation period is 21 days after the first dose. In each dose escalation cohort, the first subject will be treated and then after a minimum of 24 hours, the next 2 or 3 subjects will be treated provided that there is no safety concern related to the treatment in the first subject. For subjects in each dose group, the dose level and dosing interval at which AK104 is administered remain unchanged.

Table 3-1 3 + 3 Dose Escalation Procedure

Number of subjects with DLTs	Actions Taken
None of 3 subjects	Escalate to next dose level
1 of 3 subjects	3 additional subjects at the current dose level resulting in a total of 6 evaluable subjects in the cohort
2 of 3 subjects	The MTD has been exceeded. Lower intermediate doses or the cohort prior to expansion are evaluated according to dose escalation procedure (a maximum of 6 evaluable subjects).
1 of 6 subjects	Escalate to next dose level
More than 2 of 6 subjects	The MTD has been exceeded. Lower intermediate doses or the cohort prior to expansion are evaluated according to dose escalation procedures (a maximum of 6 evaluable subjects)

Abbreviations: DLT, dose-limiting toxicity; MTD, maximum tolerated dose.

Note: In a cohort, the current dose is considered to exceed the MTD if 2 of 3 evaluable subjects, or ≥ 2 of 6 evaluable subjects experience DLTs. The MTD is defined as the highest dose level at which the incidence of DLTs is < 33.3%. If the MTD is not reached in a lower dose level cohort, 6 evaluable subjects in this cohort will be enrolled in the highest dose level cohort.

After 3 subjects in a dose escalation cohort complete the DLT evaluation period (28 days/21 days after the first dose), dose escalation will continue to the next dose escalation cohort when all available safety data as well as safety data of the previous cohorts and any available PK / PD data obtained have been reviewed and unanimously agreed by the DEC. Dose escalation cohort: If 5 DLT-evaluable subjects out of 6 enrolled subjects at any dose have completed the safety assessment of 28 days/21 days after first dose of AK104 and have no DLT, the study can proceed to the next dose escalation cohort and/or to the dose expansion enrollment. If there are the delayed AEs that occur out of the DLT evaluation period and meet the defined DLT criteria, the DEC will consider these events during data reviews for dose escalation decisions.

If the MTD is not reached in the lower dose level cohort, 3 (if no DLTs are observed) or 6 evaluable subjects (if 1 DLT is observed) will be enrolled in the highest dose level cohort. The sponsor may discontinue dose escalation until the MTD is determined based on PK/PD, toxicity, or antitumor effect.

The intermediate dose of AK104 may also be evaluated at the discretion of the sponsor in agreement with the DEC based on the available safety data. Conditions requiring selection of an intermediate dose level may include unexpected toxicities that do not meet the definition of a DLT but are considered clinically significant by the investigator and the sponsor. If an intermediate dose level is selected, the dose escalation corresponds to the next dose level listed or another dose level with an increase not exceeding 100% of the current dose level studied (as agreed with the DEC) according to the 3 + 3 dose escalation procedure. Further dose escalation will be based on toxicities observed at each dose level, and may be performed as per the original protocol, or may include additional intermediate steps based on cumulative safety data. If ≥ 2 of 6 evaluable subjects in Cohort 1 have DLTs, the study will be suspended, and the decisions to determine whether to continue the safety study at the lower dose level, or change the dosing frequency, or discontinue the study,will be made after PK data are available from the integrated assessment and analysis of safety data by the investigator and the sponsor. If ≥ 2 of 6 DLT-evaluable subjects have DLTs in the Cohort 2, the initial dose will be reduced to half of the previous dose (2 mg/kg Q2W).

In future protocol amendments, alternative treatment schedules of AK104 may be explored based on available data on PK, PD, safety and antitumor response. 3-6 subjects in the Cohort 1 of the alternative regimen may receive the highest Q2W dose level of AK104 that does not exceed the MTD at the same or lower dosing rate (mean mg/kg/week). If the MTD is not reached in this study, the sponsor and DEC will discuss whether to continue the dose escalation in future protocol amendments.

3.1.1.2. Dose expansion period (Part 2 of PhaseIb)

For the dose group with no DLT observed or DLT observed in fewer than 2 out of 6 subjects after tolerability observation for 3-6 subjects completing dose escalation, the group should be expandedafter discussion with the sponsor and DEC based on preliminary safety and efficacy data. A maximum of 40 subjects will be included into the 6 mg/kg Q2W dose

expansion group for Cohort 1 (including subjects enrolled in the dose escalation period), and a maximum of 40 subjects into 450 mg/kg dose group.

A maximum of 18 subjects (including subjects enrolled in the dose escalation period) will be included in dose expansion group for Cohorts 2, 3, 4 and 5, respectively. If DLT occurs in ≥ 2 out of 6 subjects, the dose group will not be expanded. The sponsor may, at its discretion or after consultation with DEC, decide to discontinue enrollment in certain dose expansion cohorts after reviewing available clinical data and considering subjects' benefit/risk ratio. If the dose expansion Cohorts 2, 3, 4 and 5 are allowed to enroll subjects at the same time, the subjects will be randomly assigned to one of the dose expansion cohorts.

During the dose expansion period, the safety of subjects will be monitored using the criteria applied in the dose escalation period. If ≥ 33% of subjects of the cohorts experience safety events meeting the DLT criteria defined in the dose escalation period of the study during the treatment period, the interruption of enrollment into the dose expansion cohort may be discussed and study data will bereviewed to determine if additional safety monitoring is required or if alternate dose or dosing intervals should be evaluated prior to further enrollment, etc.

3.1.1.3. Dose confirmation period (Phase II)

Cohort A: Cervical cancer

This cohort is planned to include 110 subjects with recurrent or metastatic squamous or adenosquamous carcinoma of the cervix, who have progressed during or after treatment with platinum-containing doublet chemotherapy in combination with or without bevacizumab, and have undergone no more than 2 lines of therapy in the recurrent or metastatic phase. AK104 6 mg/kg will be administered once every 2 weeks until the observation of disease progression or intolerable toxicity, withdrawal of informed consent, investigator's decision, and requirements of other protocol-specified criteria.

Cohort B: Esophageal squamous cell carcinoma

30 subjects with unresectable locally advanced or metastatic esophageal squamous cell carcinoma who have failed, unable to tolerate or refused standard of care, and receive a maximum of one line of treatment during the locally advanced or metastatic phase, will be enrolled into this cohort. AK104 6 mg/kg is administered once every 2 weeks until the observation of disease progression or intolerable toxicity, withdrawal of informed consent, investigator's decision, and the requirements of other protocol-specified criteria.

Cohort C: Hepatocellular carcinoma

30 patients with unresectable advanced hepatocellular carcinoma, who are diagnosed as stages B or C in accordance to Barcelona Clinic Liver Cancer (BCLC) system or stage B are not suitable for surgery or local treatment, or have progressed after surgery and/or local treatment and are not suitable for radical treatment; who are categorized into Child-Pugh grade A and a better rating of B (≤7 points); and who failed, are unable to tolerate or refuse standard of care,

and receive a maximum of one line of treatment during advanced stage, will be enrolled into the cohort; AK104 6 mg/kg will be administered every 2 weeks until the observation of disease progression or intolerable toxicity, withdrawal of informed consent, investigator's decision, and requirements of other protocol-specified criteria.

Cohort D: Gastric adenocarcinoma and gastroesophageal junction adenocarcinoma

This cohort is planned to enroll two dose groups, each of which includes up to 40 subjects (including subjects receiving the same dosing regimen in Phase Ib) with unresectable advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma (GEJ) who have not previously received systemic chemotherapy during the advanced or metastaticphase,.

The two dose groups were administered with AK104 6mg/kg Q2W combined with mXELOX, AK104 10mg/kg Q3W combined with XELOX chemotherapy, respectively.

Enrollment into any cohort at the dose confirmation period may be discontinued at the discretion of the sponsor and in consultation and agreement with the investigators if available clinical data of each cohort are reviewed and subjects' risk/benefit ratio are considered, or recruitment rate of subjects from the cohorts is much lower than expectation.

During the dose confirmation period, the safety of subjects will be monitored using the criteria applied in the dose escalation period. If ≥ 33% of subjects of the cohorts experience safety events meeting the DLT criteria defined in the dose escalation period of the study during the treatment period, the interruption of enrollment into the cohorts may be discussed and study data will bereviewed to determine if additional safety monitoring is required or if alternate dose or dosing intervals should be evaluated prior to further enrollment, etc.

3.1.2. Treatment Regimen

The AK104 intravenous infusion should be kept on the same day as the oxaliplatin intravenous infusion. AK104 will be administered intravenously at least 30 minutes before oxaliplatin intravenous injection and capecitabine administration. The administration time window is ± 3 days, but subjects must complete various examinations including vital signs, physical examination, laboratory examination, and physical status score to evaluate the safety and tolerability of continued treatment within 3 days before each administration.

3.1.2.1. AK104

During the dose escalation and dose expansion periods, subjects with solid tumors in Cohort 1 will be intravenously administered AK104 as a monotherapy at doses of 6 mg/kg Q2W, 10 mg/kg Q2W, 450 mg Q2W, respectively. Subjects in Cohorts 2, 3 and 4 will be administered with AK104 at doses of 4 mg/kg Q2W, 6 mg/kg Q2W, and 10 mg/kg Q2W in combination with mXELOX chemotherapy regimen, respectively. Subjects in Cohort 5 will be administered with AK104 at dose of 15 mg/kg Q3W in combination with XELOX chemotherapy regimen. In the dose confirmation period, subjects in Cohorts A, B and C will be treated with AK104 monotherapy at 6 mg/kg Q2W, and Cohort D will be treated with

AK104 6mg/kg Q2W combined with mXELOX, AK104 10mg/kg Q3W combined with XELOX chemotherapy.

AK104 in combination with mXELOX/XELOX chemotherapy will be administered until the investigator determines that the subject is unable to continue to benefit, the subject has confirmed disease progression, unacceptable toxicity and comorbidities that affect further treatment, investigator decision, subject withdrawal of consent, or other protocol-specified reasons which require treatment discontinuation. Subjects may continue to receive AK104 monotherapy between initial assessment of disease progression and confirmation of progression if the subject's performance status remains stable, and the subject can benefit from ongoing AK104 treatment in the opinion of the investigator, and other criteria listed in the protocol are met (see Section 3.1.2). Subjects will be permanently discontinued from AK104 monotherapy if their disease progressed again after continuing to receive AK104 monotherapy. In this study, the longest duration of AK104 treatment will be 24 months.

AK104 will be administered via infusion over 60 minutes (± 10 minutes). Ongoing monitoring of potential infusion reactions will be performed and pretreatment and/or adjustment of infusion rate for hypersensitivity will be allowed per protocol guidelines (Section 3.1.4.1 and Table 3-2). For subjects who cannot tolerate a 60-minute infusion or may have infusion-related risks, the infusion time may be extended to a maximum of 120 minutes (±15 minutes). Dose modification is not allowed for AK104, but dose delay is allowed for a maximum of 12 weeks, calculated from the last planned administration time, and otherwise treatment should be discontinued. In general, the dose delay does not exceed 3 days, and if the delay exceeds 3 days but less than 7 days in special cases, the actual dosing time is recommended to be used for the calculation of the time window of next dose, and then time window of subsequent dosing thereafter is still within 3 days. If the delay exceeds 7 days, it is recommended that this dose be discontinued and the next dose should be continued on the scheduled dosing date.

Refer to Section 3.1.4 for guidance on treatment modification and toxicity management of infusion-related reactions, immune-related AEs, and non-immune-related AEs related to the study drug AK104.

3.1.2.2. mXELOX and XELOX Regimens

Cohorts 2-4, mXELOX regimen

- Oxaliplatin: 85 mg/m^2 intravenous infusion for 2-6 hours on Day 1 of each 2-week cycle;

Capecitabine: 1000 mg/m^2 orally, twice daily on Days 1 to 10 of each 2-week cycle. Before the end of treatment of the first 10-12 cycles (every 2 weeks as one cycle), mXELOX chemotherapy can only be discontinued due to disease progression, unacceptable toxicity, or withdrawal of informed consent. After Cycles 10-12, if there is no confirmed disease progression or unacceptable toxicity, the subject and investigator can decide whether to continue the maintenance of capecitabine until disease progression, unacceptable toxicity, or

subject`s request or at the discretion of investigator.

Cohort 5, XELOX regimen

- Oxaliplatin: 130 mg/m^2 intravenous infusion for 2-6 hours on Day 1 of each 3-week treatment cycle;
- Capecitabine: 1000 mg/m^2 orally, twice daily on Days 1 to 14 of each 3-week treatment cycle.

Chemotherapy (XELOX) can only be discontinued due to disease progression, unacceptable toxicity, or withdrawal of consent prior to the end of Cycle 6 (every 3 weeks as one cycle). After Cycle 6, if there is no confirmed disease progression or unacceptable toxicity, the subject and investigator can decide whether to continue the maintenance of capecitabine until disease progression, unacceptable toxicity, or subject`s request or at the discretion of investigator.

Capecitabine will be supplied at the specification of 500 mg per tablet in this study. The dose will be adjusted according to the drug specification, and the dosage should be as close as possible to that of the 500 mg tablet (500 mg fold). See the investigational drug management manual for details.

If a subject experiences vomiting within 30 minutes after administration of capecitabine, the subject will be considered to take no drugs, which will be recorded in the subject diary card and the eCRF. Subjects who experience vomiting are not re-administered. However, capecitabine may be re-administered if vomiting occurs within a short period of time after administration and the capecitabine tablet taken is clearly visible. Under no circumstances should capecitabine be administered more than twice daily.

Treatment of the toxicities associated with the listed drugs, including dose delay, dose reductions, and dose modifications, should be based on the instructions in oxaliplatin and capecitabine's prescription information and standard clinical practice. In general, dose delay of oxaliplatin does not exceed 3 days, and if the delay exceeds 3 days but less than 7 days in special cases, the actual dosing time is recommended to be used for the calculation of the time window of next dose, and then time window of oxaliplatin subsequent dosing thereafter is still within 3 days. If dose delay of oxaliplatin exceeds 7 days, it is recommended that this dose be discontinued and the next dose should be continued on the scheduled dosing date. Refer to Section 3.1.5 for toxicity management and dose modifications associated with standard chemotherapy.

3.1.2.3. Treatment after Disease Progression

For subjects who receive AK104 or AK104 in combination with mXELOX/XELOX chemotherapy, they will be allowed to continue AK104 monotherapy if they are assessed as clinically beneficial and tolerant to study drug after they are considered to have disease progression by the investigator as per RECIST v1.1 criteria (see Section 3.2.3 Rationale for Treatment after Disease Progression). Subjects must discontinue AK104 treatment if there are

evidences of further disease progression (as assessed by the investigator based on imaging) or unlikely clinical benefit.

During the treatment with AK104, subjects who is judged as disease progression firstly (initial disease progression) according to RECIST v1.1 criteria must meet all of the following criteria if they continue to receive AK104 monotherapy:

- The investigator assesses that continued treatment of AK104 monotherapy can still provide clinical benefit to subjects (eg, immune-related partial response (irPR); immune-related stable disease (irSD)) (Appendix 2); the assessment of clinical benefit should take into account whether the subject has experienced clinical deterioration and is unlikely to benefit further from continued treatment.
- The subject can tolerate the study drug.
- ECOG PS score is stable.
- Management of serious complications requiring urgent intervention (eg, metastases to the CNS) will not be delayed.
- Before continuing with AK104 monotherapy, the subject needs to be fully informed and the investigator needs to elaborate on all foreseeable risks or discomfort and his or her alternative treatments.

Subjects should discontinue study treatment with AK104 if there is further evidence of continued disease progression (i.e. confirmed disease progression as assessed by the investigator) or it is unlikely to benefit from clinical treatment. A scan to confirm progression should be performed at the next follow-up and within not less than 4 weeks of the initial assessment of disease progression (without clinical deterioration).

Progression can be confirmed if the following criteria (based on irRECIST) are met:

1) The overall tumor burden (sum of the diameters of target lesions and the diameters of measurable new lesions) in 2 consecutive tumor assessments is ≥ 20%, and the absolute increase from the lowest value is 5 mm increased compared to the lowest post-dose overall tumor burden (including the baseline sum if it is the smallest sum in the study);
2) and/or compared to non-target lesions or non-measurable new lesions when the first progression is observed, there is significant definite progression (worsening) in non-target lesions or non-measurable new lesions (including additional non-measurable new lesions) when the imaging assessments are performed to confirm progression.

New lesions will be considered measurable at the time of initial progression if the longest diameter is at least 10 mm (excluding pathological lymph nodes, which should have a minimum of 15 mm in the short diameter). Any new lesion considered non-measurable at the time of initial progression may become measurable if its longest diameter increases to more

than 10 mm (excluding pathological lymph nodes, whose short diameter must increase to more than 15 mm) , and therefore needs to be included in tumor burden measurements.

3.1.3. Dose limiting toxicity

The MTD is defined as the highest dose level at which DLTs are observed in < 33% of subjects enrolled in the cohort. If DLTs are observed in < 33% of subjects at all dose levels evaluated, the MTD is not considered to have been reached.

For subjects in dose escalation cohorts, dose-limiting toxicity will be evaluated 28 days (DLT evaluation period for Cohorts 1-4) or 21 days (DLT evaluation period for Cohort 5) after the first dose. Subjects who receive < 80% of the expected dose of AK104 or AK104 in combination with mXELOX/XELOX chemotherapy during the DLT evaluation period according to the protocol without experiencing a DLT (eg, because the infusion had to be interrupted due to an infusion reaction), or subjects who do not complete safety follow up through the DLT evaluation period for reasons other than DLTs will be considered as non-evaluable subjects in the DLT evaluation, and will be replaced by other subjects who will be treated at the same AK104 dose level.

AEs will be graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) version 5.0. A DLT refers to any Grade 3 or higher investigational product-related toxicity that occurs during DLT evaluation period and is defined as follows:

A. Hepatic non-hematologic DLTs (related to study drug)

Any of the following events will be considered a hepatic DLT:

- Grade 4 AST or ALT.
- AST or ALT ≥ 5 x but ≤ 8 x upper limit of normal (ULN) and the disease severity dose not resolve decreases to Grade 2 within 5 days of optimal medical treatment including systemic corticosteroids.
- AST or ALT > 8 x ULN or total bilirubin (TBil) > 5 x ULN will be considered a DLT regardless of duration or reversibility.
- Concurrent AST or ALT > 3 × ULN and total bilirubin (TBil) > 2 × ULN (i.e. Hy's law) occur without alternative etiology.

B. Non-dermatologic, non-hepatic, non-hematologic DLTs (related to study drug)

Any Grade ≥3 non-dermatological, non-hepatic, non-hematologic toxicities (non-laboratory) related to investigational drug considered DLTs, **with the following exceptions**:

- Grade 3 fatigue lasting for ≤ 7 days.
- Grade 3 diarrhea, nausea, or vomiting that resolves to ≤ Grade 2 within 72 hours of maximized symptomatic treatment (Grade 3 diarrhea, nausea, or vomiting will not be considered a DLT without maximized symptomatic treatment).

- Grade 3 endocrine abnormalities, with or without treatment with systemic corticosteroids and/or hormone replacement therapy, and without symptoms.
- Grade 3 inflammatory reactions due to local anti-tumor response (e.g., inflammatory response in metastatic disease sites, lymph nodes).
- Grade 3 or 4 hyperglycemia that resolves to ≤ Grade 2 within 72 hours.
- Grade 3 amylase or lipase elevations alone which are not associated with clinical signs, symptoms, or radiographic features suggestive of pancreatitis.
- Grade 3 fever alone, which is unrelated to hemodynamic impairment (eg, clinical or laboratory evidence of hypotension with impaired end-organ perfusion).
- Grade 3 infusion-related reactions (first occurrence and no steroid prophylaxis).
- Grade 3 electrolyte changes without complicated related clinical adverse events.

Any other Grade 3 non-hematologic, investigational drug-related laboratory abnormality will be considered a DLT if medical intervention is required or the abnormality results in hospitalization or the abnormality lasting for > 1 week.

C. Dermatological DLTs (related to study drug)

- Grade 3 rash, which do not improve after 4 weeks of infusion delay, i.e. do not resolve to ≤ Grade 1.
- Grade 4 rash of any duration.

D. Hematologic DLTs (related to study drug)

- Grade 4 neutropenia lasting for > 3 days (despite optimal treatment).
- Grade 4 thrombocytopenia lasting for > 3 days (despite optimal treatment).
- Grade 3 thrombocytopenia with bleeding tendency.
- Grade 3 or higher neutropenia with fever (> 38.5 °C).
- Grade 3 or higher hemolysis (ie, requiring drug intervention such as blood transfusion or steroids).

3.1.4. Management and Dose Modifications for Study Drug AK104-Related Toxicities

Guidelines for treatment modification and toxicity management of infusion-related reactions, immune-related AEs, and non-immune-related AEs related to the study drug AK104 are provided in Table 3-2, Table 3-3, and Table 3-4, respectively. Additionally, treatment modification and toxicity management of other immune-related AEs not mentioned in the tables above can be found in guidelines for management of immune-related adverse events in patients treated with checkpoint inhibitors at home and abroad: including but not limited to American Society of Clinical Oncology Clinical Practice Guidelines (**Brahmer JR, 2018**).

3.1.4.1. Infusion-related reactions

The study personnel will monitor subjects for infusion-related reactions, including but not limited to, fever, chills, stiffness, rash, pruritus, flushing, headache, dizziness, syncope, sweating, shortness of breath, cough, wheezing, angioedema, hypertension, hypotension, hypoxemia, arthralgia, myalgia and other local or systemic symptoms or signs.

In this study, primary prophylaxis of infusion-related reactions (in subjects without infusion-related events) is not allowed during this study in order to avoid confusion of potential safety signals regarding the assessment of infusion reactions. Appropriate secondary prophylaxis (i.e., prevention of infusion-related reactions after the initial occurrence) should be performed at the discretion of the investigator. Prophylaxis may be performed following study site procedures or as recommended: acetaminophen/paracetamol and/or antihistamines (eg, diphenhydramine) may be administered within approximately 30 minutes prior to the start of subsequent infusions, and/or corticosteroids or equivalents may be administered according to institutional standards and guidelines. Non-sedating antihistamines (eg, cetirizine) may be considered for subjects with recurrent infusion-related reactions. After these treatments above, steroid therapy should be considered if symptoms persist. In accordance with institutional standards and guidelines, pethidine/morphine sulfate and promethazine or their equivalents may be administered intravenously at the discretion of the investigator prior to the start of subsequent infusions. Reuse of pretreatment medications may be required during infusion; therefore, this should be taken into account when deciding the dose of the pretreatment medication. Any major reactions and use requirements related to steroids must be discussed with the medical monitor.

In the event of ≤ Grade 2 infusion-related reactions, the infusion rate of AK104 may be reduced by 50%, or the infusion may be interrupted until resolution of the event, and the infusion can be resumed at 50% of the initial infusion rate until the infusion is completed. Once the infusion rate of AK104 is reduced by 50% or the infusion is interrupted due to an infusion-related reaction, all subsequent infusions must be continued at the reduced infusion rate. If a subject experiences a second Grade 2 infusion-related reaction under proper preventive medication and/or at a slower infusion rate, the investigator is advised to consider whether to discontinue AK104 based on the subject's performance of infusion-related reactions and disease status.

If a subject experiences a Grade 3 or 4 infusion-related reaction at any time, treatment with AK104 must be discontinued and no further AK104 treatment will be administered to the subject. Supportive care should be provided according to standard medical practice. If a subject experiences symptoms or signs of systemic allergic reactions or severe type I hypersensitivity reactions during administration of AK104, the subject will be treated with appropriate medications and medical equipment, which should be readily available at all study sites. Steroids, epinephrine, anti-allergic agents (antihistamines) or equivalent drugs should be readily available. Subjects should be instructed to report any delayed reactions immediately to the investigator.

AK104 treatment modifications and recommended toxicity management guidelines for AK104 infusion-related reactions are shown in Table 3-2. In the event of multiple concurrent low-grade AEs (discontinuation will not be required when occurring alone), AK104 will be discontinued at the discretion of the investigator.

Table 3-2 Recommendations on Treatment Modification for Infusion-related Reactions Caused by AK104

NCI-CTC AE Grade	Treatment Modifications for AK104
Grade 1-Mild Minor, transient reaction; infusion interruption not required; and intervention not required.	Reduce the AK104 infusion rate by 50% and closely monitor for any worsening. The total infusion time of AK104 should not exceed 120 minutes.
Grade 2-Moderate Treatment or infusion interruption required and responds promptly to symptomatic treatment (eg, antihistamines, NSAIDs, narcotics or IV fluids); and prophylactic medications required within 24 hours.	Interrupt the AK104 infusion. When the infusion-related reaction has resolved or decreased in severity to at least Grade 1, resume the infusion at 50% of the previous infusion rate and closely monitor for any worsening. If a subject experiences a second Grade 2 infusion-related reaction under proper preventive medication and/or at a slower infusion rate, the investigator is advised to consider whether to discontinue AK104 based on the subject's performance of infusion-related reactions and disease status
Grade 3 or Grade 4 – Severe or life-threatening Grade 3: delayed symptom relief (e.g., no rapid response to symptomatic medications and/or infusion interruption); recurrence after symptom improvement; and hospitalization indicated for clinical sequelae. Grade 4: Life-threatening; urgent treatment required.	Stop the AK104 infusion immediately and disconnect infusion tube from the subject. Subjects must be withdrawn from AK104 treatment immediately and must not receive any further AK104 treatment.

Abbreviations: IV, intravenous; NCI-CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events; NSAID, non-steroidal anti-inflammatory drug.

3.1.4.2. Immune-related adverse event

An irAE is defined as an AE which is associated with drug exposure and consistent with an immune-mediated mechanism of action, with no clear alternative etiology. Based on the

mechanism of action of AK104 targeting PD-1 and CTLA-4 leading to T cell activation and proliferation, it is possible to observe irAEs during the conduct of this study. Potential irAEs may be similar to those seen with anti-CTLA-4 and anti-PD-1/L1 drugs, which may include but are not limited to the following types of toxicities: infusion-related reactions (pyrexia, rash, pruritus, hypotension, dyspnea, chest discomfort, wheezing, tachycardia and stiffness), skin toxicity (rash, pruritus and vitiligo), endocrine toxicity (hyperglycemia, Hypothyroidism, hyperthyroidism, primary hypoadrenalism and hypophysitis), hepatotoxicity (hepatitis, increased aspartate aminotransferase, increased alanine aminotransferase increased bilirubin), gastrointestinal toxicity (diarrhea and colitis), pulmonary toxicity (pneumonia and pulmonary sarcoidosis), rheumatoid/boneiliac toxicity (rheumatoid arthritis, myositis and myalgia), neurotoxicity (myasthenia gravis, aseptic meningitis, encephalitis, transverse myelitis and Guillain-Barre syndrome), hematotoxicity (autoimmune hemolytic anemia, aplastic anemia, immune thrombocytopenia and acquired hemophilia), nephrotoxicity (nephritis and renal insufficiency), cardiotoxicity (myocarditis, pericarditis and cardiovascular abnormalities), and ocular toxicity (uveitis and scleritis). The investigator should monitor subjects for signs and symptoms of irAEs. In the absence of an alternative etiology (e.g., infection or disease progression), an immune-related etiology associated with clinical signs or symptoms should be considered.

The investigator will assess the severity of immunotoxicity that occurs during treatment according to NCI CTCAE v5.0, and adjust AK104 treatment to manage irAEs according to the subject's medical history, symptoms or signs. The general recommendations are as follows:

- For Grade 1 irAEs ,treatment modification is generally not required.
- For Grade 2 irAEs, the treatment of AK104 will be delayed until Grade 2 resolves to ≤ Grade 1 or baseline:
  - (1) If toxicity worsens or suggests worsening, treat it as a Grade 3 or 4 event.
  - (2) If toxicity improves, resumption of AK104 treatment at an appropriate scheduled treatment visit may be considered.
  - (3) If a Grade 2 irAE does not resolve to ≤ Grade 1 or baseline within 12 weeks after symptomatic treatment, the investigator will consider the need to permanently discontinue AK104 based on the subject's clinical condition (and communication with the sponsor's medical monitor if necessary).
- For Grade 3 irAEs, the treatment of AK104 will be delayed until Grade 3 resolves to ≤ Grade 1 or baseline:
  - (1) If toxicity improves, the investigator may consider whether to permanently discontinue AK104 treatment based on the type of individual toxicity, occurrence, and outcome process of the toxicity.
  - (2) If a Grade 3 irAE does not resolve to ≤ Grade 1 or baseline within 12 weeks after symptomatic treatment (without the following exceptions), permanent

discontinuation of AK104 treatment is recommended.

- In the event of a Grade 4 irAE, AK104 treatment must be permanently discontinued.
- The allowed maximum duration of AK104 drug interruption is 12 weeks. If AK104 can not be resumed within 12 weeks, it will be judged whether to permanently discontinue AK104 based on the type, grade, occurrence and development of the toxicity reaction of individual subject; subjects who discontinue AK104 treatment will enter the follow-up phase. The following two conditions are excluded: corticosteroids for irAEs, AK104 dose interruption for more than 12 weeks due to the glucocorticoid dose reduction process, or AK104 interruption for more than 12 weeks due to treatment of AEs that are possibly unrelated or unrelated to AK104. In these cases, discussion with the medical monitor is required to decide whether AK104 treatment can be continued.

To determine whether it is an immune-related adverse event, the most important thing is to identify the cause and characteristics of the event. The investigator should thoroughly evaluate the subject (e.g., symptoms and signs, previous diseases, concomitant medications, imaging characteristics, etc.) and support the diagnosis of irAEs using serological, immunological and histological examinations, as appropriate, to exclude any alternative etiology (e.g., disease progression, concomitant medications, infectious disease, etc.). In general, the recommended management and handling of irAEs are as follows:

- In the absence of a clear alternative etiology, all events may be considered inconclusive or possibly immune-related. With the occurrence, development and management of the event, whether there is an alternative etiology is determined as clearly as possible.
- Symptomatic and topical treatment for low-grade (Grade 1 or 2, unless otherwise specified) events should be considered.
- Treatment with systemic corticosteroids (e.g, prednisone or intravenous equivalent) for persistent low-grade or serious (≥ Grade 3) events should be considered.
- If symptoms recur or worsen during the gradual reduction of corticosteroids, increase the dose of corticosteroid until symptoms stabilize or improve and then the gradual reduction of corticosteroids will be given again at a slower rate.
- For events that do not respond to systemic steroids, after discussion with the medical monitor, consideration should be given to the use of more potent immunosuppressants, i.e. TNF antagonists (e.g., infliximab, and mycophenolate mofetil).
- Discontinuation of study drug is not mandatory for Grade 3 or 4 inflammatory reactions due to local tumor response (e.g., inflammatory response in metastatic disease sites, and lymph nodes).
- If it is necessary to consult experts in other fields (e.g., immunologists), this

information should be obtained in a timely manner in order to make appropriate treatment decisions.

- In the event of multiple concurrent low-grade AEs (discontinuation will not be required when occurring alone), AK104 will be discontinued at the discretion of the investigator.

AK104 treatment modification recommendation and recommended toxicity management guidelines for irAEs are presented in Table 3-3. For irAEs not covered in Table 3-3 or some specific toxicities with special circumstances, please refer to the domestic and foreign guidelines for toxicity management of immunotherapy, such as the American Society of Clinical Oncology Clinical Practice Guidelines (**Brahmer JR, 2018**), the European Society of Clinical Oncology Clinical Practice Guidelines (**Haanen JBAG, 2017**), and the Chinese Society of Clinical Oncology Management Guidelines for Toxicities from Immune Checkpoint Inhibitors, etc. For sites with extensive experience in immune research or treatment management, where there are discrepancies in the specific procedures of toxicity management guidelines from immune checkpoint inhibitors, they may be performed in accordance with site standard clinical practice, provided that the investigator monitors and processes as promptly as possible to maximize the safety of subjects; if necessary, the investigator may discuss with the sponsor and manage the toxicity accordingly.

Table 3-3 AK104 Treatment Modification Recommendation and Recommended Toxicity Management Guidelines for irAEs

Gastrointestinal irAE (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue treatment with AK104. Opioids/anesthetics may mask the symptoms of perforation. Infliximab should not be used in cases of perforation or sepsis.)		
Diarrhea/Colitis (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Asymptomatic; clinical or diagnostic observations only; no therapeutic intervention required temporarily (Grade 1 diarrhea ≤ 4 times/day)	• Continue AK104 treatment • Antidiarrheal drugs (e.g., loperamide) may be used appropriately when the subject has no signs of infection but not as a strongly recommended regimen	• Closely observe for 24-48 h for changes and progression • Perform tool routine and hematology if symptoms persist • Light diet is recommended during acute diarrhea • Instruct subjects to report immediately if worsening occurs • Avoid high fiber/lactose diet <u>If symptoms worsen:</u> • Treat as Grade 2 or 3-4
Grade 2	• Suspend AK104 treatment;	• Gastrointestinal X-ray shall be

Abdominal pain; mucus or bloody stool (Grade 2 diarrhea frequency 4-6 times/day) ADL are not interfered with IV fluids for < 24 hours. Colitis: abdominal pain; hematochezia	• Feces and blood tests: C-reactive protein, erythrocyte sedimentation rate, fecal calprotectin, lactoferrin, imaging and endoscopy optional can be chosen • If only diarrhea is present, observe for 2-3 days. If symptoms do not improve, prednisone 1 mg/(kg.d) (or equivalent dose of methylprednisolone) is administered; antidiarrheal drugs are not recommended • If there are symptoms of diarrhea and colitis (abdominal pain +/-bloody stools), immediately use prednisone 1 mg/(kg.d) (or equivalent dose of methylprednisolone): • If symptoms do not improve within 48 h, increased dose of prednisone 2 mg/(kg.d) (or equivalent dose of methylprednisolone) is administered • If the subject's symptoms improve, the hormone therapy is gradually stopped after more than 4-6 weeks: • Resume treatment with AK104 when the hormone is reduced to ≤ 10 mg/day and the subject is asymptomatic (grade ≤ 1) • Colitis reappears on resumption of AK104 treatment: ≤ Grade 2: Suspend AK104 temporarily	performed for signs of colitis; • Appointment for colonoscopy and biopsy; If symptoms improve to Grade 1: • Resume treatment with AK104 (the dose of corticosteroid shall be reduced before resuming if glucocorticoids have been given) • If symptoms last > 5 to 7 days or recur: • 0.5-1.0 mg/kg/day methylprednisolone or equivalent • When symptoms improve to Grade 1, gradually reduce the dose of corticosteroid for at least within 1 month. Consider prophylactic antibiotics for opportunistic infection prophylaxis, and then resume AK104 as per protocol. <u>If symptoms worsen or persist for > 3 to 5 days after oral corticosteroids:</u> • Treat as Grade 3 to 4

	≥ Grade 3: Discontinue AK104	
Grade 3 to 4 Grade 3: Severe abdominal pain; change in bowel habits; drug intervention indicated; peritoneal irritation sign (Grade 3 diarrhea ≥ 7 times/day) Grade 4: Life-threatening symptoms; urgent intervention indicated	• Grade 3: Delay AK104 treatment: resume AK104 treatment when the hormone is reduced to ≤ 10 mg/day and the subject has no obvious symptoms (Grade ≤ 1). Hospitalization is recommended • Grade 4: Discontinuation of AK104 and hospitalization • Blood and stool infection tests, inflammatory markers, imaging, endoscopy, and gastrointestinal counseling • Immediately inject prednisone 1-2 mg/(kg.d) (or equivalent dose of methylprednisolone) via IV. If symptoms improve, refer to the guidelines for improvement of subjects with Grade 2 symptoms. • If IV corticosteroid is resistant or symptoms do not improve, prednisone 2 mg/(kg.d) (or equivalent dose of methylprednisolone) is given for 3 days • Consider other anti-inflammatory drugs, such as infliximab 5 mg/kg, and re-dose 2 weeks later if needed	Abdominal and pelvic enhanced CT is recommended for signs of colitis; <u>If symptoms improve:</u> • Continue glucocorticoid therapy until symptoms resolve to Grade 1 and then gradually reduce the dose at least within 1 month <u>If symptoms last > 3 to 5 days, or recurrent after improvement:</u> • Add infliximab 5 mg/kg (if no contraindications) Note: Infliximab should not be used in cases of perforation or sepsis.
Skin irAEs (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue with AK104)		
Macule/Papule Grade (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**

Grade 1 Macules/papules cover < 10% of total body surface area with or without symptoms (e.g., pruritus, burning, or tightness)	• Continue AK104 treatment • Oral antihistamines. Cetirizine/Ciminol 10 mg/day (non-sedated); hydroxyzine 10-25 mg QID or at bedtime • Topical steroid hormone therapy. Classical hormones type I (clobetasol propionate, betamethasone dipropionate ointment) are used for the body; type V/VI hormones (aclomethasone, desonide, hydrocortisone 2.5% ointment) are used for the face	<u>If symptoms last > 1 to 2 weeks or recurrent:</u> • Consider skin biopsy • Delay AK104 treatment • Consider methylprednisolone 0.5 - 1.0 mg/kg/day IV or its equivalent. If symptoms improve, gradually reduce the dose of corticosteroids within at least 1 month. Consider prophylactic antibiotics to prevent opportunistic infection, and resume AK104 treatment <u>If symptoms worsen:</u> • Treat as Grade 3 to 4 Hematology, liver and kidney function tests if necessary Consider referral to dermatology for skin biopsy
Grade 2 Macule/papules cover 10% -30% of total body surface area with or without symptoms (e.g., pruritus, burning, or tightness); limiting of instrumental ADL	• Continue AK104 treatment • Non-urgent skin recommendations • Oral antihistamines: cetirizine/ciminol 10 mg/day (non-sedative); hydroxyzine 10-25 mg QID or at bedtime • Topical steroid hormone therapy (refer to Grade 1 reaction)	
Grade 3 to 4 Macules/papules cover > 30% of total body surface area with or without symptoms (e.g., pruritus, burning, or tightness); and limiting of instrumental ADL	Grade 3: Delay AK104 treatment Grade 4: discontinuation of AK104 • Consider skin biopsy • Dermatology consultation • Exclude systemic anaphylaxis: CBC, complete biochemical set • Oral antihistamines: cetirizine/ciminol 10 mg/day (non-sedative); hydroxyzine 10-25 mg QID or at bedtime • Systemic steroid hormone therapy • Prednisone 0.5 - 1 mg/(kg.d)	If Grade 3 rash does not improve to ≤ Grade 1 or baseline within 30 days after a temporary delay in AK104, then • Permanently discontinue AK104 treatment. If symptoms improve to Grade 1: • Gradually reduce the dose of corticosteroids at least 1 month and increase prophylactic antibiotics to prevent opportunistic infections • Resume AK104 (for Grade 3 events that have improved to Grade 1) Hematology, liver and kidney function tests if necessary

	IV (or equivalent dose of methylprednisolone) until rash reaction ≤ Grade 1	
Pruritus Grade (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Mild or localized; local intervention indicated	• Emollients (mainly creams or ointments without fragrance: Classical hormones type I (clobetasol propionate, betamethasone dipropionate ointment) are used physically; type V/VI hormones (aclomethasone, desonide, hydrocortisone 2.5% ointment) are used on the face. • Concomitant oral antihistamines: cetirizine/clenemin 10 mg daily (non-sedative); hydroxyzine 10-25 mg QID or at bedtime.	Hematology, liver and kidney function tests if necessary
Grade 2 Intense or extensive; intermittent; skin damage due to scratching (e.g., edema, papule, desquamation, lichenization, exudate/crusting); effective oral drug intervention; limiting of instrumental ADL	• Continue AK104 treatment under intensive antipruritic therapy • Classical steroid hormones type I (clobetasol propionate, betamethasone dipropionate ointment) are used for the body; type V/VI hormones (aclomethasone, desonide, hydrocortisone 2.5% ointment) are used for the face. Concomitant oral antihistamines: cetirizine/ciminol 10 mg daily (non-sedative); hydroxyzine 10 - 25 mg QID or at bedtime • Oral steroids	• Refer to dermatology for consultation and consider referral to dermatology • Discontinuation may be considered for some serious subjects • Hematology, liver and kidney function tests if necessary

	• Prednisone 0.5 - 1 mg/(kg.d) (or equivalent methylprednisolone) for at least 2 weeks and gradually discontinued	
Grades 3 or 4 Strong or extensive; Persistent; Limited self-care or lethargy; oral corticosteroids or immunosuppressive therapy	• Suspend AK104 • Oral corticosteroids: prednisone/methylprednison 0.5 - 1 mg/(kg.d) for at least 2 weeks and gradually discontinued . • Oral GABA agonists (pregabalin, gabapentin 100 - 300 mg TID). For treatment of refractory pruritus, aprepitant or omalizumabmay be considered (e.g., elevated IgE levels)	• Refer to dermatology for consultation and consider referral to dermatology • Hematology, liver and kidney function tests if necessary • Biopsy if necessary
Cardiovascular irAEs		
Cardiovascular system (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Mild transient reaction; infusion not interrupted; intervention not required	• Continue treatment with AK104 and consider drug interruption as appropriate • ECG, myocardial zymogram (CK, CK-MB, etc.), markers of myocardial injury (cTnI, cTnT, NT-proBNP, and BNP, etc.), and inflammatory markers (ESR, CRP, and WBC) • Analyze medical history and potential risks, and determine whether there is alternative etiology	• Pay attention to clinical monitoring, timely understand the symptoms and signs of the subjects, and remind the subjects to seek medical attention in time
Grade 2 Treatment or infusion	• Suspend AK104 • ECG, myocardial zymogram	• Subjects without risk factors or past medical history must be closely

interrupted, symptomatic management (e.g., antihistamines, NSAIDs, narcotics, and IV infusion); prophylactic medication within 24 hours	(CK, CK-MB, etc.), markers of myocardial injury (cTnI, cTnT, NT-proBNP, and BNP, etc.), and inflammatory markers (ESR, CRP, and WBC) • Consider timely hormone therapy according to clinical condition • Consultation with cardiology department as appropriate and timely supportive care as soon as possible • If underlying cardiac disease and risk factors are present: ✓ Invite the cardiology department to actively manage underlying diseases (e.g., heart failure and atrial fibrillation) ✓ Active control of risk factors for heart disease (including hypertension, hyperlipidemia, uninterrupted smoking, and diabetes) • Analyze medical history and potential risks, and determine whether there is alternative etiology	followed up and monitored after resumption of AK104 treatment • For subjects with risk factors or past medical history, the investigator determines whether to resume AK104 treatment. Once AK104 is resumed, the subjects must be closely followed up and monitored
Grade 3 Prolonged (e.g., prompt symptomatic management not required, and/or infusion temporarily interrupted); recurrence of symptoms after initial management; hospitalization to	• Discontinue AK104 • ECG, myocardial zymogram (CK, CK-MB, etc.), markers of myocardial injury (cTnI, cTnT, NT-proBNP, and BNP, etc.), and inflammatory markers (ESR, CRP, and WBC) • Consultation with cardiology department immediately and timely supportive treatment	• Close follow-up and monitoring of outcomes and changes • Hormone reduction and discontinuation, which must be based on various tests and the subject's clinical condition

manage sequelae **Grade 4** Life-threatening consequences, urgent treatment required	as soon as possible • Cardiac color ultrasonography or MRI • ECG monitoring, and ICU ward if necessary • Methylprednisolone is given 0.5-1 g/day for 3-5 days, gradually reduced within 4-6 weeks after recovery of cardiac function to baseline	
Lung irAE (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue treatment with AK104. Evaluated by imaging and lung condition consultation.)		
Pulmonitis (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Asymptomatic; radiographic changes only, confined to a single lobe or < 25% of the lung parenchyma, requiring only clinical or diagnostic observations	• Delay of AK104 treatment as appropriate • Reexamination of chest CT and pulmonary function (at least every 3 weeks) before each cycle of AK104 treatment ✓ If imaging shows improvement, no further CT examination is required. Close follow-up and resumption of treatment; ✓ If progression is shown in imaging, the treatment regimen will be upgraded and AK104 treatment will be interrupted; ✓ If no changes are shown in imaging, consider continued treatment and close follow-up until new symptoms develop. • Treatment regimen will be upgraded if symptoms worsen • Consider consulting a	• Reimaging at least every 3 weeks <u>If symptoms worsen:</u> Treat as Grade 2 or 3-4 • Self-symptom monitoring every 2-3 days, and reexamination of oxygen saturation (personal pulse oximetry can be used); • Weekly follow-up visit to hospital to track changes in symptoms, chest physical examination, repeated oxygen saturation and chest CT • Sputum test as appropriate to exclude pathogen infection

	physician of respiratory and Infectious department	
Grade 2 Symptomatic; instrumental ADL limited; medical intervention indicated; new or worsening symptoms, including: shortness of breath, cough, chest pain, fever, and hypoxia; multiple lobes and 25-50% of lung parenchyma involved; instrumental ADL impacted; and drug intervention required	• Conduct chest high-resolution CT, hematology, liver and kidney function, electrolytes, and lung function analysis • Suspent AK104 until reduced to ≤ Grade 1 • Methylprednisolone 1 - 2 mg/(Kg.d) (IV or oral equivalent) • 2-3 days of glucocorticoids/supportive care: If improvement of symptoms ≤ Grade 2, corticosteroids are given at low dose and gradually reduced in 4-6 weeks. If symptoms do not improve or worsen, treat as Grade 3-4 reactions; if infection cannot be completely excluded, addition of empiric anti-infective therapy should be considered • Reexamination of chest CT 3-4 weeks later • If clinical symptoms and improvement in imaging ≤ Grade 1, AK104 may be used after assessment	• If not improved or worsened after 2 weeks: • Treat as Grade 3 to 4 • Nasal swab, sputum culture and drug sensitivity test, blood culture and drug sensitivity test, urine culture and drug sensitivity test are performed to exclude pathogen infection • Monitor the following items every 1-3 days: medical history and physical examination, oxygen saturation (resting and active) • Weekly reexamination of chest CT, blood test, and pulmonary function • Consider consulting a physician of respiratory and Infectious department • The lungs are evaluated and bronchoscopy and bronchoalveolar lavage are performed. Biopsy is considered for atypical lesions.
Grade 3 to 4 Grade 3: Severe new symptoms with all lobes or > 50% of the lung parenchyma involved; severe symptoms; limited personal independence; oxygen required;	• Conduct chest high-resolution CT, hematology, liver and kidney function, electrolytes, and lung function analysis • Permanently discontinue AK104 treatment • Hospitalization: Consider ICU monitoring and nursing • Empiric anti-infective therapy is indicated if infection has	• Nasal swab, sputum culture and drug sensitivity test, blood culture and drug sensitivity test, urine culture and drug sensitivity test are performed to exclude pathogen infection • The lungs are evaluated and bronchoscopy and bronchoalveolar lavage are performed. Biopsy is considered for atypical lesions. <u>If improved to baseline after 48 hours:</u>

hospitalization required Grade 4: Life-threatening dyspnea, acute respiratory distress syndrome (ARDS); urgent intervention required (e.g., intubation)	not been excluded; consult respiratory department or infectious department if necessary • Methylprednisolone 2 mg/(Kg.d) IV for 2-3 days glucocorticoid/equivalent supportive treatment, and lung ventilation as appropriate; • Add prophylactic antibiotics to prevent opportunistic infections	• Reduce the dose of corticosteroids to 1 mg/(Kg.d) and gradually reduce the dose in at least 4-6 weeks If not improved or worsened after 48 hours: • Add other immunosuppression (e.g., infliximab (5 mg/kg), cyclophosphamide, IVIG, or mycophenolate mofetil (1 g twice daily) or IV gamma immunoglobulin) . Drug Re-evaluation: • Grade 3: Discuss and weigh the subject's risk/benefit to determine the subject's medication regimen until improvement in symptoms and imaging • Grade 4: Discontinue AK104
Liver irAEs (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue treatment with AK104. Consider imaging of obstruction)		
Liver function (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Normal baseline values: AST/ALT >3.0× ULN TBIL > ULN -1.5 × ULN Abnormal baseline values: AST/ALT > 1.5-3.0 × baseline TBIL > 1.0-1.5 × baseline	• Continue AK104 treatment	• Continue monitoring of liver function <u>If symptoms worsen:</u> • Treat as Grade 2 or 3-4
Grade 2 Normal baseline values: AST/ALT >3.0- 5.0× ULN TBIL >1.5-3.0× ULN	• Suspend AK104 • Exclusion of viral hepatitis, autoimmune disease, biliary obstruction, new metastasis, or thrombosis • Liver biopsy is optional	<u>If returned to baseline:</u> • Resume routine monitoring and AK104 treatment <u>If the increase persists for > 5 to 7 days or worsens:</u> • 0.5 - 1 mg/kg/day of

Abnormal baseline values: AST/ALT > 3.0-5.0 × Baseline TBIL > 1.5-3.0 x baseline	• AK104 may be resumed when corticosteroids are reduced to 10 mg/day (toxicity grade ≤ 1)	methylprednisolone or equivalent. Gradually reduce the dose of glucocorticoids in at least 1 month when LFTs return to Grade 1 or baseline. Consider prophylactic antibiotics to prevent opportunistic infections, and resume AK104 treatment; Liver biopsy if necessary
Grade 3 to 4 **Grade 3:** Normal baseline values: AST/ALT >5.0- 20.0× ULN TBIL >3.0-10.0× ULN Abnormal baseline values: AST/ALT > 5.0-20.0 × Baseline TBIL > 3.0-10.0 × baseline **Grade 4:** Normal baseline values: AST/ALT ＞ 20.0×ULN TBIL ＞10.0×ULN Abnormal baseline values: AST/ALT > 20.0 × Baseline TBIL > 10.0 x baseline	• Discontinue AK104 (AK104 may be delayed if AST/ALT ≤ 8 × ULN and total bilirubin ≤ 5 × ULN) • Increase monitoring frequency to once every 1-2 days • 1.0 to 2.0 mg/kg/day IV methylprednisolone or equivalent (in the presence of Grade 4 hepatitis, the recommended starting dose is 2.0 mg/kg/day IV methylprednisolone or equivalent) • Add prophylactic antibiotics to prevent opportunistic infections • Consider mycophenolic acid if drug resistance occurs after 3 days • Gradually reduce the dose of corticosteroids in 4 weeks or more if liver enzymes improve	If recovered to Grade 2: • Gradually reduce the dose of corticosteroids for at least 1 month If not improved, worsened, or rebounded within > 3 to 5 days: • Add 1 gram of mycophenolate mofetil, BID • If no response within an additional 3-5 days, consider other immunosuppressants according to local guidelines • Consult a liver specialist of gastroenterology department • Consider MRI/CT of the liver if clinically warranted, • Consider liver biopsy
Renal irAE (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue with AK104)		
Creatinine increased (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1	• Continue AK104 treatment	• Weekly monitoring of creatinine levels

Creatinine > ULN-1.5 x ULN	• Search for possible causes	and proteinuria If returned to baseline: • Routine creatinine monitoring is resumeed per protocol If symptoms worsen: • Treat as Grade 2 - 3 or 4
Grade 2 Creatinine > 1.5-3.0 × baseline value; Creatinine > 1.5-3.0 × ULN	• Suspend AK104 • Monitor creatinine and urine protein every 2 to 3 days • Consider renal biopsy • Nephrology consultation • Exclude renal failure due to other causes and administer methylprednisolone 0.5 to 1.0 mg/kg/day or equivalent	If recovered to Grade 1: • Gradually reduce the dose corticosteroids within at least 1 month. Consider prophylactic antibiotics to prevent opportunistic infections, resume AK104 as per protocol and start routine creatinine monitoring. If the increase persists for > 7 days or worsens: • Treat as Grade 4 • Consider addition of azathioprine/cyclophosphamide/cyclosporine/infliximab/mycophenol • When making a decision, consider the time difference between the onset of the event and the effect of treatment • For subjects with persistent or recurrent creatinine increase, AK104 treatment should be discontinued Grade 4: Dialysis recommended
Grade 3 Creatinine > 3.0 × baseline value; Creatinine > 3.0-6.0 × ULN	• Discontinue AK104 • Require inpatient hospitalization or prolongation of existing hospitalization • Monitor creatinine and urine protein every 24 hours • Nephrology consultation • Exclude renal failure due to other causes and administer methylprednisolone 0.5 to 1.0 mg/kg/day or equivalent	
Grade 4 Creatinine > 6.0 × ULN	• Discontinue AK104 • Dialysis recommended • Require inpatient hospitalization or prolongation of existing hospitalization • Monitor creatinine and urine protein every 24 hours • Nephrology consultation • Methylprednisolone 1.0 to 2.0 mg/kg/day IV or equivalent • Consult a nephrologist • Consider renal biopsy	

Neurological irAEs (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue with AK104)		
Encephalopathy/Leukoencephalopathy/Reversible Posterior Leukoencephalopathy Syndrome (PRES)		
Neurotoxicity grade (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Asymptomatic or mildly symptomatic; intervention not indicated	• Continue AK104 treatment and initiate etiology diagnosis	• Ongoing monitoring of subjects If symptoms worsen: • Treat as Grade 2 or 3-4
Grade 2 Moderate symptoms; Adjuvant ADL only	• Delay AK104 treatment • Treat symptoms according to local guidelines • Consider methylprednisolone 0.5 to 1.0 mg/kg/day IV or equivalent	If improved to baseline: • resume AK104 If symptoms worsen: • Treat as Grade 3 to 4
Grade 3 to 4 Grade 3: Severe symptoms; limiting self-care ADL Grade 4: Life-threatening, urgent intervention required	• Discontinue AK104 • Neurology Consultation • Treat symptoms according to local guidelines • Consider methylprednisolone 1.0 to 2.0 mg/kg/day IV • or equivalent • Add prophylactic antibiotics to prevent opportunistic infections	If improved to Grade 2: • Gradually reduce the dose of corticosteroids withinat least 1 month If symptoms worsen or atypical manifestations occur: • Consider IVIG or other immunosuppressive therapy according to local guidelines
Endocrine irAEs (Non-inflammatory causes should be excluded. If a non-inflammatory cause is determined, treat accordingly and continue treatment with AK104. Consider perimetry, endocrinology consultation, and imaging.)		
Hypothyroidism (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Asymptomatic;	• No dose modifications	Monitor TSH and free T4 every 4-6 weeks For Grade 1 (including asymptomatic

clinical or diagnostic observations only; no treatment required		subjects with TSH elevation): • Perform appropriate endocrine function tests to monitor subjects • If TSH < 0.5 × LLN or TSH > 2 × ULN, or consistently out of range in 2 subsequent measurements, add free T4 levels in subsequent cycles as clinically indicated and consider endocrine consultation
Grade 2 Symptomatic; thyroid hormone replacement therapy required; affecting instrumental ADL	• AK104 may be discontinued until the event resolves to ≤ Grade 1	• If toxicity worsens, treat as Grade 3 or 4 • If toxicity improves to baseline, treatment should be performed at the next scheduled treatment date
Grade 3 Severe symptoms; limited self-care ability; hospitalization required	• Suspend AK104 for irAEs ≥ Grade 3 • Rusume AK104 treatment when symptoms resolve to Grade 2 or better • Standard thyroid hormone replacement therapy will be used: Adequate doses (1.6 mcg/Kg) can be achieved in young and normally healthy subjects, but a reduced dose of 25-50 mcg will be used for the elderly or subjects with a history of cardiovascular disease	• Replacement therapy will be performed in the event of isolated hypothyroidism without interruption of treatment or use of glucocorticoid • Serum TSH and free T4 will be measured every 6-8 weeks, • Medication will be adjusted based on the results. If TSH is above the reference range, thyroid hormone will be increased by 12.5 - 25 mcg • For the treatment with maintenance dose, the assessment should be performed annually or whenever the subject feels unwell
Grade 4 Life-threatening; urgent intervention required		
Hyperthyroidism (Refer to NCI-CTCAE V5.0)	**Treatment**	**Follow-up**
Grade 1 Asymptomatic; clinical or diagnostic observations only; no treatment required	• Suspend AK104 for irAEs ≥ Grade 3 • Standard of care regimen for hyperthyroidism will be performed • If the subject experiences hypothyroidism (decreased	• Thyroiditis is a self-limited disorder that consists of 2 stages. During the hyperthyroidism stage, β-blockers (e.g., atenolol 25 - 50 mg/day, titrable when BP is acceptable and heart rate < 90 beats/min) can be used if symptoms are present. Symptoms will be closely
Grade 2 Symptomatic; thyroid		

hormone suppression indicated; affecting instrumental ADL	free T4/T3, even if TSH is not elevated), thyroid hormone therapy will be used. • Grave's disease will be treated according to the guideline.	assessed and blood T4 will be measured every 2 weeks.
Grade 3 Severe symptoms; limited self-care ability; hospitalization required		
Grade 4 Life-threatening; urgent intervention required		

In general, for other immune-related AEs: AK104 will be suspended for intolerable/persistent Grade 2 AEs, suspended or permanently discontinued for Grade 3 AEs, and permanently discontinued for Grade 4 AEs or recurrent Grade 3 AEs.

If sustained clinical improvement is observed, subjects receiving IV steroids may switch to an equivalent dose of oral corticosteroids (e.g., prednisone) at the start of dose reduction or earlier. Low bioavailability of oral corticosteroids should be considered when switching to equivalent oral corticosteroids.

ADL, activities of daily living; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CT, computed tomography; irAE, immune-related adverse event; IV, intravenous; IVIG, intravenous immunoglobulin;LFT, liver function test; LLN, lower limit of normal; MRI, magnetic resonance imaging; NCI-CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events; T4, thyroxine; TSH, thyroid stimulating hormone;ULN = upper limit of normal.

3.1.4.3. Non-immune-related Adverse Events

All non-immune-related AEs will be graded according to NCI CTCAE version 5.0. For AK104-related Grade 3 or higher non-immune-related AEs assessed by the investigator, AK104 administration may be suspended and clinical management will be given. If the adverse event resolves or improves, the investigator may determine whether to continue the study drug. For laboratory test abnormalities that are clearly not attributable to AK104 or not clinically significant, missed dose will not be required. Dose reductions will not be allowed. Non-immune-related reactions will be managed according to the clinical practice at each site.

Table 3-4 AK104 Treatment Modification Recommendation and Recommended Toxicity Management Guidelines for Non-Immune-Related Reactions

Event Grade	Dose Modifications and Toxicity Management
Any grade	• Note: Dose adjustment is not recommended for AEs not related to study treatment (i.e., events due to underlying disease) or clinically insignificant laboratory abnormalities.
Grade 1	• Dose adjustment is not generally recommended • Treatment will be performed according to the clinical standard of the study site

Event Grade	Dose Modifications and Toxicity Management
Grade 2	• Dose adjustment is not generally recommended • Treatment will be performed according to the clinical standard of the study site
Grade 3	• Suspend study drug until the event resolves to ≤ Grade 1 or baseline • For AEs that decrease to ≤ Grade 2 within 7 days or resolve to ≤ Grade 1 or baseline within 14 days, the resumption of study drug may be considered as soon as possibly. • Treatment will be performed according to the clinical standard of the study site.
Grade 4	• Study drug may be discontinued at the investigator's clinical discretion and in consultation with the medical monitor (Note: for Grade 4 laboratory abnormalities, treatment discontinuation will be considered based on medical history and concomitant clinical signs/symptoms). • Treatment will be performed according to the clinical standard of the study site.

3.1.5. Toxicity Management and Dose Modifications Related to Standard Chemotherapy

Treatment of the toxicities associated with the listed drugs, including dose delay, dose reductions, and dose modifications, should be based on the instructions in oxaliplatin and capecitabine's prescription information and standard clinical practice. If, in the opinion of the investigator, toxicity is due to only one drug under study treatment, the administration of the drug should be delayed or dose adjusted. If there is no contraindication, the administration of other chemotherapeutic agents should be continued. If toxicity is considered by the investigator to be related to only one chemotherapeutic agent, no dose adjustment of other chemotherapeutic agents will be required. Subjects who temporarily discontinue chemotherapy due to adverse reactions may continue treatment with AK104 as originally planned after discussion with the sponsor or designated medical monitor if they meet the requirements for the use of AK104. Subjects who discontinue chemotherapy due to toxicity should not withdraw completely from all study treatment and should continue to complete treatment with AK104.

Any subject who requires a dose reduction of chemotherapy will continue to receive the dose reduction in subsequent treatment cycles. Any subject who has had 2 dose reductions will be recommended to consider discontinuing the drug if a third dose reduction is required due to toxicity. Treatment may be postponed to allow sufficient time for the subject to recover from the toxicity caused by the study drug. For concomitant conditions that have already occurred at baseline, the dose will be adjusted according to the corresponding change in toxicity grade if deemed appropriate by the investigator. For example, if the asthenia of a subject is Grade 1 at baseline and increases to Grade 2 during treatment, it is a change from Grade 1 and will be considered a Grade 1 toxicity for dose modification purposes. When severe toxicities of different severity grades occur simultaneously, dose modifications should be made according to the highest grade observed. The study sites are allowed to reasonably and promptly process the manifestations of toxicity in different subjects based on the drug labeling and clinical

consensus, and combined with clinical diagnosis and treatment routine and practical experience, provided that the investigator monitors and observes the subjects as closely as possible to maximize the safety of subjects; if necessary, the investigator may discuss with the sponsor to manage the toxicity and make dose modifications accordingly.

3.1.5.1. Dose Modifications for Hematotoxicity

If the subject' s absolute neutrophil count (ANC) is ≥ 1.5 x 10^9/L and the platelet count is ≥ 100 x 10^9/L at the start of the treatment cycle, a new 2-week cycle may be initiated. Otherwise, treatment shall be delayed until blood parameters have recovered. At initiation of subsequent treatment cycles, dose modifications should be based on the lowest hematology count in prior treatment cycles. If relevant tests do not meet the requirements, treatment may be delayed to allow sufficient recovery time. Detailed guidance on dose modification for hematologic toxicities after subject's recovery is provided in Table 3-5.

Table 3-5 Dose Modification of Capecitabine (X) in Combination with Oxaliplatin (O) for Hematologic Toxicities During the Planned Treatment Period

Toxicity[a]	**Dose Modifications for Capecitabine and Oxaliplatin at Restart of Treatment**
ANC ≥ 1.0 and < 1.5 (x 10^9/L) and/or Platelet count ≥ 50 and < 75 (x 10^9/L)	First occurrence: X: 100% dose, O: 85 mg/m^2 or 130 mg/m^2 Second occurrence: X: 75% dose, O: 65 mg/m^2 or 100 mg/m^2 Third occurrence: X: 50% dose, O: 50 mg/m^2 or 75 mg/m^2 Fourth occurrence: permanent discontinuation
ANC ≥ 0.5 and < 1.0 (x 10^9/L) and/or Platelet count ≥ 25 and < 50 (x 10^9/L)	First occurrence: X: 75% dose, O: 65 mg/m^2 or 100 mg/m^2 Second occurrence: X: 50% dose, O: 50 mg/m^2 or 75 mg/m^2 Third occurrence: permanent discontinuation
ANC <0.5(x 10^9/L) and/or Platelet count < 25 (x 10^9/L)	First occurrence: X: 50% dose, O: 50 mg/m^2 or 75 mg/m^2 Second occurrence: permanent discontinuation
Grade 3 neutropenic fever (ANC < 1.0 (x 10^9/L) with fever ≥ 38.5 °C	First occurrence: X: 50% dose, O: 50 mg/m^2 or 75 mg/m^2 Second occurrence: permanent discontinuation
Grade 4 neutropenic fever (ANC < 1.0 (x 10^9/L) with fever ≥ 38.5 °C, and with life-threatening sepsis	Drug permanently discontinued

a. Nadir of prior cycles. X = Capecitabine, O = Oxaliplatin

If a unscheduled assessment performed during the treatment demonstrates hematologic toxicity, capecitabine administration must be interrupted during this cycle, and capecitabine should be reduced during subsequent cycles as shown in Table 3-6.

Table 3-6 Dose Modification of Capecitabine (X) for Hematologic Toxicities During the Treatment Cycle

Toxicity[a]	Dose Modification of Capecitabine During the 1-10/1-14 Days of Treatment
ANC ≥ 1.0 and < 1.5 (x 10^9/L) and/or Platelet count ≥ 50 and < 75 (x 10^9/L)	Suspend until the recovery to grade 0-1. Take the original dose in the cycle, and do not make up the missing dose.
ANC ≥ 0.5 and < 1.0 (x 10^9/L) and/or Platelet count ≥ 25 and < 50 (x 10^9/L)	Suspend until the recovery to grade 0-1. Take 75% of original dose in the cycle, and do not make up the missing dose.
ANC <0.5(x 10^9/L) and/or Platelet count < 25 (x 10^9/L)	Discontinue the treatment unless recovery to grade 0-1 in the opinion of the investigator. Take 50% of original dose in the cycle, which is the best for the subjects.

3.1.5.2. Dose Modifications for Non-Hematologic Toxicities

Recommendations for dose modification of capecitabine apply to toxicities related to capecitabine and not to toxicities associated with combination therapy.

In the event of Grade 1 non-hematologic toxicity, the original dose may be continued. In the event of a Grade 2, 3, or 4 non-hematologic toxicity, capecitabine treatment must be interrupted or discontinued immediately. The interruption of capecitabine treatment should be regarded as missing of treatment time, and the missing dose will not be made up. If relevant tests do not meet the requirements, treatment may be delayed to allow sufficient recovery time as follows:

If a Grade 2 non-hematologic adverse reaction occurs, interruption of study drug is recommended for the first, second, and third occurrence until recovery to Grade 0-1, and the starting dose for the next cycle will be 100%, 75%, and 50% of the original dose, respectively. Permanent discontinuation will be considered for the fourth occurrence.

If a Grade 3 non-hematologic adverse reaction occurs, drug interruption is recommended for the first and second occurrence until recovery to Grade 0-1, and the starting dose for the next cycle will be 75% and 50% of the original dose, respectively. Permanent discontinuation will be required for the third occurrence.

If a Grade 4 non-hematologic adverse reaction occurs first, and if the investigator believes that continuation of treatment is beneficial to the subject, the drug will be suspended until recovery to Grade 0-1, and the starting dose in the next cycle will be 50% of the original dose. Permanent discontinuation will be required for reappearance.

Recommendations for dose modification of oxaliplatin will apply to toxicities related to oxaliplatin and not to toxicities associated with combination therapy. Modification of oxaliplatin for non-hematologic toxicities are presented in Table 3-7 and for neurotoxicities in

Table 3-8.

Table 3-7 Dose Modification of Oxaliplatin for Non-hematologic Toxicities

Toxicities	Grade	Dose Modification
Allergic reaction	3 or 4	Permanently discontinue treatment
Respiratory symptoms suggest pulmonary fibrosis	Any	Interruption of treatment and investigation of clear reason
New interstitial lung disease	Any	Permanently discontinue treatment
Nausea and vomiting after prophylactic use of antiemetics	3	Q2W: 65 mg/m^2; Q3W: 100 mg/m^2
Nausea and/or vomiting	4	Q2W: 65 mg/m^2; Q3W: 100 mg/m^2
Diarrhoea	3 or 4	Q2W: 65 mg/m^2; Q3W: 100 mg/m^2
Stomatitis	3 or 4	Q2W: 65 mg/m^2; Q3W: 100 mg/m^2
Skin toxicity	3 or 4	Prolong the rest of chemotherapy until recovery to Grade 0-1, without dose modification

Table 3-8 Dose Modification of Oxaliplatin for Neurotoxicities

Toxicities	Grade	Duration		
		1-7 days	> 7 days	Continued between cycles[a]
Paresthesia/Dysesthesia[b] No functional impact	1	Dose unchanged	Dose unchanged	Dose unchanged
Paresthesia/Dysesthesia affects function but does not affect self-care activities of daily living（ADL）	2	Dose unchanged	Dose unchanged	Q2W: 65 mg/m^2; Q3W: 100 mg/m^2
Paresthesia/dysesthesia with pain, affecting function and affecting ADL	3	Dose unchanged	Q2W: 65 mg/m^2; Q3W: 100 mg/m^2	Permanently discontinue treatment
Paresthesia/dysesthesia disabling or life threatening	4	Permanently discontinue treatment	Permanently discontinue treatment	Permanently discontinue treatment
Acute (during infusion or within 2 hours after the end of infusion) laryngeal dysesthesia[b]		Prolongation infusion timeto 6 hours for this or next infusion [c]	NA	NA

a. Not recovered until start of next cycle

b. Probably caused by cold

c. Can be pretreated with benzodiazepine

3.1.6. Study Duration

For each cohort, the study will last until the death of all subjects of the cohort or at least 12 months after the last subject is enrolled, whichever occurs first.

3.1.7. Duration of Subject Participation

The number of treatment cycles in this study will not be fixed. Subjects who achieve CR, PR, or SD will be treated with AK104 for up to 24 months without withdrawal of informed consent, confirmed progression, and unacceptable toxicity.

All subjects in the Cohorts 2, 3, 4, 5 from Phase Ib, and Cohort D from Phase II will receive 10-12 cycles/6 cycles of mXELOX/XELOX (except for disease progression, unacceptable toxicity, or withdrawal of consent) in combination with AK104 according to the cohort and treatment regimen, every 2 weeks as a cycle for Cohorts 2, 3 and 4, every 3 weeks as a cycle for Cohort 5In Cohort D, every 2 weeks as a cycle for the AK104 6mg/kg combined with mXELOX dosing regimen, and every 3 weeks as a cycle for the AK104 10mg/kg combined with XELOX dosing regimen. After Cycles 10-12 (in combination with mXELOX) or Cycle 6 (in combination with XELOX), maintenance treatment of capecitabine in combination with AK104 may be optionally continued until disease progression (PD) or unacceptable toxicity or withdrawal for other reasons, whichever occurs first.

Subjects who are still receiving study treatment at the end of the study or beyond the longest treatment duration of 24 months may continue treatment through additional extension studies or other forms agreed upon by the sponsor if stable or relieved disease response is achieved in response evaluation and the adverse reactions are tolerable. Refer to Section 3.1.2 for treatment regimens.

3.1.8. Criteria for Study Termination

During the operational phase of the study, the sponsor may terminate the study at any time after all subjects complete the trial or based on the DEC`s opinion and the actual clinical development.

3.2. Justification for Study Design

3.2.1. Justification for Medication Regimen Design and Determination of Escalating Initial Dose

The escalating initial dose of AK104 as a single agent of 6 mg/kg Q2W in the escalation phase of this study (Cohort 1) is determined to be based on safety data from a Phase Ia dose escalation trial conducted in Australia. In a completed Phase Ia tolerability clinical study of AK104 in patients with advanced solid tumors, the escalating initial dose of AK104 alone was 0.2 mg/kg. Thereafter, dose escalation was performed at 0.5 mg/kg, 1 mg/kg, 2 mg/kg, 4 mg/kg, 6 mg/kg Q2W, and the safety of the 10 mg/kg Q2W dose was planned to continue after protocol amendment. Six subjects have been currently enrolled in the 6 mg/kg Q2W dose group and no drug-related dose-limiting toxicities have been observed. Only 1 case of drug-related Grade 2 rash and 1 case of drug-related Grade 3 hypokalemia have been

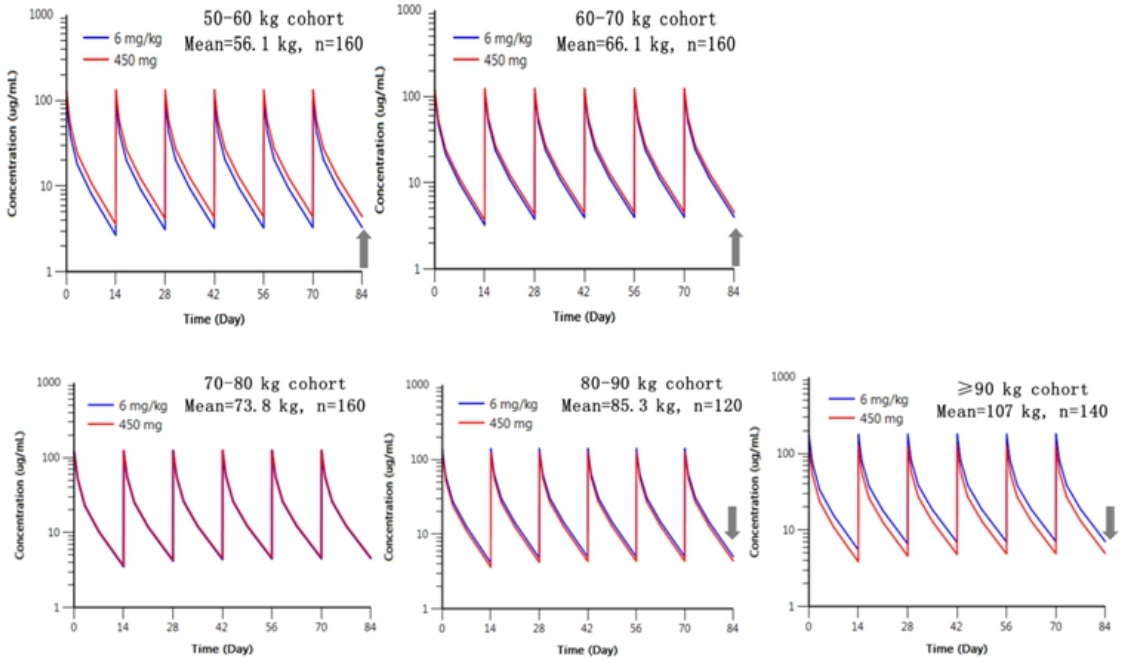
observed (complete recovery within 3 days), so AK104 is safe and tolerated as a single agent (data cutoff October 9, 2018).

The safety profile of anti-PD-1/PD-L1 antibody in combination with chemotherapy is consistent with that expected for individual agents, based on the results of several clinical studies of anti-PD-1/PD-L1 monoclonal antibodies in combination with standard chemotherapy currently published. No known overlapped and significant toxicity, or drug-drug interactions were observed between anti-PD-1/PD-L1 antibodies and chemotherapeutic agents in these trials. The mechanism of standard chemotherapy differs from that of PD-1/PD-L1 inhibitor or AK104 (a bispecific antibody targeting both anti-PD-1 and CTLA-4), so does in the management of adverse events.

In combination with the mXELOX chemotherapy regimen, the initial dose of AK104 (Cohort 2) was 4 mg/kg Q2W (lower than the currently conducted Phase Ia dose escalation trial in Australia, MAD 6 mg/kg Q2W). In the group of 4 mg/kg as initial dose, the first subject was treated first. If the first subject had no treatment-related safety concerns after treatment, the next second or third subject would be treated at least 24 hours later. According to the 3 + 3 dose escalation principle, the dose was escalated following AK104 alone 6 mg/kg Q2W, and AK104 in combination with mXELOX chemotherapy regimen in which AK104 doses will be 4 mg/kg Q2W, 6 mg/kg Q2W, and 10 mg/kg Q2W.

In the AK104-101 study, the increase in exposure was proportional to the dose increase at doses of 0.2-10 mg/kg, consistent with a linear pharmacokinetic profile. No dose-limiting toxicities occurred in the 6 subjects treated with 10 mg/kg Q2W. Dose escalation studies showed good safety with AK104 at 0.2-10 mg/kg, which provides a good margin of safety for AK104. On the other hand, population pharmacokinetic simulation results (Figure 3-2) showed that the exposure to a fixed dose of 450 mg was similar to that of 6 mg/kg (both dosing intervals were every 2 weeks). Therefore, this study is planned to explore the efficacy, safety, pharmacokinetics and other characteristics of AK104 450 mg fixed dose administered every 2 weeks in subjects with advanced solid tumors.

Figure 3-2 Similar drug exposure at 6 mg/kg and 450 mg

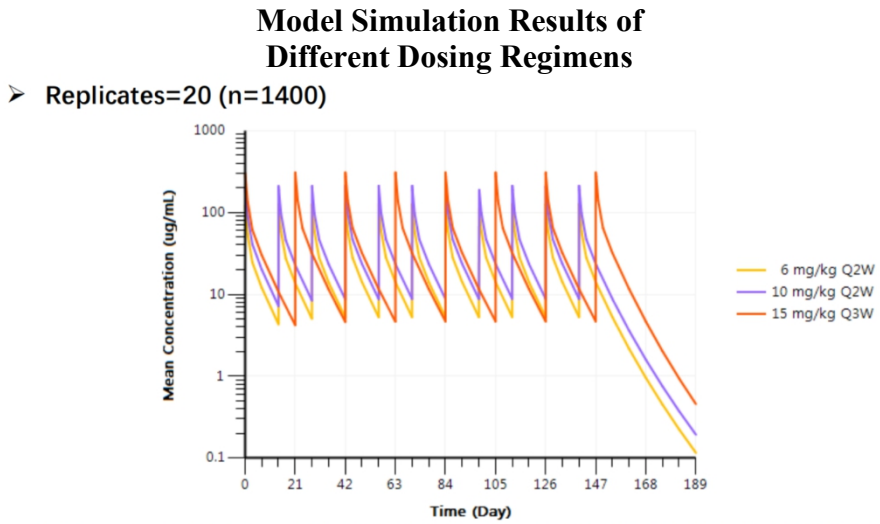


Rationale for AK10415 mg/kg Q3W (updated on December 15, 2020)

The escalating initial dose of AK104 alone was 0.2 mg/kg Q2W, which was sequentially escalated to 15 mg/kg Q3W and 25 mg/kg Q3W after multiple dose levels, and the MTD was not reached. As of November 20, 2020, 12 subjects had been treated with 15 mg/kg Q3W, and the incidence of TRAEs was 75.0%. Two subjects experienced ≥ Grade 3 treatment-related adverse reactions, 1 had infusion-related reaction, and 1 subject had diarrhea and immune-mediated enteritis. Five subjects received 25 mg/kg Q3W, and the incidence of TRAEs was 40.0%. One subject had ≥ Grade 3 treatment-related adverse reaction, i.e. immune-mediated hepatitis. In 15 mg/kg Q3W evaluable subjects, the ORR was 45.6% (5/11), and 1 subject with MSI-H colorectal cancer was CR.

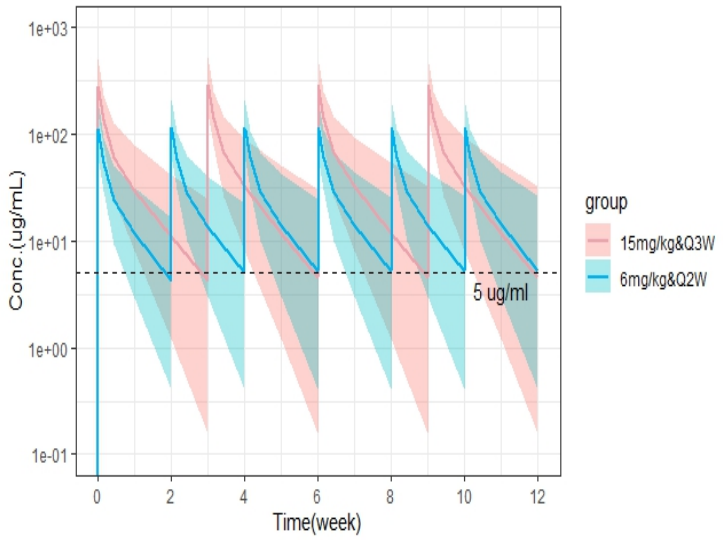
Based on the model simulation results of different AK104 dosing regimens, the trough serum concentrations (Ctrough) were comparable between the 6 mg/kg Q2W and 15 mg/kg Q3W dosing regimens. The predicted concentration values and measured values from subjects receiving 6 mg/kg Q2W were compared.,most of the measured values fell within the P5 and P95 quantiles of the predicted values and were generally evenly distributed on both sides of the median line, indicating good predictive performance of the model (Figure 3-3). The AUC of 15 mg/kg over 6 weeks was approximately 1.6-fold of that of 6 mg/kg, and there was overlapping exposure between the two dosing modalities over a 6-week interval, supporting the 15 mg/kg Q3W dosing regimen where safety permits (Figure 3-4).

Figure 3-3 Model simulation results of different dosing regimens



Result: Ctrough was comparable between the two dosing regimens, 6 mg/kg Q2W and 15 mg/kg Q3W (predicted from the model on February 17, 2020).

Figure 3-4 Comparison of AUC of AK10415 mg/kg Q3W and 6 mg/kg Q3W over 6 weeks



Cohort D Rationale for selection of AK104 10 mg/kg Q3Wcombined with XELOX

As of May 2021, the study of AK104 incombination with mXELOX/XELOX in the first-line treatment of advanced gastric cancer has enrolled 87 untreated patients with advanced gastric cancer, 73 of which are evaluable, with an overall ORR of 65.8% and a DCR of 93.2%, of which 2 patients reached CR. 18 subjects received AK104 10 mg/kg combined chemotherapy once every two weeks, the incidence of grade 3 TRAE (TRAE is defined as an adverse event related to AK104 or chemotherapy) was 44.4%, and the incidence of grade ⩾3 irAE was 20.0%; ORR had reached 83.3% among the 18 subjects who had undergone tumor evaluation at least once, and the median PFS had not been reached. The results indicated that tumor response may improve as the dose of AK104 increases, accordingly. The safety profile did not show significant differences in 4 mg/kg Q2W, 6 mg/kg Q2W, 10 mg/kg Q2W of

AK104 combined with mXELOX and 15 mg/kg Q3W of AK104 combined with XELOX, in the first-line treatment of gastric cancer. Different doses of AK104 combined with chemotherapy did not increase the TRAE related to AK104. While the ORR was significantly improved (83.3% vs. 68.8%), compared with AK104 4 mg/kg Q2W and AK104 10 mg/kg Q2W combined with chemotherapy.

Based on the model simulation results of different AK104 dosing regimens, the steady-state exposure were consisitent between the 6 mg/kg Q2W and 10 mg/kg Q3W dosing regimens, and the C_{min} showed no significant difference. The 5%-95% quantile concentration range of the two are also consistent. Therefore, chose AK104 10 mg/kg Q3W combined with XELOX as the first-line treatment dosing regimen for advanced gastric cancer or gastroesophageal junction adenocarcinoma in Cohort D is reasonable.

3.2.2. Justification for Selection of ORR as Primary Endpoint

A recently published KEYNOTE-059 study is a series of Phase II studies of Pembrolizumab monoclonal antibody in the treatment of advanced gastric cancer, including 3 cohorts: Cohort 1 is Pembrolizumab monotherapy for patients with advanced gastric cancer who failed second-line or higher chemotherapy; Cohort 2 is the first-line treatment of Pembrolizumab in combination with FP/XP for advanced gastric cancer; Cohort 3 is the first-line treatment of Pembrolizumab monotherapy in PD-Ll-positive advanced gastric cancer. Cohort 1 included 259 patients with ORRs of 16% and 6% for PD-L1-positive and negative patients, respectively, with a median response time of up to 14.2 months once patients responded. Based on the results of Cohort 1, FDA approved Pembrolizumab as a single agent for the treatment of patients with recurrent locally advanced or metastatic gastric adenocarcinoma who are PD-L1 positive and who have received at least second-line chemotherapy. Cohort 2 included 25 patients with advanced gastric cancer treated as first-line therapy. The overall ORR is 60%, respectively 69% and 38% for PD-L1 positive and negative patients, with median PFS of 6.6 months and median OS of 13.8 months. Cohort 3 included 31 patients with PD-L1 positive advanced gastric cancer treated as first-line therapy, with ORR 26%, median PFS 3.3 months, and median OS 20.7 months.

ORR is selected as the primary endpoint and duration of response as the key secondary endpoint for this study. A key feature of immunotherapy is that once a patient has relieved, the response is sustained and durable, and it can be expected to reflect clinical benefit.

3.2.3. Justification for Treatment after Disease Progression

Responses to immunotherapy may differ from typical responses observed with cytotoxic chemotherapy (**Wolchok JD, 2009**), including clinical objective responses and/or stable disease that may occur after disease progression, and new lesions that do not represent disease progression with immunotherapy, from which clinical benefit is still possible. It is now thought that this phenomenon may be due to the following two reasons. First, worsened inflammation within the tumor may lead to an increase in tumor volume, as evidenced by enlargement of measurable diseases and the appearance of new, visible, and non-measurable diseases. Over time, the malignant and inflammatory parts of the mass may shrink, resulting

in significant imaging response and improvement in clinical signs. Second, in some patients, the antitumor immune response is initiated slowly, and its early tumor suppression is less effective than tumor growth kinetics. Over time, antitumor activity will predominate and manifest as an improvement in imaging response and clinical signs. In a Phase II study, a subgroup analysis of patients who continued to receive nivolumab beyond disease progression was performed (**Escudier B, 2017**). Among 142 advanced RCC patients who continued treatment with nivolumab after disease progression and with pre- and post-progression tumor measurements, 13% of them had a subsequent reduction in tumor burden of ≥ 30%.

Therefore, subjects will be allowed to continue AK104 monotherapy if they are assessed as clinically beneficial and tolerant to study drug after the investigator has initially assessed disease progression as defined by RECIST v1.1. Subjects must discontinue AK104 treatment after evidence of further progression appears.

3.2.4. Justification for Collection of Archived or Fresh Tumor Samples and Biomarker Blood Samples

Screening suitable patients for treatment through biomarkers is a way to improve the efficacy. The development of predictive diagnostic assays could help screen patients who are more likely to benefit from this treatment. PD-L1 expression in tumor tissues is a potential biomarker for predicting response to anti-PD-1 therapy. Therefore, in some clinical studies on anti-PD-1 /PD-LI therapy for gastric adenocarcinoma, PD-L1 positive tumor tissue is been taken as one of the inclusion criteria, or only the outcomes of PD-L1 positive tumor patients are taken as the study endpoint. FDA has approved Pembrolizumab as a single agent for the treatment of patients with recurrent locally advanced or metastatic gastric adenocarcinoma who are PD-L1 positive and have received at least second-line chemotherapy.

Compared with PD-Ll expression status, MSI status is more enriched in the effective population. FDA has approved Pembrolizumab for the treatment of malignant solid tumors with MSI-H or dMMR. The positive rates of MSI-H and dMMR in advanced gastric cancer are very low, which is less than 10%. For the large non-MSI-H and pMMR populations, better predictors in addition to PD-L1 status are required. Because of the complexity of the immune system, finding an independent indicator that perfectly predicts the efficacy of anti-PD-1 or anti-CTLA-4 seems difficult. On the one hand, efforts will be made to find specific populations such as MSIH and dMMR, for example, patients with high tumor mutation burden and EBV-positive gastric cancer may be effective subtypes. On the other hand, different efficacy predictors will be combined, such as the establishment of immunogenetic signature to achieve better predictive effect. Researchers in The Cancer Genome Atlas (TCGA) have classified gastric cancer into four subtypes: EBV-positive, MSI, genetically stable and chromosomally unstable.

Other potential pharmacodynamic variables will be further assessed based on recent progression, including, but not limited to the detection of changes in CD4 + Ki67 + and CD8 + Ki67 + proliferating cells in blood from baseline over time, and detection of PD-1 or CTLA-4 receptor occupancy in blood before and after treatment. Further evidence can be

provided to study the biological activity of AK104 in humans.

4. Study Population

4.1. Enrollment

Relevant information about this study will be provided to each subject and all questions raised will be answered until the subject is satisfied and the ICF will be signed and dated. These will be completed before any study-specific procedures are performed (except in Table 6-1). Further information on the informed consent procedure is provided in Section 13.4. Once the investigator or designee obtains written informed consent from a subject and determines that all inclusion and exclusion criteria are met, the subject will be considered enrolled in the study. For subjects receiving study treatment, data obtained at all study visits will be recorded on the case report form. For subjects who do not meet the inclusion/exclusion criteria and/or do not receive study drug, only a small number of data (demographics, reason for withdrawal, and AEs) will be recorded on the case report form. Once a subject fails screening or has decided to withdraw from the study prior to receiving study drug, no additional data will be collected for that subject.

4.1.1. Inclusion Criteria

Subjects must meet all of the following inclusion criteria to be enrolled in the study:

1) Written informed consent must be voluntarily signed.

   - Subjects must sign and date an IRB/IEC-approved informed consent form in accordance with guidelines from the competent authority and the test facility. Informed Consent Form must be signed prior to the conduct of any protocol-related procedures (not part of the subject's routine medical care).
   - Subjects must be willing and able to comply with the scheduled visits, treatment regimens, laboratory tests, and other requirements in the study.

2) Males or females aged ≥ 18 to ≤ 75 years at the time of signing informed consent.

3) Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1.

4) Expected survival ≥ 3 months.

5) For different cohorts, subjects must meet the following requirements accordingly:

   5-1) Cohort 1: solid tumors

   a) Histologically or cytologically confirmed unresectable advanced solid tumors.

   b) Failed to standard of care for advanced disease, or no available standard of care, or intolerance to standard of care, or refused to accept standard of care.

   5-2) Cohort A: cervical cancer

   a) Histologically or cytologically confirmed recurrent or metastatic cervical cancer, by non-radical surgical resection and/or radical chemoradiotherapy; with the pathological type of squamous cell carcinoma or adenosquamous cell

carcinoma.

b) Must have received platinum-containing doublet chemotherapy with or without bevacizumab at the recurrent or metastatic stage, with radiologically confirmed disease progression during or after treatment.

c) No more than 2 lines of therapies for recurrent or metastatic disease and disease progression confirmed by imaging during or after the most recent treatment. Chemotherapy in neoadjuvant chemotherapy, adjuvant chemotherapy, or chemoradiotherapy in the non-recurrent or metastatic stage will not be counted as a single line of therapy.

5-3) Cohort B: esophageal squamous cell carcinoma

a) Histologically or cytologically confirmed unresectable locally advanced or metastatic esophageal squamous cell carcinoma.

b) Subjects who failed to standard of care, or who were unable to tolerate standard of care or refused to accept standard of care, and have received a maximum of one line of therapy; if disease progression occurs during neoadjuvant/adjuvant chemotherapy or radical chemoradiotherapy or within 6 months after the completion of treatment, subject is considered to have received first-line treatment.

5-4) Cohort C: hepatocellular carcinoma

a) Histologically or cytologically confirmed unresectable advanced hepatocellular carcinoma.

b) With the Barcelona Clinic Liver Cancer (BCLC) stage C, with BCLC Stage B and not suitable for surgery or local treatment, or with progresses after surgery and/or local treatment and not suitable for radical therapy.

c) Child-Pugh liver function classification: class A and better gradeB (≤7 points).

d) Failed to standard of care, or intolerance to standard of care, or refuse to acceptstandard of care for advanced disease, and received a maximum of one line of therapy.

5-5) Cohorts 2, 3, 4, 5 and Cohort D: gastric adenocarcinoma or gastroesophageal junction adenocarcinoma

a) Histologically or cytologically confirmed unresectable advanced or metastatic gastric adenocarcinoma or gastroesophageal junction (GEJ) adenocarcinoma. Note: GEJ adenocarcinoma is defined as a tumor centered within 5 cm proximal and distal to the anatomical location of the cardia as described in the Siewert Classification System.

b) Subjects have not previously received systemic chemotherapy for advanced or

metastatic gastric adenocarcinoma or GEJ adenocarcinoma.

c) The subjects who have previously received adjuvant chemotherapy, neoadjuvant chemotherapy, or radical chemoradiotherapy for progression disease will be eligible, provided that the disease progression occurs > 6 months after the completion of the last treatment.

6) At least one measurable tumor lesion as per RECIST v1.1 criteria; subjects in Cohort A should have measurable disease confirmed by IRRC. A a measurable lesion that has been previously treated with local therapies such as radiotherapy can be considered a target lesion provided that it has radiographically confirmed progression.

7) All subjects must provide archived or freshly obtained tumor tissue samples after a definite diagnosis, and approximately 10 (at least 5 if approved by the medical monitor) unstained FFPE pathology slides (preferably freshly obtained tumor tissue samples). Fine-needle aspiration biopsy or burr biopsy samples require sufficient tumor tissue for biomarker testing. Cell smears centrifuged from pleural/abdominal/pelvic/pericardial effusion drainage, bone lesions without soft tissue components or decalcified bone tumor samples are not acceptable. Tumor lesions for fresh biopsy should not be used as RECIST v1.1 target lesions unless no other lesions are suitable for biopsy and the longest length of lesion used for biopsy is ≥ 2 cm. For the subjects in Cohorts 2, 3, 4, 5 from Phase Ib, and in Cohort D from Phase II, archived or freshly obtained tumor tissue samples within 6 months before the first dose of AK104 at the advanced or metastatic stage must be provided; the subjects in Cohort A from Phase II must provide tumor tissue samples for the site or central laboratory before the first dose. The freshly obtained tumor tissue samples are provided whenever possible (preferentially at the recurrent or metastatic stage); for the subjects in the Cohort 1 from Phase Ib and Cohorts B, C from Phase II, if archived tumor tissue samples are not available and the subjects have great risk for a fresh biopsy as determined by the investigator, they may be exempted from collection of archived tumor tissue samples after discuss with the medical monitor.

8) Adequate organ function, which will be determined by the following requirements:

   a) Hematology (subjects will not be allowed to receive blood transfusion or growth factor support within 7 days of initiating study treatment):

      i. Absolute neutrophil count (ANC) ≥ 1.5×10^9/L (1,500/mm^3);

      ii. Platelet count ≥ 100×10^9/L (100,000/mm^3); for Cohort C, platelet count ≥ 75×10^9/L (100,000/mm^3);

      iii. Hemoglobin ≥ 9.0 g/dL.

   b) Kidney:

      i. Serum creatinine ≤ 1.5 × ULN or creatinine clearance* (CrCl) ≥ 50

mL/min

* CrCl will be calculated using the Cockcroft-Gault formula (**Cockcroft DW, 1976**)

CrCL (mL/min) = {(140 - age) × body weight (kg) × F}/(SCr (mg/dL) × 72)
where F = 1 for men and 0.85 for women; SCr = serum creatinine.

c) Liver:

i. TBil ≤ 1.5 × ULN; TBil ≤ 3 × ULN for Cohort C and subjects with confirmed/suspected Gilbert' s disease;

ii. AST and ALT ≤ 2.5 × ULN; AST and ALT ≤ 5 × ULN for Cohort C and subjects with liver metastases;

iii. Serum albumin ≥ 28 g/L.

d) Coagulation:

i. International normalized ratio (INR) and activated partial thromboplastin time (APTT) ≤1.5 × ULN (unless participant is receiving anticoagulant therapy, and coagulation parameters (PT/INR and APTT) are within the expected range of anticoagulant treatment at screening).

9) If a female subject of childbearing potential has sexual intercourse with an unsterile male partner, the subject must use an acceptable method of contraception from the start of screening and must agree to continue this method of contraception for 120 days after the last dose of study drug; it should be discussed with the attending physician whether contraception will be discontinued after this time point. Periodic abstinence, contraceptive rhythm method, and withdrawal are not acceptable methods of contraception.

a) Females of childbearing potential are defined as those who are not surgically sterile (i.e., bilateral tubal ligation, bilateral oophorectomy, or complete hysterectomy) or postmenopausal (defined as 12 months with no menses without an alternative medical cause).

b) Highly effective methods of contraception are defined as those that result in a low failure rate (i.e. less than 1% per year) when used consistently and correctly. Acceptable methods of contraception are shown in Table 4-1. Not all methods of contraception are highly effective. In addition to barrier contraception (e.g., male condom plus spermicide), female subjects must also use hormonal contraception (e.g., contraceptives) alone to ensure that pregnancy does not occur.

c) If a woman is pregnant or is suspected to be pregnant and she or her partner is participating in the study, her attending physician must be notified immediately.

10) If a male subject who is not sterile has sexual intercourse with a female partner of childbearing potential, the subject must use an effective method of contraception (see Table 4-1) from Day 1 to Day 120 after the last dose of study drug. Female partners of male subjects are strongly recommended to use effective methods of contraception throughout the study.

Table 4-1 Effective Methods of Contraception

Barrier contraception	Hormonal contraception
• Male condom + spermicide • Copper T-type intrauterine device • T-type intrauterine device containing progesterone • Levonorgestrel-releasing intrauterine device (e.g., mirena)	• Implant • Hormonal contraceptive injection • Compound contraceptives • Low-dose oral contraceptive pills • Contraceptive patch

4.1.2. Exclusion Criteria

Subjects who meet any of the following criteria are not eligible to participate in this study:

1) For different cohorts, subjects are not eligible to participate in the study if they meet the following corresponding criteria:

1-1) Cohort A: cervical cancer

a) Subjects with other histopathological types (e.g. adenocarcinoma, small cell carcinoma, clear cell carcinoma, and sarcoma, etc.) other than squamous cell carcinoma and adenosquamous cell carcinoma.

b) Clinically significant hydronephrosis that cannot be relieved by nephrostomy or urethral stenting at the discretion of the investigator.

1-2) Cohort B: esophageal squamous cell carcinoma

a) Subjects with other histological types of esophageal cancer other than squamous cell carcinoma.

b) Subjects with high risk of bleeding or perforation due to obvious invasion of esophageal tumors into adjacent organs (e.g., stomach, large arteries, and trachea, etc.), or who have developed bleeding, perforation or fistula.

c) Subjects who have just experienced esophageal or tracheal stent placement.

1-3) Cohort C: hepatocellular carcinoma

a) Subjects with other histological types known as fibrolamellar HCC, sarcoid HCC, intrahepatic cholangiocarcinoma or mixed hepatocellular carcinoma-cholangiocarcinoma;

b) Subjects who have tumor thrombus that has invaded the main trunk and branches of great vessels such as portal vein, superior mesenteric vein or inferior vena cava, and caused complications such as portal hypertension and jaundice,

or who are judged by the investigator to be ineligible for this study due to related clinical risks;

c) Subjects who have received local-regional treatment of the liver (i.e., transarterial chemoembolization, transcatheter embolization, hepatic arterial infusion, radiotherapy, radioembolization, or ablation) within 4 weeks prior to the first dose of study drug.

1-4) Cohorts 2, 3, 4, 5; Cohort D: gastric adenocarcinoma or GEJ adenocarcinoma

a) Subjects of gastric or GEJ adenocarcinomas withHER2 positive.

b) Subjects with squamous cell or undifferentiated or other histological types of gastric or GEJ cancer.

2) Subjects with active malignancy within 2 years prior to the first dose, in addition to the tumor present at enrollment. Subjects with locally curable tumor (manifested as cured), such as cutaneous basal cell carcinoma, cutaneous squamous cell carcinoma, superficial bladder cancer, and carcinoma in situ of the breast, will not be excluded.

3) Subjects who have previously participated in the study of an investigational drug, received study treatment or used investigational device within 4 weeks prior to the first dose.

4) Subjects who have enrolled in another clinical study also, unless it is an observational, non-interventional clinical study or follow-up period for an interventional study (the period from the first dose of AK104 to the last dose in the previous clinical study should be more than 4 weeks or 5 half-lives of the previous study drug).

5) Subjects who have received systemic treatment of Chinese patent drugs with antitumor indications, Chinese herbal medicine with antitumor effects, or immunomodulatory agents (thymosin, interferon, and interleukin) within 2 weeks prior to the first dose.

6) Subjects who have received the last systemic or curative anti-tumor therapy within 4 weeks prior to the first dose, including radiotherapy, chemotherapy, targeted therapy (within 2 weeks prior to the first dose for small molecule targeted therapy), and immunotherapy. Palliative radiotherapy for the control of symptoms is allowed to be completed at least 2 weeks prior to the first dose and no additional radiotherapy is planned for the same lesion.

7) Subjects who have previously received any anti-PD-1, anti-PD-L1, anti-CTLA-4 antibody, or any other antibody or drug therapy targeting T-cell costimulation or checkpoint pathway, e.g. ICOS or agonists (e.g., CD40, CD137, GITR, and OX40, etc.).

8) Subjects with active, known or suspected autoimmune disease, or with a history of

autoimmune disease, with the following exceptions: vitiligo, alopecia, Graves' disease, psoriasis, or eczema that does not require systemic treatment within recent 2 years, hypothyroidism that is asymptomatic or only requires a stable dose of hormone replacement therapy (due to autoimmune thyroiditis), type 1 diabetes that only requires stable dose of insulin replacement therapy, asthma that has resolved completely in childhood and requires no intervention in adulthood, or the diseases that will not recur in the absence of external triggers (see Appendix 3 for a list of autoimmune diseases).

9) Subjects with active or previously documented inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis, or chronic diarrhea). Subjects who can't swallow or have malabsorption syndrome, uncontrolled nausea, vomiting, diarrhea, or other gastrointestinal disorders that severely affect drug intake and absorption.

10) Subjects who require systemic treatment with corticosteroids (> 10 mg/day of prednisone or equivalent) or other immunosuppressive agents within 14 days prior to the first dose. with the following exceptions:

- Inhaled, ophthalmic, or topical steroids and corticosteroids at doses not exceeding the therapeutic dose of 10 mg/day prednisone are allowed in the absence of active autoimmune disease.
- Physiologic doses of systemic glucocorticoids that do not exceed 10 mg/day prednisone, or equivalent doses of other glucocorticoids.
- Glucocorticoids that are used as prophylaxis for hypersensitivity reactions (e.g., before CT).

11) Known history of immunodeficiency.

12) Known history of allogeneic organ transplantation and allogeneic hematopoietic stem cell transplantation.

13) Subjects who have experienced major surgery (e.g., laparotomy, thoracotomy, organ resection, etc.), and severe trauma, etc. (replacement of intravenous infusion line is acceptable) within 28 days prior to the first dose; subjects who have underwent surgery to improve or reduce the risk of tumor complications (e.g., nephrostomy or urethral stenting in subjects with cervical cancer, etc.) within 14 days prior to the first dose; or subjects who have not fully recovered from the previous surgery described above. Subjects who plan to undergo a major surgery within 30 days after the first dose (at the discretion of the investigator) . Local surgery (e.g., placement of systemic ports, core needle biopsy, and prostate biopsy) is allowed provided that the procedure is completed at least 24 h before the first dose of study drug.

14) Subjects with history of gastrointestinal perforation, gastrointestinal fistula, and female genital fistula (e.g., vesicovaginal fistula, urethrovaginal fistula, and

vesicocervical fistula, etc.) within 6 months prior to the first dose; if the perforation or fistula has been treated with resection or repair and the disease has recovered or resolved at the discretion of the investigator, the subjects can be enrolled.

15) Known interstitial lung disease that is symptomatic or may interfere with the detection or management of suspected drug-related pulmonary toxicity.

16) Known active pulmonary tuberculosis (TB). Subjects with suspected active TB should undergo the chest x-ray and sputum testing to exclude this condition combined with clinical symptoms and signs.

17) Serious infection within 4 weeks prior to first dose, including but not limited to complications requiring hospitalization, sepsis, or severe pneumonia.

18) Active infection requiring systemic treatment.

19) Patients with untreated chronic hepatitis B, or HBV carriers with chronic hepatitis B virus (HBV) DNA greater than 1000 IU/mL, or patients with active hepatitis C should be excluded. Inactive HBsAg carriers, patients with hepatitis B who have been treated and in stable condition (HBV DNA < 1000 IU/mL), and patients with cured hepatitis C can be enrolled. Subjects who are HCV antibody positive are eligible to participate in the study only if they have a negative HCV RNA test.

20) Subjects with HIV-positive.

21) Metastases to the central nervous system, leptomeningeal metastases, spinal cord compression, or leptomeningeal disease confirmed by imaging or pathological examination.

22) Subjects with uncontrolled pleural effusion, pericardial effusion, or ascites requiring repeated drainage.

23) Presence of clinically active hemoptysis, active diverticulitis, abdominal abscess, and gastrointestinal obstruction.

24) History of clinically significant hemorrhage symptom or clear hemorrhagic diathesis within 1 month prior to the first dose, such as digestive tract hemorrhage, gastirc ulcer hemorrhage or vasculitis.

25) Presence of any of the following cardiovascular and cerebrovascular diseases or cardiovascular and cerebrovascular risk factors:

- Myocardial infarction, unstable angina, pulmonary embolism, acute/persistent myocardial ischemia, cerebrovascular accident, transient ischemic attack, or other arteriovenous thrombotic, embolic or ischemic events that are clinically significant/ requiring pharmacological intervention within 6 months prior to the first dose.
- Symptomatic congestive cardiac failure (New York Heart Association functional

class III or IV) within 6 months prior to the first dose.

- Unstable arrhythmias or atrioventricular block II/III degree requiring drug intervention occurred within 6 months prior to the first dose.

26) Unresolved toxicities during prior antitumor therapy are defined as the toxicities that do not resolved to National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) (NCI CTCAE v5.0) Grade 0 or 1, or to the levels specified in the inclusion/exclusion criteria, with the exception of alopecia. Subjects with irreversible toxicity that are not expected to worsen after study drug administration (e.g., hearing loss) may be included in the study after consultation with the medical monitor. Subjects with long-term toxicity due to radiotherapy that cannot be recovered at the discretion of the investigator may be included in the study after consultation with the medical monitor.

27) ≥ Grade 2 peripheral neuropathy as defined by NCI CTCAE v5.0.

28) Live or attenuated vaccines inoculated within 30 days prior to the first dose, or live or attenuated vaccines planned to be inoculated during the study.

29) Known history of severe hypersensitivity reactions to other monoclonal antibodies.

30) Subjects with known contraindications to oxaliplatin and capecitabine chemotherapy (see instructions for oxaliplatin and capecitabine (not applicable to Cohort 1 from Phase Ib and Cohorts A, B, and C)).

31) Subjects with known history of allergy or hypersensitivity to AK104, oxaliplatin or other platinum compounds (not applicable to Cohort 1 from Phase Ib, and Cohorts A, B, and C), and capecitabine or any of its components (not applicable to Cohort 1 from Phase Ib, and Cohorts A, B, and C).

32) Subjects with known dihydropyrimidine dehydrogenase (DPD) deficiency (not applicable to Cohort 1 from Phase Ib, and Cohorts A, B, and C).

33) Pregnant or lactating women.

34) Any condition that, in the opinion of the investigator, may result in a risk when receiving study drug, or would interfere with the study drug evaluation of the interpretation of subject safety or study results.

4.2. Subject Withdrawal from Treatment

All subjects may discontinue study treatment or withdraw from the study at any time. If a subject withdraws from study treatment or the study, the reason for withdrawal will be recorded in the eCRF. The investigator will complete the observation period and report the results in as much detail as possible until withdrawal from the study, including the date of the last treatment and the reason for withdrawal.

4.2.1. Reason for Withdrawal

4.2.1.1. Withdrawal from Study Treatment

Reasons for Withdrawal/Discontinuation of Study Treatment:

- AEs (any AE that meet the criteria for discontinuation of study drug as defined in Section 3.1.4, DLTs as defined in Section 3.1.3, or contraindications that investigator or sponsor believes to be inadvisable for further dosing)
- Lost to follow-up
- Death
- Major protocol deviations (e.g., non-compliance of subjects believed by the investigator or sponsor, pregnant or plan to be pregnant, requirements for study population not met, etc.)
- Subject's withdrawal of informed consent
- Start a new anti-tumor therapy
- Study discontinued by the sponsor
- Disease progression (imaging progression); disease progression should be confirmed without significant clinical deterioration. (See Section 3.1.2.3)
- Others (e.g., at the discretion of the investigator, the presence of clinical progression)

For subjects who discontinue study treatment, please refer to Section 6.3 for end of treatment and follow-up procedures. Subjects who discontinue study treatment due to an AE will be followed up by the investigator as far as possibly until the AE recovers or relieves.

4.2.2. Withdrawal from Study

Withdrawal from the study includes discontinuation of all follow-up procedures. Reasons for withdrawal from the study include:

- Death
- Lost to follow-up
- Subject's withdrawal of informed consent of study participation (including follow-up period)
- Study discontinued by the sponsor

4.2.2.1. Withdrawal Procedure

Study treatment follow-up will end 30 days after the last dose and 90 days after the last dose of AK104. After the last dose, subjects will undergo safety follow-up for tracking the relief of adverse event, followed by survival follow-up. Survival assessments will be performed every

3 months after the discontinuation of treatment drug. The procedures described in Section 6.3 will be performed at this visit.

4.2.3. Replacement of Subjects

During the dose escalation period (Part 1, Phase Ib), non-evaluable subjects in the DLT assessment will be replaced as appropriate. For the definition of DLT evaluable subjects, see Section 11.1.2. No subjects will be replaced in the dose expansion cohort (Part 2, Phase Ib) and the dose confirmation cohort (Phase II).

4.2.4. Re-screening Procedure for Subjects

Re-screening (only re-screening once) will be allowed for subjects who failed to meet eligibility criteria at initial screening. At the time of re-screening, subjects must re-sign the informed consent form and obtain a new subject identification number (SID).

5. Study Drugs and Other Treatments

5.1. Investigational Drug

Akeso Biopharma Co., Ltd. will provide sufficient quantities of study drugs to the investigators or provide through designated distribution centers: the investigational product AK104 Injection and other study drugs (oxaliplatin and capecitabine).

AK104 will be provided as a sterile solution. AK104 is an injection (solvent type), and the strength is 125 mg/10 mL/vial. Each glass vial contains 10 mL of investigational drug at a concentration of 12.5 mg/mL. Each vial is intended for single use only and should not be used to treat more than one subject.

AK104 should be used before the expiry date indicated on the package. AK104 should be stored at 2-8 °C and protected from light. Freezing and vigorous shaking are prohibited. AK104 does not contain any antimicrobial preservatives, therefore, care must be taken to ensure the sterility of the prepared solutions.

For standard chemotherapeutic drugs, marketed products will be purchased uniformly by the sponsor, and their management (i.e., handling, storage, administration, and disposal) will follow relevant local guidelines and/or instructions. For further details of relevant chemotherapeutic agents, please refer to the manufacturer' s instructions. Medications used to treat infusion-related reactions (e.g., diphenhydramine, acetaminophen, and corticosteroids) are not investigational medicinal products and are not provided by the sponsor. These medicinal products are marketed products that will be purchased by the study site.

5.1.1. Methods of Assignment of Subjects to Treatment Groups and Blinding

5.1.1.1. Subject identification number

Each subject will be assigned a unique subject identification (SID) number. The SID number will be used to identify subjects throughout the study and will be entered in all study documents. The SID will consist of 8 digits, with the first 4 digits being the study site number and the last 4 digits being the sequential number assigned to the study site at screening. If a subject discontinues from the study at any time, his/her study number may not be reassigned to another subject.

5.1.1.2. Randomisation

During the dose expansion period, if the dose expansion cohorts 2, 3, 4 and 5 are allowed to enroll subjects at the same time, the subjects will be randomly assigned to one of the dose expansion cohorts.

5.1.1.3. Blinding

All three phases of the study were open-label and therefore were not blinded.

The decision to enroll a particular subject and the order of subject enrollment should not be affected by the extent to which the investigator is aware of the dose level assigned to the

subject.

5.1.2. Treatment Compliance Assessment Method

According to the protocol, AK104 and oxaliplatin will be administered to subjects participating in the clinical study at the study site under the supervision of the investigator or designee. The time of dosing will be recorded in the eCRF. Treatment compliance will be monitored with drug dispensing records, medical records of subjects and eCRFs. The subjects were administered capecitabine orally twice daily, and were required to record the dosage and administration timing of capecitabine administered daily in writing and submit it to the study coordinator of the study site at the next visit, and return all used and unused drug containers at the end of treatment for compliance evaluation.

5.1.3. Preparation

5.1.3.1 Preparation of AK104

The dose of study drug used for administration must be prepared by the investigator or the authorized personnel of the study site using aseptic techniques.

Each vial selected to contain the dose formulation should be inspected. If there are any defects in the investigational product, the investigator and the monitor in the study site should be informed immediately. Refer to the Product Complaints section (Section 5.1.9) for further clarification.

For the AK104 drug product, it is recommended that doses over a dose range of 1.0 mg/kg to 6 mg/kg are formulated to use an intravenous infusion bag containing 100 mL of normal saline (0.9% (w/v) sodium chloride injection), and the 10 mg/kg and 15 mg/kg doses are formulated in the intravenous infusion bag containing 250 mL of normal saline (0.9% (w/v) sodium chloride injection) . First, an intravenous infusion diluent equal to the dose volume of the AK104 formulation to be administered is withdrawn from the infusion bag. The AK104 dose from one or more vials is then added to the infusion bag. After swabbing the injection port of the infusion bag, AK104 by injection was added, the infusion bag was inverted gently to allow the solution to mix completely. Do not shake the infusion bag vigorously. The final solution was visually inspected. If the infusion solution is not clear or the contents contain precipitation, the solution should be discarded (as described in Section 5.1.8) and recorded in the drug accountability log.

AK104 does not contain preservatives and intravenous infusion bags containing AK104 must be used immediately after dilution/mixing. If the AK104 diluent cannot be used immediately and requires storage, the total storage time from the prepared diluent to completion of infusion should not exceed 24 hours in the refrigerator (2 °C to 8 °C), which includes a maximum of 4 hours at room temperature.

Prior to dosing, all prepared AK104 intravenous infusion bags must be labeled with the following: SID, subject initials, expiry date and time, and infusion personnel initials. Further instructions regarding the preparation and administration of the study drug will be provided to

the study site pharmacy in the form of a manual.

5.1.3.2 Preparation of Chemotherapeutic Agents

For the preparation of oxaliplatin, please refer to the product labeling of the drug. Capecitabine is an oral formulation and does not require preparation.

5.1.4. Administration of Therapeutic Drugs

Both AK104 and oxaliplatin are intravenous injections and must be administered in an outpatient clinic or ward of the study site, and the drug administration process must be performed by qualified or experienced investigators, and the drugs must not be taken out of the study site for use. Capecitabine is an oral formulation and subjects are required to record the dosage and time of capecitabine administered daily in writing.

The AK104 intravenous infusion should be kept on the same day as the oxaliplatin intravenous infusion. AK104 will be administered intravenously at least 30 minutes before oxaliplatin intravenous injection and capecitabine administration. The administration time window is ± 3 days, but subjects must complete various examinations including vital signs, physical examination, laboratory examination, and physical status score to evaluate the safety and tolerability of continued treatment within 3 days before each administration. Refer to Section 3.1.2 for AK104 in combination with mXELOX/XELOX.

5.1.4.1. AK104 Injection

AK104 will be administered via intravenous infusion over 60 minutes ± 10 minutes. For subjects who cannot tolerate a 60-minute infusion or may have infusion-related risks, the infusion time may be extended to a maximum of 120 minutes ±15 minutes. Adequate epinephrine, intravenous diphenhydramine hydrochloride or other anti-allergic drugs, and resuscitation equipment should be prepared before AK104 infusion for immediate use of serious allergic reactions. After infusion, IV access should remain open for dosing as needed.

After dilution in the IV bag, the entire contents of the IV bag should be administered by intravenous infusion over 60 minutes using an in-line filter of 0.2 μm or 0.22 μm. After completion of the infusion, the IV line was flushed with the same volume of saline (approximately 10 mL or as routine volume performed at the study site) as perfusion volume of the infusion device to ensure that full infusion was administered to the subject.

Since the compatibility of AK104 with other intravenous infusion drugs and solutions other than normal saline (0.9% (w/v) sodium chloride injection) is not known, AK104 solutions should not be infused using an IV line that infuses other solutions or drugs. The date, start time, interruption time, and completion time of AK104 administration must be recorded in the source document.

5.1.4.2. Oxaliplatin Injection

Oxaliplatin is dissolved in 250 mL to 500 mL of 5% dextrose solution, 85 mg/m^2 (mXELOX) or 130 mg/m^2 (XELOX) , intravenously infusion for 2-6 hours on Day 1 of each 2/3-week

cycle. Please refer to the product labeling of the drug for precautions. Avoid exposure to cold air and cold water during oxaliplatin treatment (3-5 days after infusion).

5.1.4.3. Capecitabine Tablets

1000 mg/m^2, orally twice daily, swallowed with water within 30 minutes after a meal, Days 1 to 10 (mXELOX)/Days 1 to 14 (XELOX) of each 2/3-week cycle. Capecitabine will be supplied at the strength of 500 mg per tablet in this study. The dose will be adjusted according to the drug strength, and the dosage should be as close as possible to that of the 500 mg tablet (500 mg fold). See the investigational drug management manual for details. Treatment with capecitabine was initiated on the same day after oxaliplatin administration. Please refer to the product labeling of the drug for precautions.

5.1.5. Monitoring during Dosing

Subjects will be monitored during and after the completion of AK104 infusion, and vital signs will be assessed within 30 minutes before start of infusion, 30 minutes (± 5 minutes) during the 1-hour infusion period, after the end of infusion (+ 10 minutes), and 30 minutes (± 10 minutes) after the end of infusion and before oxaliplatin infusion. On Day 1 of Cycle 1, additional vital sign measurements will be performed 3 hours (± 30 minutes) after the end of the first infusion. For subsequent dosing, a 3-hour observation period was not required unless the subject experienced an infusion-related reaction. Additional monitoring and vital sign assessments will be performed at the discretion of the investigator as per standard clinical practice or clinical indications.

Subjects may develop the same allergic reactions to dosing as all antibodies. Therefore, it is essential to provide appropriate drugs and medical equipment immediately to treat acute allergic reactions, and the study staff must be trained to recognize and in treat systemic allergic reactions.

Subjects may develop the same allergic reactions to study drug AK104 as all antibodies. Therefore, it is essential to provide appropriate drugs and medical equipment immediately to treat acute allergic reactions, and the study staff must be trained to recognize and in treat systemic allergic reactions. Subjects will be monitored for signs and symptoms of infusion-related reactions and serious systemic allergic reactions. Refer to Section 3.1.4.1 and Table 3 – 2 for toxicity management guidelines of AK104 infusion reactions.

5.1.6. Labeling and Packaging

Akeso Biopharma Co., Ltd . shall uniformly provide labeling and packaging for clinical study drugs for clinical trials.

The labeling of the investigational product AK104 will be performed in accordance with the relevant regulatory guidelines of GCP (Good Clinical Practice), and the labeling should include, but not be limited to the following content: clinical approval number, protocol number, drug name, study drug number, packaging specification, manufacturing batch number, expiry date, dosage and administration, drug storage conditions, sponsor name, and should be

marked with "clinical study use only".

5.1.7. Storage

The study drug should be stored in their original container and as indicated on the label. If the storage conditions on the label and in other materials of the study drug (such as the pharmacy manual) are inconsistent, the storage conditions on the label should prevail.

AK104 is stored at 2 °C to 8 °C and protected from light. Avoid exposure to temperatures beyond the recommended range and avoid vigorous shaking. If the storage temperature is above 8 °C or below 2 °C, a temperature excursion will need to be reported and the sponsor will be notified promptly and the impact of this excursion on the product will be evaluated by the sponsor. The storage conditions of chemotherapeutic drugs are described in the product instructions.

Records of the actual storage conditions during the study (e.g., date and time, continuous temperature records of the freezer used to store trial supplies, or maintenance records of instruments such as temperature alarm systems used in conjunction with temperature recorders) must be maintained.

5.1.8. Drug Accountability

The medications provided in this study may only be used under the direction of the protocol. The investigator/institution is responsible for establishing a study treatment drug (including investigational product) management process to ensure that:

- Delivered study drugs are correctly received by authorized personnel.
- Such deliveries are recorded.
- Study treatment medications are handled and stored safely and appropriately as specified on the label.
- Only study treatment medications are assigned to study subjects per protocol.
- Unused products should be destroyed after contacting the clinical research associate. Unused study drug supplies are returned to the sponsor for destruction or destroyed at the study site in accordance with drug management SOPs.

At the end of the study, the use and destruction records must be consistent with the delivery records. Use records should include the identification of the person dispensing the study treatment medication and the doseand date dispensed. This record complements any drug accountability information recorded in the case report form. Any discrepancies must be explained in appropriate forms.

All study drug-related forms must be inspected by an authorized representative or designee of the sponsor and inspectors from regulatory authorities . The investigator or appropriate authorized person is responsible for the accountability of all used and unused study drugs at the study site.

5.1.9. Reporting Drug Complaints

Any deficiencies in the study drug must be immediately reported to the Sponsor's Product Complaints Department by study site and further notified to the monitor in the study site. All deficiencies will be notified to the sponsor and further investigated with Product Complaints Department. During the investigation of product complaints, all study drugs must be stored under labeling conditions unless otherwise indicated.

Contact Information for Reporting Product Complaints:

Email:	productcomplaints@akesobio.com
Tel.:	+86-760-89873999
Fax:	+86-760-89873900
Company	Akeso Biopharma Co., Ltd.
Sector	Product Complaints Department
Address	Building A1, No. 6, Shennong Road, Torch Development Zone, Zhongshan City, Guangdong Province, China

5.2. Concomitant Medications and Treatments

Medications taken within 30 days prior to screening and all concomitant medications taken during the study will be recorded in the eCRF. If appropriate supportive care is deemed necessary, the investigator may prescribe concomitant medications or treatments, with the exception of "excluded" medications listed in Section 5.2.2.

5.2.1. Permitted Concomitant Medications and Treatments

1) Supportive treatment or symptomatic treatment: antibiotics, analgesics, nutritional infusion, psychotherapy, any other supportive treatments for control of non-tumor-related essential symptoms, or any symptomatic treatment for adverse events (e.g., granulocyte colony-stimulating factor (G-CSF) and other hematopoietic growth factors may be given for the treatment of acute toxic reactions such as neutropenic fever if clinically indicated or at the discretion of investigator).

2) Pretreatment for infusion-related reactions and/or hypersensitivity reactions during the study treatment.

3) Hormone therapy for non-tumor-related diseases (such as insulin and hormone replacement therapy for the treatment of diabetes).

4) Palliative radiotherapy for known metastatic sites (e.g., radiotherapy for pain control of bone metastases) is permitted, provided that target and non-target lesions are not included in the radiation field, and the palliative radiotherapy does not affect tumor response assessment or interrupt study treatment for more than 2 weeks.

5) If bisphosphonates and RANK-L inhibitors have been started 2 weeks before the first dose

of AK104 for bone disease, they can be continued during the study period. Whenever possible, only these medications are allowed to initiate treatment of hypercalcemia associated with neoplastic disease during the study or for other relevant indications that are allowed after discussion with the medical monitor.

7) Immunosuppressive drugs are permitted to treat AEs related to the study drug; or for subjects who are allergic to enhancers.

8) Inhaled, topical and intranasal corticosteroids are permitted.

9) After discussion with the medical monitor, corticosteroids may be allowed for the short-term treatment of underlying or concurrent conditions, or during the palliative radiotherapy.

10) Oral contraceptives.

5.2.2. Prohibited Concomitant Medications and Treatments

Subjects will be instructed not to take any medication, including over-the-counter drugs, without prior consultation with the investigator.

The following medications were prohibited during the study: if a subject received any of the following medications during the study, the sponsor must be notified.

1) Any other investigational drug.
2) Any other anti-tumor therapy: including, but not limited to chemotherapy, antibody therapy, targeted therapy, endocrine therapy, cell therapy, radiotherapy (exception for palliative radiotherapy, see Section 5.2.1), vitamin A-like treatment, and Chinese patent medicine or Chinese herbal medicine with anti-tumor effect, etc.
3) Immunosuppressive drugs include, but are not limited to systemic corticosteroids, prednisone at doses exceeding 10 mg daily or equivalent, methotrexate, azathioprine, and TNF-α antagonists (exception for immunosuppressive medications, see Section 5.2.1).
4) Live vaccines.
5) Concomitant use of dietary supplements, non-investigator-prescribed medications, and alternative/complementary therapies is discouraged but not prohibited, and should be discussed with the medical monitor.

5.2.3. Concomitant Medications Used with Caution

Considering that AK104 is a therapeutic bispecific antibody, the potential for drug-drug interactions between the investigational drug AK104 and small molecule drugs is considered low. Since AK104 is expected to be degraded to amino acids and recycled to other proteins, it is unlikely to have an effect on drug metabolizing enzymes or transporters. For concomitant medications used with the chemotherapeutic agents oxaliplatin and capecitabine with caution, refer to their product labeling.

1) Coumarin Anticoagulants Used with Caution

Changes in coagulation parameters and/or bleeding have been reported in patients treated with capecitabine with concomitant anticoagulants of coumarin derivatives such as warfarin and phenprocoumon. These conditions occur within days to months of capecitabine treatment, with some patients occurring within 1 month of capecitabine discontinuation. Anticoagulation parameters (INR or PT) should be monitored routinely in patients receiving capecitabine and oral coumarin derivative anticoagulants, and the dose of the anticoagulants should be adjusted accordingly.

2) Substrates of Cytochrome P-4502C9

Substrates of cytochrome P-4502C9 (diclofenac, phenytoin sodium, piroxicam, S-warfarin, and tolbutamide) should be used with caution during treatment.

6. Study Procedures

6.1. Enrollment/Screening Period

Table 6-1 shows all procedures performed during the screening period.

Table 6-1 Screening Procedure Schedule

Study Period	Screening
Procedures	Days -28 to -1
Written Informed Consent Form/Assignment of SID Number[a]	X
Demographic data and medical history, smoking history, alcohol history[b]	X
Prior tumor history and prior treatment history[c]	X
Inclusion/exclusion criteria	X
Physical examination, height, weight	X
Vital signs	X
12-lead ECG	X
Echocardiography[d]	X
ECOG performance status	X
AE/SAE assessment	X
Pregnancy test, serum β-hCG[e]	X
Concomitant Medications/Treatments [f]	X
Hepatitis B, C; HIV[g]	X
Safety Laboratory Tests	
Urinalysis (see Section 9.4)	X
Blood chemistry (see Section 9.4)	X
Hematology (see Section 9.4)	X
Thyroid function tests (TSH, free T3 and free T4)	X
Coagulation (see Section 9.4)	X
Collect blood samples for biomarker evaluation	
Whole blood for flow cytometry (Ki67) [h]	X
Whole blood for receptor occupancy analysis[h]	X
Biopsy	
Biopsy of archived tumor samples or freshly obtained tissues[i]	X
Tumor Disease Assessment	
Imaging (CT with contrast or MRI with contrast preferred)[j]	X
Assessment of tumor lesion per RECIST 1.1[k]	X

Abbreviations: AE, adverse event; β-hCG, subunit of beta human chorionic gonadotropin; CT, computed tomography; ECOG, Eastern Cooperative Oncology Group; HIV, human immunodeficiency virus; MRI, magnetic resonance imaging; RECIST, Response Evaluation Criteria in Solid Tumors; SAE, serious adverse event; SID, subject identification; TSH, thyroid stimulating hormone

a. A written informed consent must be obtained before any study-specific procedures, for radiological tumor assessment, tumor tissue biopsy, except those already performed within a specified period prior to the first dose. The study allows subjects who have previously failed screening to be re-screened (only once) , and the re-screened subjects must re-sign the informed consent and re-register to obtain a new subject number at the time of re-screening.
b. Demographic data: including date of birth, gender, ethnicity; past medical history (collect all previous medical history other than this indication that started before signing the ICF), drug allergy history, smoking history, alcohol history.
c. Prior tumor history and prior treatment history: including histological/cytological diagnosis, pathological classification, histological grade, clinical stage (2018 FIGO stage for Cohort A from Phase II), time of initial diagnosis, and time of recurrence or metastasis, etc.; prior surgical history, drug history (chemotherapy, targeted therapy, endocrine therapy, etc.), and radiotherapy history, and the start and end date of treatment regimen, best tumor response, and reason for discontinuation should be recorded as fully as possible.
d. Screening only and subsequent testing as clinically indicated.
e. Only for women of childbearing potential, if serum pregnancy tests are not available, urine pregnancy tests are available.
f. Concomitant medications/therapies: Record medications and non-drug therapies for diseases other than those for indication of study drug within the first 30 days prior to screening.
g. Testing will be performed by the central laboratory of the study. Hepatitis B virus screening includes HBsAg, HBsAb, and HBcAb (HBV DNA by hepatitis B virus DNA load test if necessary) and hepatitis C antibody (qualitative testing for hepatitis C RNA if antibody test is positive) and HIV-Ab antibodies.
h. A total of approximately 6-9 subjects will be selected for whole blood sampling for flow cytometric analysis (Ki67) and receptor occupancy (RO) test in each dose group in each cohort of the dose escalation and expansion period (Phase Ib).
i. All subjects should provide 10 unstained slides of tumor tissue specimens that are archived after the confirmation of diagnosis or fleshly obtained (a minimum of 5 slides, with the approval by the medical monitor). In the Cohorts 2, 3, 4, 5 from Phase Ib and, Cohort D from Phase II, slides which are archived within 6 months prior to the first dose or freshly obtained are acceptable (preferably fresh tumor tissue samples) . If an archived tumor tissue sample is not suitable for use, a fresh tumor biopsy at baseline will be obtained prior to the subject 's first dose of study drug (see Section 8.4.1 for instructions for tumor tissue collection). Subjects in Cohort A from Phase II should provide the most recent tumor tissue samples as far as possible (preferably at the recurrent or metastatic stage); For the subjects in the Cohort 1 from Phase Ib and Cohorts A, B, C from Phase II, if archived tumor tissue samples are not available and the subjects have great risk for a fresh biopsy as determined by the investigator, they can be exempted from collection of archived tumor tissue specimens after discuss with the medical monitor.
j. Contrast-enhanced CT or contrast-enhanced MRI is the preferred method for tumor assessment in this study. Tumor assessments must include neck, chest, abdomen, and pelvis throughout the study. Contrast-enhanced MRI or contrast-enhanced CT of the head is required at baseline assessment to determine brain metastases; re-examination of the head will be performed if abnormalities are suspected by the investigator or clinically indicated during the course of the study. The presence of

bone metastases is also required at baseline assessments. Skeletal scans (e.g., bone scan) are not required to be repeated if they have been performed within 42 days prior to the first dose. If bone metastases are present at screening or if clinically indicated during the study treatment phase, skeletal imaging should be repeated at least when progression of bone metastases is suspected or when a complete response is identified in target lesion. The investigator will make tumor assessments per RECIST V1.1. All lesion sites that are known to be RECIST v1.1-compliant as judged by the investigator should be recorded as far as possible at screening and reassessed at each subsequent tumor assessment. To assess response, baseline and post-treatment test methods should be consistent to facilitate comparisons. Assessments should be performed by the same assessor whenever possible.

k. Baseline assessments should be completed within 28 days prior to the first dose and ideally as close as possible to the time of the first dose. The results of tumor assessments performed prior to obtaining informed consent and within 28 days prior to the first dose can be used for screening assessments rather than repeating such assessments. Skeletal scans (e.g., bone scan) are not required to be repeated if they have been performed within 42 days prior to the first dose..

6.2. Treatment Period

Every 2 weeks is one cycle during the treatment period of AK104 monotherapy and AK104 in combination with mXELOX . For AK104 in combination with mXELOX, chemotherapy (mXELOX) can only be discontinued due to disease progression, unacceptable toxicity, or withdrawal of consent prior to the end of Cycles 10-12 (every 2 weeks as one cycle). After Cycles 10-12, if there is no confirmed disease progression or unacceptable toxicity, the subject and his/her investigator can decide whether to continue the maintenance of capecitabine until disease progression, unacceptable toxicity, or subject`s request or at the discretion of investigator.

Every 3 weeks is one cycle during the treatment period of AK104 in combination with XELOX . For AK104 in combination with XELOX, chemotherapy (XELOX) can only be discontinued due to disease progression, unacceptable toxicity, or withdrawal of consent prior to the end of Cycle 6 (every 3 weeks as one cycle). After Cycle 6, if there is no confirmed disease progression or unacceptable toxicity, the subject and his/her investigator can decide whether to continue the maintenance of capecitabine until disease progression, unacceptable toxicity, or subject`s request or at the discretion of investigator.

The investigational product AK104 is administered every 2/3 weeks until any of the following occurs: the subject is unable to continue to benefit as determined by the investigator, confirmed disease progression, unacceptable toxicity, concomitant diseases that affect further treatment, investigator decision, subject withdrawal of consent, or other reasons as specified in the protocol. Treatment with AK104 can be given over a maximum of 24 months.

The on-treatment study procedures are presented in Table 6- 2.

Assessments should be performed according to the study procedures shown in the table. When scheduled time points of vital signs, 12-lead ECG and blood sampling are the same,

assessments should be performed in the following order: 12-lead ECG, vital signs, and then blood sampling.

If the following assessments have been completed within 3 days prior to the first dose of study drug, repeated assessments are not required on Day 1: 12-lead ECG, hematology, blood chemistry, urinalysis, cardiac function and pregnancy tests; if the following assessments have been completed within 1 week prior to the first dose of study drug, repeated assessments are not required on Day 1: thyroid function tests, coagulation.

Table 6-2 On-treatment Procedure Schedule

Procedures/Study Days[a]	Treatment cycle of AK104 monotherapy or in combination with mXELOX/XELOX (2 weeks/3 weeks as 1 cycle)										
	Cycle 1 D1	**Cycle 2 D1**	**Cycle 3 D1**	**Cycle 4 D1**	**Cycle 5 D1**	**Cycle 6 D1**	**Cycle 7 D1**	**Cycle 8 D1**	**Cycle 9 D1**	**D1 of Cycle 10 and subsequent cycles[t]**	**End of Treatment Visit[s]**
AK104 alone or in combination with mXELOX (Q2W)	**D1**	**D15**	**D29**	**D43**	**D57**	**D71**	**D85**	**D99**	**D113**	**D127+**	
AK104 in combination with XELOX (Q3W)	**D1**	**D22**	**D43**	**D64**	**D85**	**D106**	**D127**	**D148**	**D169**	**D190+**	
Time Window		**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**0-7**
Inclusion/exclusion criteria reconciliation	X										
Study Procedures and Test Items											
Physical examination, body weight[b]	X	X	X	X	X	X	X	X	X	X	X
ECOG performance status	X	X	X	X	X	X	X	X	X	X	X
12-lead ECG [c]	X	X	X	X	X	X	X	X	X	X	X
ECHO	Clinically indicated or determined by the investigator										
Vital signs[d]	X	X	X	X	X	X	X	X	X	X	X
Concomitant medications/treatments	X	X	X	X	X	X	X	X	X	X	X
AE/SAE assessment	X	X	X	X	X	X	X	X	X	X	X
Safety Laboratory Tests											
Hematology (see Section 9.4)	X	X	X	X	X	X	X	X	X	X	X
Blood chemistry (see Section 9.4)	X	X	X	X	X	X	X	X	X	X	X
Thyroid function: TSH, fT4, fT3	X		X		X		X		X (every 2 cycles)		X
Cardiac function[e]	X	X	X	X	X	X	X	X	X	X	X
Urinalysis (see Section 9.4) [f]	X	X	X	X	X	X	X	X	X	X	X
Coagulation (see Section 9.4)	X		X		X		X		X (every 2 cycles)		X
HBV, HCV test	Clinically indicated or determined by the investigator										
Pregnancy test [g]	Clinically indicated or determined by the investigator										

Procedures/Study Days[a]	Treatment cycle of AK104 monotherapy or in combination with mXELOX/XELOX (2 weeks/3 weeks as 1 cycle)										
	Cycle 1 D1	**Cycle 2 D1**	**Cycle 3 D1**	**Cycle 4 D1**	**Cycle 5 D1**	**Cycle 6 D1**	**Cycle 7 D1**	**Cycle 8 D1**	**Cycle 9 D1**	**D1 of Cycle 10 and subsequent cycles[t]**	**End of Treatment Visit[s]**
AK104 alone or in combination with mXELOX (Q2W)	**D1**	**D15**	**D29**	**D43**	**D57**	**D71**	**D85**	**D99**	**D113**	**D127+**	
AK104 in combination with XELOX (Q3W)	**D1**	**D22**	**D43**	**D64**	**D85**	**D106**	**D127**	**D148**	**D169**	**D190+**	
Time Window		**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**0-7**
Pharmacokinetics and Immunogenicity											
AK104 PK (intensive sample collection, see Table 8-1)[h]	X	X	X	X	X	X	X			X (every 3 cycles)	
AK104 PK (sparse sample collection, see Table 8-2)[i]	X		X		X		X			X (every 3 cycles)	
Immunogenicity[j]	X		X		X		X			X (every 3 cycles)	
Biomarker Evaluation											
Whole blood for flow cytometry analysis (Ki67)[k]	X	X	X		X						
Whole blood for receptor occupancy analysis[k]	X	X	X		X		X			X (Cycle 10 only)	**X**
Tumor tissue biopsy (optional)[l]	At the time of first tumor response of PR or CR, or at PD of disease progression										
Study Drug Administration											
AK104 administration[m]	X	X	X	X	X	X	X	X	X	X	
Oxaliplatin (mXELOX)[n]	X	X	X	X	X	X	X	X	X	X	
Oxaliplatin (XELOX)[n]	X	X	X	X	X	X					
Capecitabine (mXELOX)[O]	Day 1-10	D15-D24	D29-D38	D43-D52	Day 57-66	D71-D80	D85-D94	D99-D108	Day 113-122	Day 127-136	
Capecitabine (XELOX)[O]	Day 1-14	D22-D35	D43-D56	D64-D77	Day 85-98	D106-D119	D127-D140	D148-D161	Day 169-182	Day 190-203	
Medication compliance of capecitabine[p]		X	X	X	X	X	X	X	X	X	
Tumor Disease Assessment											
Imaging tests	X (once every 6 weeks [Phase Ib] for the first 54 weeks or once every 8 weeks [Phase II] for the first 56 weeks and once										X

Procedures/Study Days[a]	Treatment cycle of AK104 monotherapy or in combination with mXELOX/XELOX (2 weeks/3 weeks as 1 cycle)										
	Cycle 1 D1	**Cycle 2 D1**	**Cycle 3 D1**	**Cycle 4 D1**	**Cycle 5 D1**	**Cycle 6 D1**	**Cycle 7 D1**	**Cycle 8 D1**	**Cycle 9 D1**	**D1 of Cycle 10 and subsequent cycles[t]**	**End of Treatment Visit[s]**
AK104 alone or in combination with mXELOX (Q2W)	**D1**	**D15**	**D29**	**D43**	**D57**	**D71**	**D85**	**D99**	**D113**	**D127+**	
AK104 in combination with XELOX (Q3W)	**D1**	**D22**	**D43**	**D64**	**D85**	**D106**	**D127**	**D148**	**D169**	**D190+**	
Time Window		**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**±3**	**0-7**
	every 12 weeks thereafter)										
Tumor assessments are performed as per RECIST 1.1/irRECIST[q, r]	X (once every 6 weeks [Phase Ib] for the first 54 weeks or once every 8 weeks [Phase II] for the first 56 weeks and once every 12 weeks thereafter)										X

Abbreviations: AE, adverse event; D, days; ECOG, Eastern Cooperative Oncology Group; PK, pharmacokinetics; RECIST, Response Evaluation Criteria in Solid Tumors; SAE, serious adverse event; TSH, thyroid stimulating hormone. W = weeks

Note: All samples or procedures required to be collected on the dosing day were performed prior to dosing unless otherwise stated. In addition to the test items and time points listed in the table, the investigator may add items that are necessary at any time.

a. Procedures/Study Days: Nominal assessments and laboratory tests on Day 1 will be performed prior to dosing. If the following assessments have been completed within 3 days prior to the first dose of study drug, repeated assessments are not required on Day 1: 12-lead ECG, hematology, blood chemistry, urinalysis, and pregnancy tests; if the following assessments have been completed within 1 week prior to the first dose of study drug, repeated assessments are not required on Day 1: thyroid function tests, coagulation.

b. A targeted physical examination should be performed based on symptoms and prior to administration of study drug.

c. 12-lead ECGs are performed for subjects within 3 days prior to each dose at the study site. Additional ECGs and monitoring should also be performed in a timely manner if clinically indicated and/or cardiac function tests suggest relevant abnormalities. All 12-lead ECGs are usually obtained by a single examination. Subjects with a cardiovascular medical history, or at risk of cardiac function as assessed by the investigator, may be tested repeatedly. The investigator will confirm appropriate pre-dose results of ECGs.

d. Of the AK104 dosing dates, vital signs will be assessed within 30 minutes prior to initiation of AK104 infusion, at the 30th (± 5 minute) minute within 1 hour of AK104 infusion, at the end of AK104 infusion (+ 10 minutes), 30 minutes (± 10 minutes) after the end of AK104 infusion, and before oxaliplatin infusion. On Day 1 of Cycle 1, additional vital sign measurements will be performed 3 hours (± 30 minutes) after the end of the first AK104 infusion. For subsequent dosing, a 3-hour observation period is not required unless the subject experiences an infusion-related reaction. If a subject experiences an infusion-related reaction, the investigator should whether

additional vital sign measurements should be performed after the end of the AK104 infusion and after the end of each subsequent infusion based on the subject's response characteristics and clinical conditions. Of oxaliplatin only dosing dates, vital signs will be assessed within 30 minutes prior to the start of oxaliplatin infusion. Additional monitoring and vital sign assessments will be performed at the discretion of the investigator as per standard clinical practice or clinical indications.

e. Cardiac function tests are performed for subjects within 3 days prior to each dose at the study site. Items of cardiac function tests should include creatine kinase (CK), creatine kinase isoenzyme (CK-MB), cardiac troponin I (cTnI), cardiac troponin T (cTnT), brain natriuretic peptide (BNP), and/or N-terminal pro-brain natriuretic peptide (NT-proBNP) as far as possible. Additional cardiac function tests may be performed at the discretion of the investigator, depending on the subject's clinical conditions.

f. Urinalysis: If a subject shows proteinuria 2 + or higher on urine dipstick or routine analysis, a 24-hour urine collection should be provided for further analysis.

g. Pregnancy test: Only for women of childbearing potential. A serum pregnancy test will be performed and documented as negative within 3 days prior to the first dose at screening. If a serum pregnancy test is not available, a urine pregnancy test is available. Urine or serum pregnancy tests will be performed as clinically indicated during treatment and at the safety follow-up. If a urine pregnancy test is positive, a serum pregnancy test must be performed (at each study site), and the result of this serum pregnancy test should be employed.

h. Intensive PK blood sampling will be performed for subjects who enroll in the dose escalation period (Part 1 of Phase Ib) and the dose expansion period (Part 2 of Phase Ib) at selected sites, with a total of approximately 6-9 subjects per dose group, and intensive PK blood sampling time points are presented in Table 8-1 and Table 8-3. If the number of the subjects with intensively collected samples in each cohort in Phase Ib is less than 6-9, the investigator may increase the number of subjects in the cohorts or exempt from further collection after discussion with the sponsor, depending on whether there is a possible risk to affect the PK analysis. Sparse PK sampling will be performed at the time points shown in Table 8-2 and 8-4 for subjects enrolled in the other dose expansion period (Part 2 of Phase Ib) and the dose confirmation period.

i. Sample collection for immunogenicity testing of AK104: Blood samples are collected within 30 min prior to the start of the AK104 infusion for all subjects who undergo ADA blood sampling. The sampling time points are shown in Table 8-5.

j. A total of approximately 6-9 subjects will be selected for whole blood sampling for flow cytometric analysis (Ki67) and receptor occupancy (RO) test in each dose group in each cohort of the dose escalation and expansion period (Phase Ib).

k. Voluntary participation in optional biomarker studies may be encouraged. For subjects participating in optional biomarker studies, fresh tumor biopsies will be performed at the time of response assessment of tumor response (PR/CR) and/or progressive disease (PD) if tumor lesion tissue is clinically available, and it is obtained subject's informed consent. Please refer to Section 8.4.1 for instructions regarding the collection of fresh tumor tissue biopsies. Selected target lesions or measurable non-target lesions should not be used as biopsy tumor lesions planned for biomarker studies.

l. Intravenous infusion of AK104: once every 2 weeks, or once every 3 weeks, with a predefined time window of ± 3 days. With the exception of radiological examinations, all examinations should be completed within 3 days prior to dosing (see Note a for examination requirements before first dose), and the administration can be continued only after safety assessment. Refer to Section 3.1.2 for detailed dosing regimens. If the AK104 intravenous infusion is administered on the same day as the oxaliplatin intravenous infusion. AK104 should be administered intravenously at least 30 minutes before intravenous injection of oxaliplatin and administration of capecitabine.

m. Cohorts 2-4: Oxaliplatin is administered intravenously once every 2 weeks in a 2-week cycle for 10-12 treatment cycles. Cohort 5: Oxaliplatin is administered intravenously once every 3 weeks in a 3-week cycle for 6 treatment cycles. Cohort D is decided based on the recommended dose regimen.

n. Capecitabine administration: twice daily, oral. Cohorts 2-4, Days 1 to 10, in a 2-week cycle, for 10-12 treatment cycles. After Cycles 10-12, the subject and his/her investigator can decide whether to continue the maintenance chemotherapy of capecitabine. Cohort 5, Days 1 to 14, in a 3-week cycle, for 6 treatment cycles. After Cycle 6, the subject and his/her investigator can decide whether to continue the maintenance chemotherapy of capecitabine; Cohort D is decided based on the recommended dose regimen.

o. Subjects are required to document the doseand timing of capecitabine administered daily in writing and to submit them to the study coordinator of the study site at the next visit. If a subject experiences vomiting within 30 minutes after taking capecitabine, he/she will be considered to take no drugs, which will be recorded in the subject diary card and the eCRF.

p. Tumor assessments: in Phase Ib, tumor assessments should be performed every 6 weeks (± 7 days) for 54 weeks after enrollment (with respect to the date of the first AK104 administration), and every 12 weeks (± 7 days) after 54 weeks of enrollment; in Phase II, tumor assessments should be performed every 8 weeks (± 7 days) within 56 weeks of enrollment (with respect to the date of the first AK104 administration), and every 12 weeks (± 7 days) after 56 weeks of enrollment until the initiation of subsequent anti-tumor therapy, disease progression, withdrawal of informed consent, death, or until study closure, whichever occurs first. The investigator may perform additional scans or more frequent assessments if clinically indicated. Unscheduled radiographic tests may be performed when disease progression is suspected (eg, worsening symptoms). If an unscheduled assessment is performed and the subject has no progression, tumor assessments will need to follow the original plan for subsequent assessments. In the event of treatment delay, tumor assessments are performed as originally planned. A scan to confirm progression should be performed at the next follow-up visit and within not less than 4 weeks after the initial assessment of disease progression (without significant clinical deterioration). The same radiographic procedures must be followed throughout the study for each subject. Assessments should be performed by the same assessor whenever possible to ensure intrinsic consistency between visits. If bone metastases are present at screening or if clinically indicated during the study treatment phase, skeletal imaging should be repeated at least when progression of bone metastases is suspected or when a complete response is identified in target lesion.

q. For subjects who discontinue study drug for toxicity or reasons other than confirmed disease progression, tumor assessments should be performed according to the following schedule: Tumor imaging assessments are required at the End of Treatment (EOT) visit (if it is less than 4 weeks apart from the last tumor imaging assessment, tumor assessments do not have to be repeated at the EOT visit); then tumor assessments will be performed every 6 weeks or 8 weeks (± 7 days) according to the cohort within 54 weeks or 56 weeks after enrollment, and then every 12 weeks (± 7 days) until the initiation of subsequent anti-tumor therapy, disease progression, withdrawal of informed consent, death, or until study closure, whichever occurs first.

r. The End of Treatment (EOT) visit is the day on which the investigator determines that dosing will no longer be administered. If routine laboratory tests (hematology, blood chemistry, urinalysis, etc.) are performed within 7 days prior to the EOT visit, they do not have to be repeated at the End of Treatment visit. Tumor imaging assessments should be performed at the end of treatment and do not need to be repeated at the EOT visit if the interval from the last assessment is less than 4 weeks. If the EOT visit is performed at least 30 days (± 3 days) after the last dose for any reason, the first safety follow-up visit is not necessary and the end-of-treatment examinations can be considered as a part of the safety follow-up.

s. Treatment with AK104 can be given over a maximum of 24 months. Subjects who are still receiving study treatment at the end of the study may continue treatment through additional extension studies or other forms agreed upon by the sponsor if stable or relieved disease is achieved in response evaluation and tolerable for adverse reactions.

6.3. End of Treatment and Follow-up Period

Table 6-3 shows all procedures performed during end of treatment and follow-up period: (1) subjects who discontinue study drug due to toxicity in the absence of disease progression; and (2) subjects with confirmed disease progression. Subjects will continue to be followed up until death, withdrawal of consent, loss of follow-up, or end of study, whichever occurs first.

Table 6-3 End of Treatment Schedule and Follow-up Procedure

Procedure/Study Days	**Follow-up Period**		
	Safety Follow-up1 (Day 30 ± 3 after last dose) [a]	**Safety Follow-up2 (Day 90 ± 7 after last dose)** [b]	**Survival Follow-up (Every 3 months ± 14 days after last dose)**
Pregnancy test[c]	X		
AE/SAE assessment[d]	X	X	
Concomitant medications/treatments[e]	X	X	
Physical examination, body weight[f]	X		
ECOG performance status	X		
Vital signs	X		
12-lead ECG	X		
Tumor Disease Assessment			
Imaging tests	X		
Tumor assessments are performed as per RECIST 1.1/irRECIST	X[g]		
Safety Laboratory Tests			
Urinalysis (see Section 9.4) [h]	X		
Blood chemistry (see Section 9.4)	X		
Thyroid function tests (TSH, free T3 and free T4)	X		
Hematology (see Section 9.4)	X		
Coagulation (see Section 9.4)	X		
Survival Follow-up and Subsequent antitumor Therapy		X	X

AE, adverse event; ECOG, Eastern Cooperative Oncology Group; EOT, end of treatment; hCG, human chorionic gonadotropin; RECIST, Response Evaluation Criteria in Solid Tumors; SAE, serious adverse event; TSH, thyroid stimulating hormone.

a. The first safety follow-up will be conducted 30 ± 3 days after the last dose or before the initiation of new anti-tumor therapy, whichever occurs first. If the end of treatment visit is more than 30 ± 3 days from the last dose, the first safety follow-up visit is not necessary. Tumor imaging assessment is required at the first safety follow-up if tumor imaging assessment is not performed at the end of treatment and is greater than 4 weeks from the last radiological assessment at the first safety follow-up.

b. Study treatment follow-up for subjects will end 90 days after the last dose of study drug.

c. For women of childbearing potential only. Subjects will undergo urine or serum pregnancy tests at the site at the safety follow-up. If a urine pregnancy test is positive, a serum pregnancy test must be performed, and the result of this serum pregnancy test should be employed.

d. After initiation of study treatment, all AEs and SAEs will be reported (regardless of relationship to study drug) until 30 days (AE) and 90 days (SAE) after the last dose of study treatment (including chemotherapy) or initiation of other anti-tumor therapy, whichever occurs first (whether observed by

the investigator or reported by the subject), both will be recorded on the AE page of the case report form. irAEs (regardless of serious or nonserious irAEs) will be reported until 90 days after the last dose of AK104 (whether other anti-tumor therapy have been initiated). After this period, the investigator should report any SAEs that are assessed as related to AK104.

e. The collection of concomitant medications will be discontinued 90 days after the last dose, unless the subject continues to experience an irAE or SAE possibly related to AK104, in which case concomitant medications should continue to be collected until the event resolves. If the subject has started new anti-tumor therapy within 90 days, record if available.

f. A targeted physical examination should be performed based on symptoms and prior to administration.

g. For subjects who discontinue study drug for toxicity or reasons other than confirmed disease progression, tumor assessments should be performed according to the following schedule: tumor imaging assessments are required at the End of Treatment (EOT) visit (it does not have to be repeated if the interval from the last tumor imaging assessment is less than 4 weeks at the EOT visit); then every 6 weeks or 8 weeks (± 7 days) for the first 54 weeks or 56 weeks, depending on the cohort, and then every 12 weeks (± 7 days) thereafter until the initiation of subsequent anti-tumor therapy, disease progression, withdrawal of informed consent, death, or until study closure, whichever occurs first. The investigator may perform additional scans or more frequent assessments if clinically indicated. Unscheduled radiographic tests may be performed when disease progression is suspected (eg, worsening symptoms). If evolution of radiographic findings is observed in the absence of clinical deterioration, scans should be performed not less than 4 weeks later for confirmation. The same radiographic procedures must be followed throughout the study for each subject. Assessments should be performed by the same assessor whenever possible to ensure intrinsic consistency between visits.

h. A 24h urine collection should be provided for further analysis if the subject shows proteinuria 2+ or higher on urine dipstick or routine analysis.

7. Efficacy Assessment

7.1. Tumour Response Assessments per RECIST 1.1

Tumor response assessments will be based on RECIST guidelines (version 1.1) and will be performed according to the schedule in Section 6. The schedule of assessments also applies to subjects who experience progression but continue to receive AK104 treatment. Guidelines for irRECIST are presented in Appendix 1 for measurable, non-measurable, target and non-target lesions and objective tumor response criteria (CR, PR, SD, or PD). For subjects receiving the treatment, the clinical decision will be made based on the response assessed by the investigator per RECIST version 1.1, including confirmation of disease progression.

Baseline assessments should be completed within 28 days prior to the first dose and ideally as close as possible to the time of first dose. The results of tumor assessments performed prior to obtaining informed consent and within 28 days prior to the first dose can be used for screening assessments rather than repeating assessments. In Phase Ib, tumor assessments will be performed every 6 weeks (± 7 days) within 54 weeks of enrollment (with respect to the date of the first AK104 administration) and then every 12 weeks (± 7 days) thereafter; in Phase II, tumor assessments will be performed every 8 weeks (± 7 days) within 56 weeks of enrollment (with respect to the date of the first AK104 administration) and then every 12 weeks (± 7 days) thereafter until the innitiation of subsequent anti-tumor therapy, disease progression, withdrawal of informed consent, death, or until study closure, whichever occurrs first. The investigator may perform additional scans or more frequent assessments if clinically indicated. Unscheduled radiographic tests may be performed when disease progression is suspected (eg, worsening symptoms). If an unscheduled assessment is performed and the subject has no progression, tumor assessments will need to follow the original plan for subsequent assessments. In the event of treatment delay, tumor assessments are performed as originally planned. A scan to confirm progression should be performed at the next follow-up visit and within not less than 4 weeks after the initial assessment of disease progression (without significant clinical deterioration).

For subjects who discontinue study drug for toxicity or for reasons other than confirmed disease progression, tumor assessments should be performed according to the following schedule: Tumor imaging assessments are required at the End of Treatment (EOT) visit (if it is less than 4 weeks apart from the last tumor imaging assessment, tumor assessments do not have to be repeated at the EOT visit); then tumor assessments will be performed every 6 weeks or 8 weeks (± 7 days) according to the cohort within 54 weeks or 56 weeks after enrollment, and then every 12 weeks (± 7 days) until the innitiation of subsequent anti-tumor therapy, disease progression, withdrawal of informed consent, death, or until study closure, whichever occurrs first.

Contrast-enhanced CT or contrast-enhanced MRI is the preferred method for tumor assessment in this study. Prior to any imaging, study sites are advised to assess subjects for potential contrast contraindications, such as potential allergic reactions to contrast media or risk of renal impairment. If the subject is allergic or contraindicated to contrast agents of both

enhanced CT or enhanced MRI, CT or MRI is preferred based on the site of imaging examination. PET-CT is a considered examination method, which requires communication with the medical monitor in certain instances of intolerance or inapplicability. Tumor assessments must include neck, chest, abdomen, and pelvis throughout the study. Contrast-enhanced MRI or contrast-enhanced CT of the head is required at baseline assessment to determine brain metastases; re-examination of the head will be performed if abnormalities are suspected by the investigator or clinically indicated during the course of the study. The presence of bone metastases is also required at baseline assessments. Skeletal scans (eg, bone scan) are not required to be repeated if they have been performed within 42 days prior to the first dose. If bone metastases are present at screening or if clinically indicated during the study treatment phase, skeletal imaging should be repeated at least when progression of bone metastases is suspected or when a complete response is identified in target lesion. If there is evidence of metastases to any other site at screening, imaging examinations of these sites should be performed based on the clinical symptoms and signs of the individual subject and followed throughout the study. For subsequent tumor assessments, the same test method as used to assess disease sites at screening should be used throughout the study (using the same contrast regimen as possible for contrast-enhanced MRI or contrast-enhanced CT). All lesion sites that are known to be RECIST v1.1-compliant as judged by the investigator should be recorded as far as possible at screening and reassessed at each subsequent tumor assessment.

Response will be assessed by the investigator per RECIST v1.1. Assessments should be performed by the same assessor whenever possible to ensure intrinsic consistency between visits. An objective response should be confirmed by reassessment at least 4 weeks after the first documented objective response. It is recommended not to wait until the next scheduled CT scan for confirmatory tumor assessments, but to perform assessment as early as 28 days after the first documented response. Following confirmatory scans, tumor assessments will need to be performed as originally planned.

7.2. Exploratory Tumor Response Evaluation using irRECIST

Subjects will continue to receive AK104 monotherapy only if they are assessed by the investigator as clinically beneficial and tolerant to AK104 after they are assessed as disease progression per RECIST 1.1 criteria by the investigator. Subjects may discontinue AK104 treatment based on radiographic assessment or clinical benefit if there is evidence of further disease progression. For radiographic assessment of disease progression, refer to irRECIST **(Nishino M, 2013)** criteria. It differs from RECIST 1.1 in the following key aspects:

1) The appearance of new lesions represents PD per RECIST v1.1. According to irRECIST, new measurable diseases are included in tumor burden to determine immune-related progressive disease (irPD), immune-related partial response (irPR), and immune-related complete response (irCR). irCRs are excluded from new non-measurable diseases.

2) No confirmation of PD is performed per RECIST v1.1. Consecutive scans (at least 4

weeks apart) are required to determine response and irPD, assuming no clinical deterioration per irRECIST.

Guidelines for irRECIST are provided in Appendix 2 for measurable, non-measurable, target and non-target lesions, and objective tumor response criteria (irCR, irPR, irSD, or irPD).

7.3. Central Imaging Assessment

The study plans to designate a contract research organization to conduct independent imaging assessments for selected tumor types or tumor cohorts in the Phase II. Refer to the IRRC charter for specific imaging requirements and guidance on the collection of results. Radiological documentation for selected subjects, including unscheduled imaging data, will be collected by the sponsor at the request of the independent imaging evaluators. All imaging data may be stored centrally by the sponsor.

The results of the independent review will not be communicated to the investigator and the management of subjects will be based solely on assessment results per RECIST performed by the investigator.

7.4. Survival assessment

Survival assessments must be performed every 3 months after treatment discontinuation. Survival information can be obtained by contacting with the subject or subject's family via telephone or by contacting with the subject's current physician, or by reviewing clinical records from the case. Details of the first and subsequent anti-tumor treatment after discontinuation will be collected. In addition, subjects may be contacted at the time of treatment or survival follow-up to obtain complete survival data after the data cutoff for the primary analysis and all subsequent survival analyses.

8. Pharmacokinetics/Immunogenicity/Pharmacodynamics Evaluation

8.1. Pharmacokinetics

Intensive or sparse blood sampling will be collected for PK analysis at the scheduled site by the study personnel at the visit time points shown in Section 6.2. Sparse PK samples will be continuously evaluated for PK parameters of AK104 and their correlation with demographic variables, biomarkers, and clinical endpoints by a population PK approach.

Intensive PK blood sampling will be performed for subjects who enroll in the dose escalation period (Part 1 of Phase Ib) and the dose expansion period (Part 2 of Phase Ib) at selected sites, with a total of approximately 6-9 subjects per dose group, and the time points are shown in Table 8-1 and Table 8-3. If the number of intensively collected subjects in each cohort in Phase Ib is less than 6-9, the investigator may discuss with the sponsor to increase the number of subjects in cohorts or to be exempted from further collection, depending on whether there is a possible risk to affect the PK analysis. Sparse PK sampling will be performed at the time points shown in Table 8-2 and Table 8-4 for subjects enrolled in the other dose expansion period (Part 2 of Phase Ib) and the dose confirmation period. The scheduled PK sampling day may be adjusted according to the actual medication of the subject.

Table 8-1 PK Sampling Schedule for AK104 - Intensive Sampling (Q2W)

	Scheduled day	Time relative to AK104 infusion
1st dose	Day 1	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
		3 hours (± 15 minutes) after infusion
	Day 2	24 hours (± 3 hours) after infusion
	Day 3	48 hours (± 6 hours) after infusion
	Day 8	168 hours (± 1 day) after infusion
2nd dose	Day 15	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
3rd dose	Day 29	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
4th dose	Day 43	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
5th dose	Day 57	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
		3 hours (± 15 minutes) after infusion
	Day 58	24 hours (± 3 hours) after infusion

	Scheduled day	Time relative to AK104 infusion
	Day 59	48 hours (± 6 hours) after infusion
	Day 64	168 hours (± 1 day) after infusion
6th dosing	Day 71	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
7th dose and subsequent doses (every 3 treatment cycles)	Day 85 and post-Day 85	Pre-infusion (within 30 minutes prior to infusion)

Table 8-2 PK Sampling Schedule for AK104 - Sparse Sampling (Q2W)

	Scheduled day	Time relative to AK104 infusion
1st dose	Day 1	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
3rd dose	Day 29	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
5th dose	Day 57	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
7th Dose	Day 85	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
10th Dose and subsequent doses (every 3 treatment cycles)	Day 127 and post-Day 127	Pre-infusion (within 30 minutes prior to infusion)

Table 8-3 PK Sampling Schedule for AK104 - Intensive Sampling (Q3W)

	Scheduled day	Time relative to AK104 infusion
1st dose	Day 1	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
		3 hours (± 15 minutes) after infusion
	Day 2	24 hours (± 3 hours) after infusion
	Day 3	48 hours (± 6 hours) after infusion
	Day 8	168 hours (± 1 day) after infusion
2nd dose	Day 22	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
3rd dose	Day 43	Pre-infusion (within 30 minutes prior to

	Scheduled day	Time relative to AK104 infusion
		infusion)
		End of infusion (+ 5 minutes)
4th dose	Day 64	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
5th dose	Day 85	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
		3 hours (± 15 minutes) after infusion
	Day 86	24 hours (± 3 hours) after infusion
	Day 87	48 hours (± 6 hours) after infusion
	Day 92	168 hours (± 1 day) after infusion
6th dosing	Day 106	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
7th dose and subsequent doses (every 3 treatment cycles)	Day 127 and post -Day 127	Pre-infusion (within 30 minutes prior to infusion)

Table 8-4 PK Sampling Schedule for AK104 - Sparse Sampling (Q3W)

	Scheduled day	Time relative to AK104 infusion
1st dose	Day 1	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
3rd dose	Day 43	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
5th dose	Day 85	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
7th Dose	Day 127	Pre-infusion (within 30 minutes prior to infusion)
		End of infusion (+ 5 minutes)
10th Dose and subsequent doses (every 3 treatment cycles)	Day 190 and post-Day 190	Pre-infusion (within 30 minutes prior to infusion)

The PK sample sampling time points shown in Table 8-1 to Table 8-4 may be modified based on the preliminary PK data obtained. However, the maximum number of PK samples to be collected within the first 57 days of the dose escalation period will not exceed 18.

Subject samples will be analyzed for AK104 concentrations at a central laboratory using a validated bioanalytical method. Specific instructions on serum PK sample collection and

transfer will be provided in the laboratory manual.

8.2. Pharmacodynamics

The antitumor effects of AK104 are mediated by enhanced lymphocyte proliferation/amplification at tumor-draining lymph nodes and increased tumor killing activity within the tumor microenvironment. Therefore, pharmacodynamic assessments should involve systemic aspects of immune priming/activation as well as local effects of treatment on tumors.

Pharmacodynamic biomarkers will be assessed in whole blood by flow cytometry and the proliferative state in T cells will be altered by monitoring the frequency (percentage) of Ki67-positive cells within the total T cell population and within the T cell subsets. Whole blood samples will be collected at the time points shown in Section 6.2 for subjects enrolled in the dose escalation period (Part 1 of Phase Ib) and the dose expansion period (Part 2 of Phase Ib) at selected sites, with a total of approximately 6-9 subjects per dose group.

Receptor occupancy is an important method to determine whether AK104 (a bispecific antibody targeting both anti-PD-1 and CTLA-4) exerts biological activity, and the occupancy of PD-1 or CTLA-4 receptor on circulating T cells is measured as a target marker. Whole blood and flow cytometric techniques will be used to determine PD-1 or CTLA-4 receptor occupancy on the surface of lymphatic T cells from whole blood. Whole blood samples will be collected at the time points shown in Section 6.2 for subjects enrolled in the dose escalation period (Part 1 of Phase Ib) and the dose expansion period (Part 2 of Phase Ib) at selected sites, with a total of approximately 6-9 subjects per dose group.

Details of sample collection, handling and testing will be provided in the laboratory manual.

8.3. Immunogenicity

Blood samples will be collected to evaluate the response of ADA to AK104 in serum. The following bridging immunoassays will be developed and validated: screening, qualification, and titers. Once the tested sample is confirmed to be positive for ADA, the specificity of the ADA will be assessed to confirm whether the ADA is neutralizing. Samples will be collected and stored to assess the specificity of the ADA and this step will be performed after development and validation of the analysis. The scheduled immunogenicity sampling day may be adjusted according to the actual medication of the subject.

Table 8-5 Immunogenicity Sampling Schedule - Q2W and Q3W

	Scheduled day	Time relative to AK104 infusion
1st dose	Day 1	Pre-infusion (within 30 minutes prior to infusion)
3rd dose	Day 29 (Q2W)/Day 43 (Q3W)	Pre-infusion (within 30 minutes prior to infusion)
5th dose	Day 57 (Q2W)/Day 85 (Q3W)	Pre-infusion (within 30 minutes prior to infusion)

7th Dose	Day 85 (Q2W)/Day 127 (Q3W)	Pre-infusion (within 30 minutes prior to infusion)
10th Dose and subsequent doses (every 3 treatment cycles)	Day 127 (Q2W) and post-Day 127 / Day 190 (Q3W) and post-Day 190	Pre-infusion (within 30 minutes prior to infusion)

Detailed information on blood sample collection, processing, storage, and transfer for ADA evaluation are provided in the laboratory manual.

8.4. Biomarkers

8.4.1. Archived tumor samples and tumor tissue biopsies

In accordance with the inclusion criteria in Section 4.1.1, all subjects must provide archived or freshly obtained tumor tissue samples after a definite diagnosis, approximately 10 (at least 5 if approved by the medical monitor) unstained FFPE pathology slides (preferably freshly obtained tumor tissue samples).

Operating Requirements and Recommendations for Collection of Fresh Tumor Tissue Samples:

- Fresh tumor biopsies should be obtained preferentially from tumor tissues that are safely collected as determined by the investigator, rather than from the sites that are obtained by major-risk surgery (e.g., brain biopsy) . Tumor lesions for fresh biopsy should not be used as RECIST v1.1 target lesions unless no other lesions are suitable for biopsy and the lesion used for biopsy is ≥ 2 cm in length. If target lesions are used for biopsy, a biopsy is recommended prior to imaging that meets screening requirements.
- Fine needle aspirate biopsy or burr biopsy samples are often insufficient for biomarker testing, and this method should be carefully selected to ensure that tumor tissue is adequate for biomarker testing.
- Cell smears centrifuged from pleural/abdominal/pelvic/pericardial effusion drainage, bone lesions without soft tissue components or decalcified bone tumor samples are not acceptable.

Study Cohort Subject Collection Requirements:

- Subjects from Cohorts 2, 3, 4 and 5 of Phase Ib and Cohort D of Phase II must provide archived or freshly obtained tumor tissue within 6 months prior to the first dose.
- Subjects from Cohort A of Phase II will provide freshly obtained tumor tissue samples whenever possible (preferentially at the recurrent or metastatic phase).
- For the subjects in Cohort 1 from Phase Ib and Cohorts A, B and C from Phase II, it

is necessary to discuss with the medical monitor, and then they can be exempted from collection of archived tumor tissue specimens if archived tumor tissue specimens are not available and the subjects have great risk for a fresh biopsy as determined by the investigator.

One of the exploratory objectives of this study is to explore biomarkers in tumor tissues that potentially predict response to AK104, including but not limited to, PD-L1 expression, type and number of tumor infiltrating cells (TIL), and detection of mRNA expression profiling of immune-related genes. Fresh or archived tumor tissue samples provided by subjects will be used for exploratory studies to determine IHC of PD-L1, determine the type and number of tumor-infiltrating cells (TIL) by multiplex immunohistochemistry (mIHC), and detect mRNA expression level of immune-related genes.

Subjects may be encouraged to voluntarily participate in optional biomarker studies. To obtain exploratory biomarkers in tumor tissues during the disease progress or response, including but not limited to the detection of immune-related gene characteristics and changes in the number of CD8+ effector T cells; For subjects participating in optional biomarker studies, fresh tumor biopsies will be performed at the time of response assessment of tumor response (PR/CR) and/or progressive disease (PD) if tumor lesion tissue is clinically available，and it is obtained subject's informed consent. Tumor lesions scheduled for biopsy should not be used as target lesions for disease assessment. Subjects may withdraw informed consent at any time and from optional biomarker studies at any time. The shipment, storage, and handling of archived tumor samples, fresh tumor samples, and remaining tumor tissues for biomarker assessment will be managed by the central laboratory. Refer to the laboratory manual for details of sample handling.

9. Safety Assessments

9.1. Adverse event

At each follow-up, subjects will be evaluated by the investigator to determine if an AE has occurred. At each study follow-up, adverse events may be observed directly, reported voluntarily by the subject, or questioned by the subject. Subjects should be asked questions about their general condition and not about any specific symptoms. The investigator must assess the clinical significance of all laboratory test results. All abnormal post-baseline laboratory test results considered clinically significant by the investigator must be recorded on the AE page of the case report form or electronic data capture (EDC).

After the informed consent form has been signed, but prior to the administration of study drug, only SAEs should be reported. After initiation of study treatment, all AEs and SAEs will be reported (regardless of relationship to study drug) until 30 days (AE) and 90 days (SAE) after the last dose of study treatment (including chemotherapy) or initiation of other anti-tumor therapy, whichever occurs first (whether observed by the investigator or reported by the subject), both will be recorded on the AE page of the case report form. Immune-related AEs (irAEs) , regardless of serious or nonserious irAEs, will be reported until 90 days after the last dose of AK104 (whether other anti-tumor therapy has been initiated). After this period, the investigator should report any SAEs that are assessed as related to AK104.

Any findings observed during screening assessments are part of the subject' s medical history and will be recorded as part of the medical history. AEs or SAEs are reported using diagnostic medical terms; when the diagnosis is not available, symptoms and signs are reported as separate entries for AEs or SAEs until a final diagnosis is obtained. For death, the cause of death should be reported as an SAE.

All SAEs should be reported according to the investigator' s SAE reporting process described in Section 9.2. In addition, SAEs occurring after the reporting period should also be reported and handled as SAEs if it is considered to be possibly related to study drug by the investigator. A pre-planned process or hospitalization for an existing condition without serious deterioration should not be reported as an SAE (refer to the definition in Section 9.1.1).

The investigator should follow up the subject for adverse events until resolution of the event or until the stability of condition. At the end of the treatment assessment, if unresolved AEs include significantly abnormal test results, these events will be followed up until the event has resolved to baseline, the event is assessed by the investigator as stable, the subject is lost to follow-up, the subject withdraws consent, or it is confirmed that study treatment or participation is not the cause of the adverse event.

9.1.1. Definition

9.1.1.1. Adverse Events (AEs)

An AE is an unexpected medical occurrence or worsening of an existing medical event after subjects receive the investigational product in a clinical study, and a clear causal relationship

to the investigational product is not necessarily inferred. Therefore, an AE can be any adverse effects and unintended sign (including for example, an abnormal laboratory finding), symptom, or disease associated with the use of the investigational product, whether or not considered related to investigational product (ICH E2A guideline) . Clinical Safety Data Management: Definitions and Standards for the Expedited Reporting System, October 1994).

Investigators are obligated to determine which conditions or abnormal laboratory findings should be AEs based on their knowledge and experience. AEs include, but are not limited to, the following:

- Worsening of pre-existing (before entering the clinical trial) medical condition/disease (including worsening of symptoms, signs, laboratory abnormalities);
- Any new adverse medical condition (including symptoms, signs, newly diagnosed diseases);
- Clinically significant abnormal laboratory values or results.

9.1.1.2. Definition of Serious Adverse Events

AEs occurring during any study phase (screening, treatment, and follow-up) and meeting one or more of the following criteria:

- Leading to death;
- Life-threatening (Note: The term “life-threatening” in the definition of “ serious” refers to a situation in which the subject is at risk of death at the time of the event; it does not refer to a situation in which hypothetically might have caused death if it is more severe);
- Hospitalization or prolongation of hospitalization (see Section 9.1.2 for exceptions);
- Resulting in permanent or serious disability/incapacity;
- Congenital anomaly/birth defect, or
- Other medically significant events.

Note: The term "life-threatening" in the definition of "serious adverse event" refers to the possible death of the subject at the time of the event; it does not refer to a possible death if the event is assumed to be further serious (ICH E2A guideline). Clinical Safety Data Management: Definitions and Standards for the Expedited Reporting System, October 1994).

Medical and scientific judgments should be made in deciding whether expedited reporting is appropriate in other circumstances, for example, the significant medical event may not be immediately life-threatening, or may result in death, or require hospitalization but may jeopardize the subject, or may require intervention to prevent other consequences listed in the above definition. Examples include allergic bronchospasm, convulsion, and blood cachexia, or drug dependence or drug abuse.

Note:

- A process is not an AE or SAE, but the cause of the process may be an AE.
- Pre-planned surgery or hospitalization for an existing condition that did not worsen significantly is not an SAE.

9.1.1.3. Adverse event of special interest

Adverse events of special interest (AESIs) are events of special scientific and medical interest that contribute to understanding the safety profile of the investigational product AK104, and require close observation by the investigator and rapid communication with the sponsor/CRO. AESIs may or may not be serious. The expedited reporting of AESIs (only to the sponsor according to the procedures described in Section 9.2) is ongoing monitoring of these events to characterize and understand their relevance to the use of the investigational product.

AK104 is a humanized IgG1 bispecific antibody that efficiently binds both PD-1 and CTLA-4. Potential risks based on mechanism of action are associated with AK104, which activates the immune system. AESIs in this study include, but are not limited to, events with underlying inflammatory or immune-mediated mechanisms and that may require more frequent monitoring and/or intervention, such as treatment with steroids, immunosuppressants, and/or hormone replacement therapies.

Based on similar drugs and AK104 data, ≥ Grade 3 immune-related AEs (including but not limited to colitis, dermatitis, pneumonia, myocarditis, myositis, hepatitis, encephalitis, nephritis, endocrine disorders, and immune thrombocytopenia), ≥ Grade 3 infusion-related reactions and systemic allergic reactions/serious type I hypersensitivity are considered as AESIs, which are required to be reported as soon as possible in accordance with this study (only to the sponsor according to the procedures described in Section 9.1.2).

9.1.1.4. Immune-related adverse event

An irAE is defined as an AE which is associated with drug exposure and consistent with an immune-mediated mechanism of action, with no clear alternative etiology. Appropriate measures should be taken to rule out tumors, infections, metabolism, toxins, or other causes that cause suspected irAEs. Serologic, immunologic, and histological tests should be used to support the diagnosis of irAEs, as appropriate.

Based on the mechanism of action of AK104 leading to T cell activation and proliferation, it is possible to observe irAEs during the conduct of this study. Overlapping or more severe irAEs may occur when anti-PD-1/L1 or anti-CTLA-4 drugs are used alone. Potential irAEs may be similar to those seen with anti-PD-1/L1 or anti-CTLA-4 drugs, which may include but are not limited to immune-mediated colitis, dermatitis, pneumonia, hepatitis, encephalitis, nephritis, and endocrine disorders. For potential risks associated with AK104, pleease refer to 1.3.1. Patients should be monitored for signs and symptoms of irAEs. In the absence of an alternative etiology (e.g., infection or PD), an immune-related etiology associated with clinical signs or symptoms should be considered. Guidelines for the management of subjects with irAEs are described in Section 3.1.4.

9.1.1.5. AE severity

The investigator shall assess the severity of each AE and SAE reported during the study. All AEs and SAEs will be evaluated and graded according to the NCI-CTCAE v5.0. Toxicities that are not specified in the NCI-CTCAE v5.0 will be graded according to the following definitions.

- Grade 1, mild: asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
- Grade 2, moderate: smaller, non-invasive or local treatment is required; age-related instrumental activities of daily living are limited.
- Grade 3, severe: serious or medically significant but not immediately life-threatening; leading to hospitalization or prolongation of hospitalization; disabling; limiting self-care activities of daily living;
- Grade 4: life-threatening; urgent intervention indicated;
- Grade 5, AE-related death.

Note: Instrumental activities of daily living refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

Self care activities of daily living refer to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden, etc.

9.1.1.6. Causality assessment

Based on the causality judgment criteria of drugs and adverse events, the relationship between adverse events and investigational product will be classified into five grades, i.e., definitely related, possibly related, unlikely related, definitely unrelated, not determined. Of these, definitely related, possibly related and not determined are assessed as adverse drug reactions. Refer to the table below for specific judgments. The relationship between AEs and AK104 will be determined by the investigator based on his/her clinical judgment and the definition in Table 9.1:

Table 9-1 Causality Judgment Criteria of Adverse Drug Reactions

Grading	Judgment criterion
Definitely related	The event occurs in a reasonable chronological order after administration, and the event is consistent with the known type of reactions of the suspected investigation product; the event is improved after drug withdrawal and the event reappears after repeated dose.
Possibly related	The event occurs in a reasonable temporal relationship after administration, and the event is not consistent with the known type of reactions of the suspected investigation product; the subject's clinical status or other treatment regimens may also have contributed to the event.
Unlikely related	The event does not occur in a reasonable chronological order after administration, and the event is not consistent with the known type of reactions of the suspected investigation

Grading	Judgment criterion
	product; the subject's clinical status or other treatment regimens may also have contributed to the event.
Definitely unrelated	The event does not occur in a reasonable chronological order after administration; the event is not consistent with the known type of reactions of the suspected investigation product; the subject's clinical status or other treatment regimens may also have contributed to the event; the event is resolved when the disease state is improved or the other treatments are discontinued, and reappears when the other treatments are resumed.
Not evaluable	The event does not occur in an definite temporal relationship after administration; the event is similar to the known reaction type of the investigational product, and other drugs used concomitantly may also have caused the corresponding event.

When an SAE occurs, the investigator' s initial report to the sponsor/CRO safety team may contain very little information. However, it is important for the investigator to assess the causality of each event before sending the SAE case report form to the sponsor/CRO safety team. The investigator may modify his/her opinion on causality based on subsequent information and modify the SAE case report form accordingly. The causality assessment is one of the criteria used when determining regulatory reporting requirements.

9.1.1.7. Actions Taken with Study Drug

1 = None: No change in the administration dose of study drug.

2 = Permanent discontinuation: Permanently discontinue the study drug.

3 = Dose reduction: Reduce the dose of study drug (AK104 not being allowed in this study).

4 = Interruption: Temporarily discontinue the study drug.

5 = Not applicable.

9.1.1.8. Outcomes of Adverse Events

1 = Recovered/Resolved

- The subject fully recovers from the AE and no sequelaes are observed.

2 = Recovered/Resolved with sequelae

- AEs are ongoing with sequelaes.
- With definite sequelae/residual effects.

3 = Not recovered/Not resolved

- The AE itself is ongoing and still observable.

4 = Fatal.

5 = Unknown.

9.1.1.9. Other Actions Taken for the Event

1 = None.

- No treatment required.

2 = Medical treatment required.

- Prescription and/or over-the-counter medication is required for this AE.

3 = Hospitalization or prolongation of hospitalization required.

- Hospitalization or prolongation of hospitalization is required due to AEs (regardless of need for medication).

4 = Other.

9.1.2. Recording the Adverse Events

The investigator is required to fully document any adverse event, including a diagnosis (if not diagnosed, symptoms and signs are recorded including laboratory abnormalities), start and stop dates, seriousness grades of CTCAE and changes, whether the event is serious adverse event, whether the event is adverse event of special interest, actions taken with study drug, treatment given due to the AE, and outcome of the event, the causal relationship between the adverse event and the study drug judged by the investigator and whether the AE leads to withdrawal of the subject from study treatment.

For serious adverse events, the investigator should also provide the date of the AE that met the criteria for an SAE, the date of the SAE that the investigator learned of, the rationale for the AE being an SAE, date of hospitalization, date of discharge, possible cause of death, date of death, whether an autopsy is performed, causality assessment with the study procedure, causality assessment with other drugs, and other possible cause of SAEs. The investigator should also provide the rationale for judging the relevance and the description of SAEs. In the description of SAEs, it is also necessary to include the subject's number, age, gender, height and weight; the indication for treatment with the investigational drug and disease stage of the subject and relevant systemic conditions; and the clinical course such as the occurrence, development, outcome and consequence of the SAE; laboratory test results related to SAEs (test time, units and normal range must be provided); SAE-related past history, concomitant diseases, occurrence and duration, etc.; SAE-related medication history, concomitant drugs and their start, duration of treatment, dosage and administration, etc.; details of initiation, duration, dosage and administration of the investigational product treatment.

Items regarding the recording of AEs are described as follows:

Diagnosis, Symptoms and Signs

If a diagnosis has been made, the diagnosis should be recorded on the case report form rather than the individual symptoms and signs (for example, liver failure should be recorded, rather than jaundice, elevated transaminase, and asterixis). If it is not determined that symptoms and

signs are caused by the diagnosis at the time of reporting, they will be recorded as individual AE/SAE. If it is determined that symptoms and signs are caused by the diagnosis, only the diagnosis is reported individually, and symptoms and signs are included in the diagnosis. Records of symptoms and signs should be deleted for AEs, and follow-up update reports should be sent for SAEs.

Adverse Events Secondary to Other Events

In general, primary events should be recorded for adverse events secondary to other events, such as those caused by other events or those that are clinical sequelae, unless the secondary event is severe or serious adverse events. However, a secondary event with significant clinical relevance should be recorded as an independent adverse event in the case report form if its time of occurrence is different from the primary event. If the association between events is unclear, the event should be recorded separately in the case report form.

Persistent or Recurrent Adverse Events

A persistent adverse event refers to an adverse event that persists and is not alleviated between two assessment time points of the subject. Such adverse events should be recorded only once on the case report form. The initial severity of the event should be recorded and then updated in case the event is aggravated to record the highest severity of the event.

A recurrent adverse event refers to an adverse event that has been alleviated between two assessment time points, but subsequently recurs. The occurrence of the event should be recorded separately in the case report form.

Laboratory Abnormalities

Abnormal laboratory values or other abnormal assessments that are judged by the investigator to be clinically significant during the study, or that are present at baseline but significantly worsen after the start of the study, will be reported as AEs or SAEs. It is the responsibility of the investigator to review all laboratory abnormalities and to make medical judgment on whether each laboratory abnormality should be reported as an AE.

Clinically significant laboratory abnormalities or other abnormal assessments related to the disease under study will not be reported as AEs or SAEs unless judged by the investigator to be more severe than expected for the subject's condition, or if present or detected at the start of the study without worsening. It should be reported as an AE or SAE if it causes clinical signs or symptoms, requires active intervention, requires dose interruption or discontinuation, or is considered clinically significant by the investigator.

If a laboratory value or vital sign worsens with clinical signs and symptoms, such signs or symptoms will be reported as AEs and concomitant laboratory results or vital signs will be considered additional information. The reporting investigator should use clinical terms rather than laboratory terms whenever possible (e.g., anemia rather than low hemoglobin values). In the absence of clinical signs or symptoms, clinically significant worsening of non-specified parameters should be reported as AEs.

Death

All deaths occurring during the entire study, including that during the 90-day follow-up period after the last dose, or before the initiation of other anti-tumor therapy, whichever occurs first, whether related to the study drug or not, should be recorded in the death report form in the case report form and reported to the sponsor as SAEs in a timely manner.

At the time of recording an event of death, the cause of death will be recorded as an adverse event if the cause of death is clear, the outcome of the adverse event will be recorded as death and the event will be reported as an SAE; if the cause of death is unknown at the time of reporting, it should be recorded as "unexplained death" in the AE form of the CRF, and be reported as an SAE of "unexplained death" before further investigation of its exact cause.

If the death is clearly caused by tumor progression, it should be recorded on the death report form of the case report form and be notified to the sponsor in a promptly manner, by referring to the general SAEs reporting timelines and requirements for recording and reporting.

Pre-existing Medical Conditions

Symptoms/signs that have been present in the screening period of the study should be recorded and reported as adverse events only if there is an increase in severity, frequency, or a deterioration in nature (excluding deterioration of the disease under study). Changes relative to pre-existing conditions, such as "increase in headache frequency", should be reflected in the record.

Hospitalization, Prolongation of Hospitalization, or Surgery

Any adverse event that results in hospitalization or prolongation of existing hospitalization should be recorded and reported as an SAE, with the following exceptions:

- Hospitalization or prolongation of existing hospitalization scheduled according to protocol (e.g., for drug administration, efficacy assessment, etc.);
- Hospitalization due to medical condition that has been present prior to study participation but remains unchanged. For example, elective surgery/treatment that has been scheduled prior to study participation;
- Time of visit to an emergency department or other departments in the hospital is < 24 hours and the subject is not hospitalized (unless it is a significant medical or life-threatening event);
- Hospitalization for scheduled medical procedures/surgeries as per the study protocol;
- Routine medical examinations (e.g., routine colonoscopy) that require hospitalization to evaluate the baseline level/trend of health status;
- Medical/surgical hospitalizations that have been predefined prior to study entry, which are not intended for salvage therapy. However, these conditions need to be

documented;

- Hospitalization due to other conditions which are unrelated to health status and need no medical/surgical interventions (e.g., no housing, financial difficulty, temporary lack of caregivers, family environment, or management reasons);
- Hospitalization for anti-tumor therapy without any other SAEs.

However, if the pre-existing disease deteriorates during the study (e.g., requiring surgery/treatment earlier than originally planned), then the elective surgery/treatment that becomes necessary due to the deterioration will be considered as an adverse event.

Progressive disease

Progressive disease is defined as worsening of the subject's condition due to the primary tumor to which the study drug is directed. The appearance of a new lesion relative to the primary tumor, or progression of an existing lesion is considered progressive disease. In general, disease progression is not recorded as an AE or reported as an SAE, and death due to symptoms and signs of disease progression that occur within 90 days of the last dose or before the initiation of other anti-tumor therapy, whichever occurs first, should be reported with reference to the SAEs.

New tumor

The occurrence of a new primary tumor should be considered an SAE. A new primary tumor refers to a tumor that is not the main reason for the use of study drug and is found after the subject is enrolled in the study.

9.2. Reporting the Serious Adverse Events - Procedures for Investigator

9.2.1. Initial Report

The reporting of SAE should be made in accordance with regulatory requirements:

- The investigator shall fill in the Collection Form for Severe Adverse Events provided by the sponsor, sign and date it, and report it to the sponsor or its representative via mail within 24 hours. Specific contact information for reporting is listed in the Investigator File provided to each site. The original SAE report form and the sponsor’s acknowledgement email should be kept at the study site along with the case report form file.
- If the investigator does not have all information regarding an SAE, he/she is not to wait to receive additional information before notifying the sponsor or designee of the SAE and completing the form. The form will be updated when additional information is received.
- The investigator should provide other information (autopsy report, medical records of termination phase, and other necessary information) on SAEs (including reported deaths or adverse drug reactions) as required by the sponsor, medical institution, and

ethics committee.

- The sponsor should promptly study the SAEs with the investigator, take necessary measures to ensure the safety, rights and interests of the subjects, report them in a timely manner to drug regulatory authorities, and notify other investigators of clinical studies involving the same drug.
- All SAEs should be followed up as far as possible until resolution or improvement to baseline, subject death, loss of contact, or final confirmation by the investigator that the SAE is not related to study treatment.

All serious adverse events occurring during the entire study, from signing the informed consent to 90 days after the last dose (including chemotherapy) or initiation of other anti-tumor therapy, whichever occurs first, should be recorded. Investigators are required to report all serious adverse events related to study treatment regardless of whether they occur after the study. In addition, if there are unresolved adverse events or abnormal laboratory results considered related to study treatment, the subjects will continue to be followed up until resolution or return to baseline status. If a serious adverse event occurring outside the above-mentioned period is considered study drug-related, it should also be reported to the sponsor.

9.2.2. Follow-up Report

Follow-up information should be filled in a new SAE report form, and be identified as a follow-up report. Upon receipt of new information on the reported SAE, the follow-up report is uploaded within the specified time and the subsequent reporting process is the same as the initial reporting process specified in Section 9.2.1.

9.2.3. Notification of drug regulatory authorities, health authorities, investigators, clinical trial facilities and ethics committees

In accordance with national/regional regulations, the sponsor will promptly notify investigators, clinical trial facilities, ethics committees, drug regulatory authorities, and health authorities of any SUSARs.

In accordance with all applicable regulations, the investigator/sponsor reports SUSARs to the Institutional Review Board (IRB)/Independent Ethics Committee (IEC).

9.3. Other Events Requiring Immediate Reporting

9.3.1. Overdose

An overdose is defined as the dose of study drug received by the subject exceeds 20% of the dose specified in the protocol.

- AE-related overdose is recorded as AE diagnosis/symptoms on the respective AE module and the concomitant medications module in the case report form, or by using other corresponding recorded overdose information.

- Events of asymptomatic overdose are reported only on the concomitant medication module or other appropriate documentation of overdose information is performed.

All overdoses with and without AEs must be reported electronically or in writing to the Sponsor/CRO by the Investigator or other personnel in the study center (when this person becomes aware of the event) via EDC within the specified time. For applicable timeline for reporting of SAE-related overdose events, please refer to Section 9.2.

9.3.2. Concurrent Elevation of Transaminases and Bilirubin

Concurrent elevation of aminotransferases and bilirubin, whether serious or nonserious, or causally related, if in accordance with the underlying Hy's law (ALT or AST ≥ 3 × ULN and TBil ≥ 2 × ULN), should be reported to the Sponsor/CRO via EDC or SAE in written format with the investigator's judgment and detailed description of severity and causality. These events should be reported as soon as possible in accordance with the SAE reporting process specified in Section 9.2. The site will promptly initiate follow-up investigations, laboratory tests, detailed medical history collection and inquiries to determine if the results are reproducible and/or if there is objective evidence that clearly supports the results are caused by disease (e.g., gallstones and bile duct obstruction with gallbladder dilatation) or other medications other than study drug. Further testing may include testing for acute hepatitis A, B, C, and E and imaging of the liver.

9.3.3. Other Adverse Events of Special Interest

AEs of immune-related reaction ≥ Grade 3, infusion-related reaction ≥ Grade 3, and systemic anaphylaxis/serious type I hypersensitivity (as defined in Section 9.1.1.3) are required to be reported as required in Section 9.2.

9.3.4. Reporting and Follow-up of Pregnancy during Clinical Study

Any subject who becomes pregnant while on study drug or within 120 days of discontinuation of study drug must notify this to the Sponsor or Drug Safety Associate of the CRO. Although pregnancy is not technically an AE, all pregnancy events must be tracked to determine their outcome. This information is important for both drug safety and public health issues. If any subject becomes pregnant during the study, the investigator or designee is obligated to report the in-process event using the Pregnancy Report Form. The investigator should make every effort to track the subject until the end of pregnancy. If the outcome of the pregnancy meets the criteria for immediate classification as an SAE (i.e., postpartum complication, spontaneous abortion, stillbirth, neonatal death, or congenital anomaly, including abnormalities of aborted fetuses), the Investigator should follow the procedures for reporting an SAE. If a female subject becomes pregnant during the clinical trial, the subject will be excluded from the study. When the investigator becomes aware that the spouse of a male subject is pregnant, the investigator must follow up the pregnancy as much as possible. The investigator should promptly report all pregnancies to the sponsor.

9.4. Clinical Laboratory Evaluation

Safety laboratory assessments will include hematology test, blood chemistry test, coagulation test, thyroid function tests, urinalysis, pregnancy tests, etc. Refer to Section 6 for test schedule.

- Coagulation: prothrombin time (PT), international normalized ratio (INR), and activated partial thromboplastin time (aPTT).
- Hematology: including hemoglobin (Hb), packed cell volume (PCV), white blood cell count differential (WBC, including absolute count), red blood cell (RBC), blood platelet count (BPC), mean corpuscular hemoglobin concentration (MCHC) and mean corpuscular volume (MCV).
- Blood chemistry: calcium, phosphorus, magnesium, albumin (ALB), glucose (GLU), serum creatinine (Cr), uric acid (UA), total protein (TP), blood urea nitrogen (BUN) or urea (Urea), aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphate enzymes (ALP), total bilirubin (TBil) and direct bilirubin (DBil), glutamyltransferase (GGT), lactate dehydrogenase (LDH), sodium, potassium, chloride, triglycerides, cholesterol, and serum C-reactive protein (CRP), amylase and lipase.

(Note: For blood chemistry: tests for AST, ALT, ALP, and TBil must be performed concurrently and evaluated concurrently.)

- Thyroid function: TSH, free T4 and free T3.
- Items of cardiac function tests should include creatine kinase (CK), creatine kinase isoenzyme (CK-MB), cardiac troponin I (cTnI), cardiac troponin T (cTnT), brain natriuretic peptide (BNP), and/or N-terminal pro-brain natriuretic peptide (NT-proBNP) as far as possible. Additional cardiac function tests may be performed at the discretion of the investigator, depending on the subject's clinical conditions.
- Urinalysis: including acidity and alkalinity, protein, specific gravity, glucose, bilirubin, ketones, occult blood and microscopic examination (including, if necessary, white blood cells and red blood cells in urine). If the subject shows proteinuria 2+ or higher on urine dipstick or routine analysis, a 24-hour urine collection should be provided for further analysis.
- Serum β-hCG pregnancy test or urine hCG pregnancy test.
- Hepatitis B virus screening includes HBsAg, HBsAb and HBcAb (HBV DNA test if necessary) and hepatitis C antibody (qualitative hepatitis C RNA test if the antibody test is positive). Treatment period and end of treatment visits: HBV-DNA in patients with HBV infection; and HCV-RNA in patients with HCV infection.
- Human immunodeficiency virus antibody (HIV-Ab) (at screening only).

The investigator must assess the clinical significance of all laboratory results. All abnormal laboratory test results considered clinically significant by the investigator should be recorded on the AE page of the case report form. If an SAE is constituted by abnormal laboratory test results, the SAE should be reported via EDC or written SAE form and other relevant procedures should be followed up (see Section 9.2). Abnormal laboratory test results (NCI CTCAE Grade 3 or 4) that occur during clinical studies should be followed up until the repeated test results return to normal (baseline), stabilize, or are no longer clinically significant.

9.5. Vital signs

Vital sign including systolic blood pressure, diastolic blood pressure, pulse rate, respiratory rate, and body temperature will be measured during the study day as scheduled in Section 6. During the treatment period, vital signs will be measured within 30 minutes prior to initiation of AK104 administration, at the 30th (± 5) minute within 1 hour of infusion, at the end of infusion (+ 10 minutes), and 30 (± 10) minutes after the end of infusion, and before oxaliplatin infusion. On Day 1 of Cycle 1, additional vital sign measurements will be performed 3 hours (± 30 minutes) after the end of the first infusion. For subsequent dosing, a 3-hour observation period was not required unless the subject experienced an infusion-related reaction. If a subject experiences an infusion-related reaction, additional vital sign measurements will be performed 3 h after infusion and 3 h after each subsequent infusion. Additional monitoring and vital sign assessments will be performed at the discretion of the investigator as per standard clinical practice or clinical indications.

9.6. Electrocardiography and Echocardiography

- **Electrocardiogram(ECG)**

A 12-lead ECG will be recorded for all subjects during the study days described in Section 6. ECGs will be performed prior to PK blood sampling, as applicable. It is recommended that the subject has rested in the supine position for at least 5 minutes before taking a 12-lead ECG. All ECGs will be reviewed by the Principal Investigator or designated physician. If any clinically significant condition is observed on the ECG, it will be recorded as an AE by the investigator on the eCRF, which represents a change from baseline in the result. The original ECG tracing results for each timing 12-lead ECG will be saved in conjunction with the subject’s original file. At least HR and QRS, QT and PR intervals will be recorded on the eCRF. A digital copy of the ECG will be retained in the subject's original file and will be provided on request.

- **Echocardiography (ECHO)**

Echocardiography is required at screening whenclinically indicated to assess cardiac function such as left ventricular ejection fraction (LVEF). For the purpose of safety monitoring, the investigator must review all echocardiogram reports. Echocardiography reports will be retained in the subject's original files at the study site and any echocardiography abnormalities must be recorded in the eCRF.

9.7. Physical examination

Physical examinations will be performed according to the schedule in Section 6. A complete physical examination will be performed at screening, including general condition, skin, mucosa, lymph nodes, head and its organs, neck, chest, abdomen, extremities, nervous system, etc. A targeted physical examination based on symptoms should be performed during treatment and prior to study drug administration.

9.8. ECOG performance status

ECOG performance status will be determined at screening and then at each scheduled follow-up visit throughout the study (see Section 6). ECOG performance status will be assessed at the time specified in the Assessment Schedule based on the following content.

Performance status score	Performance status
0	Fully active, able to carry on all pre-disease performance without restriction.
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work.
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	Capable of only limited selfcare, confined to bed or chair for more than 50% of waking hours.
4	Completely disabled; cannot carry on any self-care; totally confined to bed or chair.
5	Death.

10. Other Assessments

NA.

11. Statistical Methodology

11.1. Analysis Sets

11.1.1. Enrollment Analysis Set

The Enrollment Analysis Set includes all subjects who have signed the ICF and been enrolled into the study.

11.1.2. DLT Evaluable Set

The DLT evaluable set includes: 1) subjects who are enrolled in dose escalation Cohorts 1, 2, 3 and 4 and experience a DLT within 28 days after the first dose, or who do not experience a DLT but have received at least 80% of the expected administration dose of AK104 alone or AK104 combined with mXELOX according to the protocol in the first 28 days (2 cycles, and every 2 weeks as a cycle) and have completed the safety follow-up through a 28-day DLT evaluation period; 2) subjects who are enrolled in dose escalation Cohort 5 and experience a DLT within 21 days after the first dose, or who do not experience a DLT but have received at least 80% of the expected administration dose of AK104 alone or AK104 combined with XELOX according to the protocol in the first 21 days (1 cycle, and every 3 weeks as a cycle) and have completed the safety follow-up through a 21-day DLT evaluation period.

11.1.3. Full Analysis Set

The full analysis set (FAS) includes all subjects enrolled in the analysis set who have received any doseof AK104 and have baseline disease assessment data with measurable disease (defined as per RECIST v1.1 criteria). The FAS for the analysis of efficacy endpoints assessed by IRRC must be confirmed by the IRRC as having measurable lesions at baseline; the FAS for the analysis of efficacy endpoints assessed by the investigator must be confirmed by the investigator as having measurable lesions at baseline.

11.1.4. Safety Analysis Set

The safety analysis set includes all enrolled subjects who have received any doseof the study drug and undergone at least one safety assessment. Analysis is based on the actual treatment received by the subject.

11.2. General Statistical Considerations

Appropriate descriptive statistics will be performed for selected population, safety, PK, PD, biomarkers, immunogenicity, and response data based on dose and time. Descriptive statistics for continuous data includes mean, median, standard deviation, and range, and categorical data are summarized using frequency counts and percentages at the same time. Data summaries may be present in graphs.

Changes from baseline to post-treatment or ratios of post-treatment to baseline are assessed only for those subjects who have both baseline and post-treatment measurements. Unless otherwise stated, the last non-missing value for a variable taken before the first dose of study

drug will be used as the baseline value. Missing or out-of-control data are generally not included in data analysis.

Safety analysis will be performed based on the safety analysis set and response-related efficacy endpoints will be analyzed on the basis of the full analysis set. The safety data of the dose escalation cohort in the dose escalation and expansion period (phaseIb) of this study will be pooled and summarized according to the dose group. In addition, safety data from the same dosing regimens in phaseIb and II studies will be pooled and summarized.

A detailed statistical analysis plan (SAP) describing the methods used in the final analysis should be prepared and completed prior to database lock. The statistical methods described in the SAP may change with the progress of the study.

11.3. Study Population Data

For subjects enrolled in the analysis set, the treatment and reason for discontinuation of treatment and study interruption will be summarized and listed.

Demographic and baseline characteristics such as age, sex, baseline ECOG performance status, prior therapy methods, and best response to prior therapy will be summarized for the enrollment analysis set, the full analysis set, and the safety analysis set. If 2 analysis sets are identical in part of the study, only one of them will be present in the table.

Study drug exposure, treatment duration, and compliance with study treatment will be summarized for the safety analysis set using descriptive statistics.

11.4. Efficacy Analysis

The following efficacy endpoints will be derived. Please refer to SAP for details.

- Objective response rate (ORR) is defined as the proportion of subjects achieving a best overall response (BOR) of confirmed CR or PR per RECIST v1.1 criteria. Best overall response is defined as the best response recorded from the start of treatment until progressive disease (sequentially CR, PR, SD, PD, or not evaluable), or best response recorded (from the start of treatment to assessment of the last evaluable tumor imaging) if PD is not achieved prior to initiation of subsequent anti-tumor therapy or study discontinuation, whichever occurs first. The best overall response requires confirmation if it is CR or PR , which is defined as confirmation of CR or PR response by repeated imaging assessments performed not less than 28 days (4 weeks) after the first observation of a CR or PR response, and there is no evidence of PD between the follow-up of the first observed CR or PR and the follow-up for confirmation of CR or PR. Patients who do not have a post-baseline response assessment (for any reason) will be considered non-responders in the best overall response assessment.
- Duration of response (DoR) is defined as the period from the first documentation of confirmed response (CR or PR) to the first documentation of progressive disease (as per RECIST v1.1 criteria) or death due to any cause, whichever occurs first. Subjects who are alive and have no progressive disease at the time of analysis data cutoff will be censored on the date of last tumor determination. DoR will be evaluated only for

subjects who have a confirmed CR or PR.

- Disease control rate (DCR) is defined as the proportion of subjects achieving a best overall response of confirmed CR or PR or SD per RECIST v1.1 criteria. Disease control rates at week 12 and 24 are defined as the proportion of subjects achieving best overall response of confirmed CR or PR or SD for at least 12 weeks and for at least 24 weeks respectively.
- Progression-free survival (PFS) is defined as the period from the start of treatment until documentation of progressive disease (per RECIST v1.1 criteria) or death due to any cause, whichever occurs first. Subjects who are alive and have no progressive disease at the time of analysis data cutoff will be censored on the date of last tumor assessment. Subjects without post-baseline assessments will be censored on the date of first dose. The 6-month PFS (6M PFS%) is defined as the proportion of subjects who have no progressive disease within 6 months after the first dose.
- Overall survival (OS) is defined as the period from the start of treatment until death due to any cause. Subjects who are alive at the time of analysis data cutoff will be censored on the last known survival date. Subjects who do not provide any follow-up information will be censored on the day of enrollment. The 1-year survival rate (1YR OS%) is defined as the proportion of subjects alive within 1 year after the first dose.

The primary endpoint of this study's dose confirmation period (phase II) is the ORR assessed per RECIST v1.1. The analysis of the primary endpoint will be performed after all subjects who meet the statistical and radiographic RECIST v1.1 assessment requirements are withdrawn from the study or complete at least 24 weeks of follow-up (if the subject have achieved PD before Week 24 , he/she will needn't to wait for a follow-up assessment at week 24). The analysis for ORR and DCR will be based on the full analysis set (FAS). For Cohort A in Phase II, its exact 2-sided 95% confidence interval can be estimated using the Clopper Pearson method and the exact 2-sided 80% confidence interval for the other cohorts can be estimated using the Clopper Pearson method. Time to event endpoints (DoR, TTR, PFS and OS) will be analyzed using the Kaplan-Meier method. PFS and OS analysis will be performed based on the safety analysis set. The graphical analysis will include a spider plotsof the tumor burden percentage of the target lesions changing from baseline over time, and a waterfall plots of the best percentage of the target lesions tumor burden changing from baseline.

11.5. Pharmacokinetics/ Pharmacodynamics /Immunogenicity Analysis

11.5.1. Pharmacokinetic Analysis

Individual AK104 concentrations will be tabulated by dose cohort along with descriptive statistics. Noncompartmental PK data analysis and planned PK sample collection will be performed for each dose cohort where data is available. The relevant descriptive statistics of the PK parameters of the non-compartmental model may include AUC, C_{max}, CL, and $t_{1/2}$. Linearity of PK parameters (AUC and C_{max}) will be explored.

A population PK model will be established in ths trial. The impact of physiologically relevant subject characteristics (covariates) and disease characteristics on PK will be evaluated. The relationship between exposure and response or safety will be evaluated. Such analysis can be

described in PK/PD analysis plans for different populations, and the result report of this exploratory analysis can be separated from the main study report.

11.5.2. Pharmacodynamic Analysis

The frequency change of CD4+ Ki67+ (as primary pharmacological biomarker) proliferative cell from baseline over time is analyzed with dose cohort and scheduled evaluation time. Other pharmacodynamic markers such as PD-1 or CTLA-4 receptor occupancy and CD8+ Ki67+ will be analyzed similarly. The relationship between exposure and pharmacodynamic biomarkers will be explored.

11.5.3. Immunogenicity Analysis

The immunogenic potential of AK104 will be assessed by summarizing the number and percentage of subjects who develop detectable ADA. The impact of ADA on PK will be assessed if data is available. Samples will be collected for evaluating the neutralizing capacity of ADAs in the future. Only subjects who have received at least one dose of AK104 and provided a baseline sample and at least one post-treatment sample will be evaluated.

11.5.4. Biomarkers and Exploratory Analysis

The relationship between PD-L1 expression as a predictive biomarker and efficacy in the archived tumor tissue samples or baseline fresh biopsy samples will be analyzed. PD-L1 expression will be assessed by the central laboratory for the determination of IHC of PD-L1. For Cohort A in Phase II, the ORR of subjects with PD-L1 positive (CPS ≥ 1) and PD-L1 negative (CPS < 1) squamous/adenosquamous carcinoma of the cervix will be assessed by IRRC based on RECIST 1.1.

If additional tumor tissue samples and data are available, other possible predictive biomarkers in the tumor tissue samples will be explored and analyzed, including but not limited to the type and number of tumor infiltrating lymphaytes (TIL) and correlation analysis between immune-related gene mRNA expression characteristics with AK104 antitumor activity.

11.6. Safety Analysis

Safety analysis will be performed using the safety analysis set according to the treatment actually received by the subjects. Safety analysis usually summarizes descriptive information appropriately in a table by dose group. The number of DLTs identified in the DLT evaluable set will be summarized for each AK104 dose group in the Phase Ib dose escalation cohort. The MTD levels will be indicated in the summary results.

11.6.1. Analysis of Adverse Events

Treatment-emergent adverse events (TEAEs) are defined as AEs that occur during treatment (during or after the first dose of study drug up to 30 days [for non-SAEs] or 90 days [for SAEs] after the discontinuation of the study drug or initiation of other anti-tumor therapy, whichever occurs first) but do not occur before treatment; or an AE that re-emerges during the treatment period and is present at baseline but stops prior to treatment; or a persistent AE with

severe deterioration after treatment relative to that before treatment. Only treatment-emergent adverse events (TEAEs) are included in the summary tables.

AEs will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) and graded according to NCI CTCAE Version 5.0. The number and percentage of subjects reporting TEAEs will be tubulated by worst CTCAE grade, system organ class, and preferred term and be graded by dose group. The number and percentage of subjects reporting treatment-emergent SAEs and TEAEs/SAEs considered related to AK104 will also be tabulated in the same manner.

Listings by subject AE (including treatment-emergent AE) data should include, but not limited to, verbatim terms, preferred terms, system organ class, CTCAE grade, and relationship to study drug. Deaths, other SAEs, and other significant AEs including those that lead to permanent discontinuation of AK104 will be listed.

11.6.2. Clinical Laboratory Evaluation Analysis

Clinical laboratory results and changes from baseline will be summarized descriptively by scheduled evaluation time and dose group, including end-of-treatment follow-up and the maximum and minimum values after treatment.

Laboratory abnormalities will be graded according to NCI CTCAE version 5.0. A two-way frequency table of baseline and worst post-treatment values based on CTCAE grade will be shown in the transitional table of clinical laboratory tests. Clinical laboratory abnormality test results for grade 3 or 4 will be listed.

11.6.3. Analysis of Vital Signs

Clinical sign measurements and changes from baseline will be summarized descriptively by scheduled evaluation time and dose group, including end-of-treatment follow-up and maximum and minimum values after treatment.

11.6.4. Electrocardiogram Analysis

ECG parameters (PR, RR, QRS, QT, QT interval corrected for heart rate using the Bazetts formula [QTcB], and QTcF) will be summarized using descriptive statistics by planned evaluation time and dose group for actual values and changes from baseline, including end-of-treatment follow-up and maximum value after treatment. QTcF is recognized as the primary adjustment method for assessing cardiac safety in patients.

The incidence of outliers of maximum absolute QTcF and QTcB intervals (new value > 450 ms, new value > 480 ms, new value > 500 ms) and maximum absolute uncorrected QT intervals (new value > 500 ms) in all post-treatment evaluations and the incidence of outliers of maximum changes of QTcF and QTcB (>30 and >60 ms) from baseline in all post-treatment evaluations will be summarized by treatment group. A "new value" is defined as a QTc abnormal value that does not exist at baseline but occurs at least once in post-baseline ECG assessments. Listings of ECG data will be provided.

11.7. Interim Analysis

For Cohort A in Phase II, an interim analysis may be planned after at least 70 subjects have been enrolled and these subjects have completed at least 16 weeks of follow-up (interval between the data cutoff date and the first dose date ≥ 16 weeks), and the results of the interim analysis will likely be used to communicate with regulatory authorities. The Sponsor dose not plan to prematurely discontinue the study based on these efficacy or safety findings.

11.8. Dose Escalation Committee (DEC) and Safety Review

During the dose escalation and expansion periods (Phase Ib), the Dose Escalation Committee (DEC) will conduct ongoing safety oversight over the study and carry out planned reviews on a regular basis to safety and other relevant data. The DEC includes, at minimum, the study chair, the medical monitor, the Sponsor's representative and other Principal Investigators or their representatives (from each site with subjects enrolled in this study). The DEC will be responsible for making decisions on dose escalation and expansion. All decisions made by the DEC will be documented in writing and shared with all participating sites.

After initiation of enrollment of subjects in the dose confirmation period (Phase II) of this study and the formation of a safety review team, the team will replace the DEC for ongoing safety oversight over the study and carry out planned reviews on a regular basis to safety and other relevant data. The safety review team includes, at minimum, the study chair, medical monitors, sponsor representatives, and other medical oncologists. Following safety review and discussion, the sponsor will make all final decisions regarding any change in study conduct.

11.9. Sample Size Calculation

Approximately 372 subjects are planned to be enrolled to participate in the dose escalation and expansion period (about 158 subjects) and the dose confirmation period (about 214 subjects) of the study. More subjects may be required for the dose escalation and expansion periods if other dose escalation cohorts or dosing regimens are to be explored.

11.9.1. dose escalation period (Part 1 of Phase Ib)

For the dose-escalation cohorts, the number of subjects to be enrolled will depend upon the toxicities observed as the study progresses. Dose escalation will follow the traditional 3 + 3 design, and at least 3 evaluable subjects and up to 6 evaluable subjects are expected to be included at each dose level. If the MTD is not achieved in the lower dose level group, 3 to 6 evaluable subjects will be enrolled in the highest dose level group. Up to 24 evaluable subjects are expected to participate in the planned dose escalation cohort. Additional subjects may be required if additional dose levels or alternative treatment regimens are to be explored. Subjects who are not evaluable for DLTs will be replaced.

11.9.2. dose expansion period (Part 2 of PhaseIb)

For the dose group with no DLT observed or DLT observed in fewer than 2 out of 6 subjects

after tolerability observation for 3~6 subjects completing dose escalation, the group could be expanded to include additional subjects based on preliminary safety and efficacy data, after discussion with the sponsor and DEC. Up to 40 subjects (including subjects enrolled in the dose escalation period) will be included into the 6 mg/kg Q2W dose expansion group and 450 mg Q2W dose expansion group for Cohort 1, respectively, and up to 18 subjects (including subjects enrolled in the dose escalation period) will be included into the dose group for Cohort 2, 3, 4 and 5 respectively.

For Cohort 1 of subjects with advanced solid tumors (tumor species not limited) treated with single AK104 at 6 mg/kg Q2W or 450 mg Q2W, the sample size in the study will be determined by practical considerations. If the incidence of an AE is 5% or more, there will be at least 87% probability that this AE will be observed in 40 subjects treated with AK104 at 6 mg/kg or 450 mg Q2W.

For AK104 in combination with mXELOX/XELOX as first-line treatment for gastric or GEJ adenocarcinoma, assuming a sample size of 18, ORR is estimated based on the proportion of subjects expected to achieve CR or PR, and its 80% CI is estimated (Table 11-1).

Table 11-1 ORR and 80% CI Estimates in 18 Subjects

Number (%) of subjects achieving CR or PR out of 18 subjects	8(44.4)	9(50)	10(55.6)	11(61.1)	12(66.7)	13(72.2)
Lower limit of 80% CI	27.9%	32.9%	38.0%	43.3%	48.8%	54.5%
Upper limit of 80% CI	62.0%	67.1%	72.1%	76.9%	81.5%	85.8%

11.9.3. Dose confirmation period (Phase II)

It is planned to enroll 110 subjects with cervical cancer in Cohort A (to ensure that there are at least 100 subjects in the FAS set), 30 subjects with esophageal squamous cell carcinoma in Cohort B and 30 subjects with hepatocellular carcinoma in Cohort C for efficacy observation. The primary objective is to evaluate the efficacy of AK104 in the treatment of cervical cancer, esophageal squamous cell carcinoma, and hepatocellular carcinoma, respectively.

For cervical cancer subjects in Cohort A, assuming there are 100 subjects in the FAS set, ORR is estimated based on the proportion of subjects expected to achieve CR or PR, and its 95% CI is estimated (Table 11- 2).

Table 11-2 ORR and 95% CI Estimates in 100 Subjects

Number (%) of subjects achieving CR or PR out of 100 subjects	15(15)	17(17)	20(20)	25(25)	30(30)
Lower limit of 95% CI	8.6%	10.2%	12.7%	16.9%	21.2%
Upper limit of 95% CI	23.5%	25.8%	29.2%	34.7%	40.0%

For esophageal squamous cell carcinoma subjects in Cohort B and hepatocellular carcinoma subjects in Cohort C, assuming a sample size of 30, the ORR is estimated based on the proportion of subjects expected to achieve CR or PR, and its 80% CI is estimated (Table 11-3).

Table 11-3 ORR and 80% CI Estimates in 30 Subjects

Number (%) of subjects achieving CR or PR out of 30 subjects	3(10)	6(20)	9(30)	12(40)
Lower limit of 80% CI	3.7%	10.9%	19.0%	27.7%
Upper limit of 80% CI	20.9%	32.5%	43.2%	53.3%

For AK104 in combination with mXELOX/XELOX, up to 40 subjects (including subjects receiving the same dosing regimen in Phase Ib) will be planned to be enrolled in each of the two dose groups respectively for the dose confirmation period in cohort D of this study for efficacy observation. The primary objective of the dose confirmation period is to preliminarily evaluate the efficacy of AK104 combined with mXELOX/XELOX as first-line therapy for unresectable advanced or metastatic gastric adenocarcinoma or GEJ adenocarcinoma. In the ORR superiority test at a one-sided alpha level of 0.05, when the sample size is 40 subjects, there will be a power of 83% to detect that the ORR evaluating AK104 combined with mXELOX/XELOX as the first-line therapy for gastric adenocarcinoma (GAC) or GEJ adenocarcinoma increases from 40% (assuming ORR of historical control mXELOX/XELOX first-line therapy for GAC or GEJ adenocarcinoma) to 60% (AK104 drug combination). ORR is estimated based on the proportion of subjects expected to achieve CR or PR and its 95% CI is estimated (Table 11-4).

Table 11-4 ORR and 95% CI Estimates in 40 Subjects

Number (%) of subjects achieving CR or PR out of 40 subjects	16(40)	20(50)	22(55)	24(60)	26(65)	28(70)
Lower limit of 95% CI	24.9%	33.8%	38.5%	43.3%	48.3%	53.5%
Upper limit of 95% CI	56.7%	66.2%	70.7%	75.1%	79.4%	83.4%

12. Data Integrity and Quality Assurance

The investigator/site is required to open the direct access to the source data/documents and allow study-related monitoring, audits, IRB/IEC review and regulatory audits. Direct access includes permission to examine, analyse, verify and reproduce any record(s) and report(s) that are essential to the evaluation of the clinical study.

12.1. Monitoring and Audit

The clinical monitor is responsible for conducting regular follow-up to the study site throughout the study to verify its compliance with the protocol, its compliance with the completeness, accuracy, and consistency of data and its compliance with ICH GCP and local regulations during the conduct of the clinical study. The clinical monitor is responsible for examining the case report forms and ensuring the integrity of the critical study documents and should have the right to access to the subject's medical records and other study-related records (as required when checking the contents of the case report form).

The clinical monitor is required to communicate to the investigators about their deviations from the protocol, SOPs, GCPs, and applicable regulations and ensure that appropriate measures designed to prevent recurrence of the identified deviations have been taken and documented.

The investigator agrees to cooperate with the clinical monitor to ensure that any issues identified during monitoring follow-up are resolved and documented.

The selected study may be audited by representatives of the sponsor in accordance with the ICH GCP and the sponsor's audit plan. To evaluate the conduct of the study and its compliance with the protocol, ICH GCP, and applicable regulatory requirements, site facilities (e.g., pharmacies, drug storage areas, laboratories, etc.) need to be inspected and study-related records (e.g., case report forms, source data, and other relevant documents) should be reviewed.

12.2. Data Collection

All data collected during the study will be recorded in the subject's individual case report form. The sponsor or its designee will provide case report forms and instructions for completing the case report forms. Any modifications made to the CRF will be automatically recorded through the "audit trail" function of the EDC system.

The case report form should be kept up to date to assist the monitor in reviewing the status of subjects throughout the study. The case report form should be completed as soon as possible after follow-up, and the questions should also be answered as soon as possible. The case report form should be completed, reviewed and signed by the investigator as soon as possible after the last subject follow-up.

Each subject who has signed the ICF and passed all screening procedures should have a completed case report form. If a subject is not treated, the reason should be recorded in the

case report form.

All external data (e.g. central laboratory, PK and PD data) will be integrated with those in the subject's case report form according to the data management plan.

The subject number and date of study participation should be recorded in the subject's medical/study file by the investigator along with the study code. The investigator should also record the following information in the medical/study file: written and oral confirmation of informed consent, subject's clinical status, date of each study follow-up, date of administration of the study drug, concomitant medications, copies of all relevant reports and laboratory test results, comments on the results, and any AEs mentioned.

The investigator will sign and date the case report form at the designated location via the electronic signature procedure of the EDC system. These signatures indicate that the investigator has examined or reviewed the data, data queries, and site notifications on the CRF and agrees with the contents.

All information and other materials used by subjects and investigators should be described in clear and understandable terms and sentences.

12.3. Data Management

The sponsor may identify each subject in the database using unique subject identifiers as defined by the sponsor.

To ensure the quality of clinical data for all subjects and sites, a clinical data management review of the subject data will be performed. The data in the case report form will be checked electronically and manually. When the data is electronically checked using internal procedural data rules within the application, the checks and queries established for data entry within the EDC system need to be edited. Post-entry validation will be performed as described in the Data Management Plan. Queries arising from the rules and those raised by the auditors will arise within the EDC application. This review process will check for consistency, omissions, and any apparent differences in subject data. In addition, compliance of the data with the protocol and GCP will also be reviewed. To address all issues that arise during the review of clinical data management, case report form queries will be raised and resolved within the EDC application.

Data obtained from external sources (eg, central laboratories) will be consistent with those in the clinical database. Serious adverse events in the clinical database will be consistent with those in the safety database.

All AEs will be coded using MedDRA. Concomitant medications selected according to the reference directory of the World Health Organization Drug Dictionary (WHO-DD) and prior tumor therapies will also be recorded on the case report form.

12.4. Study Documents and Storage

The investigator will maintain a signed list of appropriately qualified persons who are

responsible for the study. All personnels authorized to enter entries or make modifications to the eCRF are included in the signature list.

Source documents refer to original documents, data and records, from which the subject's case report form data are obtained. These documents include, but not limited to, hospital records, clinical and outpatient medical records, laboratory tests and pharmacy records, diaries, microfilms, x-rays, and correspondence. It is the responsibility of the investigator and study personnel to maintain a complete central file archiving system (Trial Master File) for all study-related (critical) documents for inspection at any time by representatives of the sponsor and/or appropriate regulatory authorities. Critical documents include:

- Subject files, containing completed eCRFs, informed consent form and copies of supporting source documents, if available. For studies using EDC, a PDF file (file in portable document format) will be generated for each subject's data (case report forms and queries) at the end of the study, which will be stored in the Trial Master File.
- Study files, containing the protocol with all amendments, the Investigator Brochure, copies of all relevant clinical documents required before initialization of study, and all correspondences to and from the IEC/IRB and the Sponsor.
- Study drug-related records, including site receipt, accountability records, final reconciliation, and corresponding correspondence.

In addition, all original documents supporting entries in the case report form must be retained and easily accessible. All critical documents will be retained by the Study Facility for at least 5 years or longer after completion of the study, if specified by other applicable regulations or requirements. When these documents are no longer required, it is the responsibility of the Sponsor to notify the Investigator/Study Facility. No study documents shall be destroyed until written consent is reached between the sponsor and the investigator. If the investigator wishes to assign the study records to another party or to another location, he/she must notify the sponsor in writing of the new responsible person and/or the new location.

12.5. Storage of Records

Subject records, source documents, monitoring follow-up logs, data correction forms, case report forms, study drug inventory listings, regulatory documents (eg, protocols and amendments, IRB/IEC correspondence and approvals, approved and signed ICFs, investigator agreements, clinical supply receipts, distribution and return records) and other study-related correspondence from and to the Sponsor must be maintained in the appropriate study file room at the study site. Source documents include all records and observations or markers of clinical manifestations, as well as all reports and records necessary to evaluate and reconstruct clinical studies. These records will be retained in the safety file for the time required by the Policy of the Study Facility or Study Site. Prior to transfer or destruction of these records, the Sponsor must be notified in writing and provided with the opportunity to continue maintaining them.

13. Ethical and Regulatory Requirements

13.1. Ethical Conduct of the Study

This study will be conducted in accordance with GCP and all applicable regulatory requirements, including, where applicable, the current version of the Declaration of Helsinki. The investigator (or sponsor, as appropriate) is responsible for ensuring that this protocol, the study site's CRF, and any other information to be presented to potential subjects (e.g., advertisements or information to support or supplement the informed consent form) are reviewed and approved by the appropriate IEC/IRB. The investigator agrees that the IEC/IRB should directly get access to all relevant documents. The IEC/IRB must be constituted in accordance with all applicable regulatory requirements. The sponsor will provide the investigator with relevant document(s)/data that are needed for IEC/IRB review and approval of the study. Before the study drug(s) can be shipped to the study site, the sponsor must receive copies of the IEC/IRB approval, the approved ICF, and any other information that the IEC/IRB has approved for presentation to potential subjects.

If the protocol, the study site's ICF, or any other information that the IEC/IRB has approved for presentation to potential subjects is amended during the study, the investigator is responsible for ensuring the IEC/IRB reviews and approves, where applicable, these amended documents. The investigator must comply with all applicable regulatory requirements regarding the use of the amended informed consent form, including that subjects enrolled after the new version of the informed consent form has been approved by the IEC/IRB should use the new version of the informed consent form. Copies of the IEC/IRB approval of the amended ICF/other information and the approved amended ICF/other information must be forwarded to the sponsor promptly.

13.2. Confidentiality of Subjects

In accordance with GCP and local regulations, all subjects participating in the study will be kept confidential by the investigator and the sponsor.

The investigator must ensure that the subject does not expose his/her name. Subjects should be identified with a unique subject identifier designated by the Sponsor in eCRFs or other documents submitted to the Sponsor and the Contract Research Organization (CRO). Documents not submitted to Akeso and CRO (eg, signed ICFs) should be kept strictly by the investigator.

As required by CFR/ICH GCP guidelines, the investigator and the study facility allow direct access to the subject's original medical records for review by authorized representatives of the company, regulatory authority and the IEC or IRB in order to verify study-related processes and data. The investigator is obligated to inform the subject that the study-related records of the subject will be reviewed by the above-mentioned representative without violating the confidentiality of the subject.

13.3. Compliance

Prior to the start of the study, the protocol, subject information and informed consent form, Investigator's Brochure, any subject diary card or written instructions to subjects, available safety information, subject recruitment process (e.g., advertisements), payment and compensation information available to subjects, and documentation demonstrating investigator qualification should be submitted to the IEC or IRB in accordance with local regulations for ethical review and approval. Names and versions of all reviewed documents should be confirmed in the written approval.

Changes that occur during the conduct of the study or planned analysis are documented in protocol amendments and/or SAPs.

When necessary, the investigator must submit all subsequent protocol amendments and changes in informed consent documents or changes in study sites, facilities or personnel to the IEC or IRB and/or the sponsor for approval. The investigator should notify the IEC or IRB of protocol deviations or site-emergent SAEs and other AE reports obtained from CROs in accordance with local procedures.

In accordance with local regulatory requirements, the sponsor's local regulatory affairs team or designee is required to ensure that all legal issues are resolved and approved by the relevant regulatory authorities prior to the start of the study, and that changes to the initial protocol or other relevant study documents are not implemented until appropriate notification or approval is received from the relevant regulatory authority.

It is the responsibility of each Principal Investigator to provide the IRB/IEC with reports of any serious and unexpected adverse drug reactions occurring in other studies conducted with the study drug. Akeso will provide this information to the Principal Investigator to ensure that the investigators can meet these reporting requirements.

According to local requirements, the sponsor will provide safety updates/reports including suspected unexpected serious adverse reactions (if relevant) to regulatory authorities, IRB/IEC and principal investigators.

13.4. Informed Consent Process

Before a subject participates in the study, it is the responsibility of the investigator to fully explain to the subject the study objectives, methods, expected benefits, and potential hazards, thereby obtaining further written consent voluntarily issued by the subject or his/her legal representative (if permitted by local regulations). A legal representative is an individual or institution that, in accordance with the requirements of law, can consent to participate in a clinical study on behalf of a subject after being authorized by the subject or law. The informed consent form should be written in the mother tongue of the potential subject population.

Investigators should comply with appropriate regulatory requirements, GCP, and ethical principles derived from the Declaration of Helsinki to obtain the subject's written consent. The informed consent form and its revisions should be approved by the IECs/IRBs prior to

being provided to potential subjects. The investigator should ensure that each subject is informed that he/she is free to voluntarily discontinue the study at any time and that each subject has the opportunity to ask questions and have time to reflect on the information obtained.

The ICF should be obtained prior to the subject's participation in the study and will be kept in the form of documents at the study site. The ICF should be signed and dated by the subject or his/her legal representative (if permitted by local regulations), and by the investigator. The original signed ICF should be kept at the study site and a copy of the signed ICF should be provided to the subject or his/her legal representative. The date on which the ICF is obtained should be documented in the medical record and case report form (CRF).

If the subject or his/her legal representative (if permitted by local regulations) is unable to read, an impartial witness should be present throughout the discussion of the informed consent. The witness should sign the ICF after the subject or his/her legal representative agrees to participate in the study and signs the ICF (if possible). Signing the ICF indicates that the witness confirms that the information in the ICF and any other written information has been fully explained to the subject or his/her legal representative, that they have understood this information clearly and that the subject or his/her legal representative have agreed to sign the written informed consent.

The ICF approved by the IRB/IEC should describe any incentive measures for subjects participating in the study and the protective measures against possible injury to the subject due to participation in the study.

13.5. Changes to Protocol and Informed Consent Form

The sponsor will communicate to the investigator any amendments to the protocol, all of which require approval from the IRB/IEC. Protocol amendments will not be implemented until IRB/IEC approval and national regulatory approval (if applicable) are available, except that changes must be implemented immediately for the safety of subjects. If the change must be implemented immediately, the details must be documented and reported to the IRB/IEC within 5 working days. The sponsor should ensure that protocol amendments are submitted to regulatory authorities in a timely manner.

If a protocol amendment requires a change to the study site's ICF, the amended ICF document will need to be approved by the sponsor and study site's IRB/IEC prior to use.

14. Financing and Insurance

14.1. Finance

Prior to initiation of the study, the Principal Investigator and/or the Study Facility will enter into a clinical study agreement with the Sponsor or CRO. The agreement will include financial information agreed upon between the parties.

14.2. Reimbursement, Compensation and Insurance

Reimbursement, compensation and insurance agreements shall be signed separately in the terms agreed upon between the parties.

15. Publication Policy

The investigator should keep the information and data related to this trial confidential and consult with the sponsor in advance and obtain the written consent of the sponsor in order to publish any conclusions or drawn from the trial. In order to protect its rights and interests, the sponsor may require the investigator not to publish relevant information until the investigational product is approved for marketing. The sponsor has the right to issue or publish information or data related to this study, or submit them to the drug regulatory authority.

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17. Appendices

Appendix 1 Response Evaluation Criteria in Solid Tumors Version 1.1 and Immune-related Response Evaluation Criteria

1. Measurability of tumour at baseline

1.1. Definition

At baseline, tumour lesions/lymph nodes will be categorised measurable or non-measurable as follows:

1.1.1. Measurable

- Tumour lesions: Must be accurately measured in at least one dimension (longest diameter in the plane of measurement is to be recorded) with a minimum size of:
  - 10 mm by CT scan (CT scan slice thickness no greater than 5 mm).
  - 10 mm caliper measurement by clinical exam (lesions which cannot be accurately measured with calipers should be recorded as non-measurable).
  - 20 mm by chest X-ray.

Measurable malignant lymph nodes: To be considered pathologically enlarged and measurable, a lymph node must be ≥ 15 mm in short diameter when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline (ie, screening phase) and in follow-up (ie, all measurements after screening), only the short axis will be measured and followed. See also notes below on 'Baseline documentation of target and non-target lesions' for information on lymph node measurement.

1.1.2. Non-measurable

All other lesions, including small lesions (longest diameter <10 mm or pathological lymph nodes with ≥10 to <15 mm short axis) as well as truly non-measurable diseases. Lesions considered truly non-measurable include: leptomeningeal disease, ascites, pleural or pericardial effusion, inflammatory breast disease, lymphangitic involvement of skin or lung, abdominal masses/abdominal organomegaly identified by physical exam that is not measurable by reproducible imaging techniques.

1.1.3. Special considerations regarding measurability of lesions

Bone lesions, cystic lesions, and lesions previously treated with local therapy require particular comments:

1.1.4. Bone lesions

- Bone scan, PET scan or plain films are not considered adequate imaging techniques

to measure bone lesions. However, these techniques can be used to confirm the presence or disappearance of bone lesions.

- Lytic bone lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, that can be evaluated by cross sectional imaging techniques such as CT or MRI can be considered as measurable diseases if the soft tissue component meets the definition of measurability described above.
- Blastic bone lesions are non-measurable.

1.1.5. Cystic lesions

- Lesions that meet the criteria for radiographically defined simple cysts should not be considered as malignant lesions (neither measurable nor non-measurable) since they are, by definition, simple cysts.
- 'Cystic lesions' thought to represent cystic metastases can be considered as measurable diseases, if they meet the definition of measurability described above. However, if non-cystic lesions are present in the same patient, these are preferred for selection as target lesions.

1.1.6. Lesions with prior local treatment

- Tumour lesions situated in an area subjected to previous loco-regional therapy (eg. TACE), are usually not considered measurable unless there has been demonstrated progression in the lesion. Lesions that have ever been irradiated should not be considered target lesions.

1.2. Specifications by methods of measurements

1.2.

1.2.1. Measurement of lesions

All measurements should be recorded in metric notation, using calipers if clinically assessed. All baseline evaluations should be performed as close as possible to the treatment start and never more than 4 weeks before the beginning of the treatment.

1.2.2. Method of assessment

The same method of assessment and the same technique should be used to characterise each identified and reported lesion at baseline and during follow-up. Imaging based evaluation should always be done rather than clinical examination unless the lesion(s) being followed cannot be imaged but are assessable by clinical exam.

CT, MRI: CT is the best currently available and reproducible method to measure lesions selected for response assessment. This guideline has defined measurability of lesions on CT scan based on the assumption that CT slice thickness is 5 mm or less. When CT scans have slice thickness greater than 5 mm, the minimum size for a measurable lesion should be twice

the slice thickness. MRI is also acceptable in certain situations (eg, for body scans).

2. Tumor response evaluation

2.

2.1. Assessment of overall tumour burden and measurable disease

To assess objective response or future progression, it is necessary to estimate the overall tumour burden at baseline and use this as a comparator for subsequent measurements. In this study, only patients with measurable disease at baseline should be included in. (See definition "1.1.2")

2.2. Baseline documentation of 'target' and 'non-target' lesions

When more than one measurable lesion is present at baseline, all lesions up to a maximum of five lesions totally representing all involved organs should be identified as target lesions and will be recorded and measured at baseline (this means in instances where subjects have only one or two organ sites involved, a maximum of two and four lesions will be recorded, respectively).

Target lesions should be selected on the basis of their size (lesions with the longest diameter), be representative of all involved organs, but in addition should be those that lend themselves to reproducible repeated measurements. It may be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement in which circumstance the next largest lesion which can be measured reproducibly should be selected.

Lymph nodes merit special mention since they are normal anatomical structures which may be visible by imaging even if not involved by tumour. As noted above, pathological nodes which are defined as measurable and may be identified as target lesions must meet the criterion of a short axis of ≥15mm by CT scan. Only the short axis of these nodes will contribute to the baseline sum. The short axis of the node is the diameter normally used by radiologists to judge if a node is involved by solid tumor. Nodal size is normally reported as two dimensions in the plane in which the image is obtained (for CT scan this is almost always the axial plane; for MRI the plane of acquisition may be axial, saggital or coronal). The smaller of these measures is the short axis. For example, an abdominal node which is reported as being 20 mm × 30 mm has a short axis of 20 mm qualifies as a malignant, measurable node. In this example, 20 mm should be recorded as the node measurement. All other pathological nodes (those with short axis ≥10 mm but <15 mm) should be considered non-target lesions. Nodes that have a short axis <10 mm are considered non-pathological and should not be recorded or followed.

A sum of the diameters (longest for non-nodal lesions, short axis for nodal lesions) for all target lesions will be calculated and reported as the baseline sum diameters. If lymph nodes are to be included in the sum, then as noted above, only the short axis is added into the sum. The baseline sum diameters will be used as reference to further characterise any objective tumour regression in the measurable dimension of the disease.

All other lesions (or sites of disease) including pathological lymph nodes should be identified as non-target lesions and should also be recorded at baseline. Measurements are not required and these lesions should be followed as 'present', 'absent', or in rare cases 'definite progression'. In addition, multiple non-target lesions involving the same organ may be recorded as a single item on the case report form (CRF), such as "multiple enlarged pelvic lymph nodes" or "multiple liver metastases".

2.3. Response criteria (RECIST 1.1)

This section provides the definitions of the objective tumour response criteria used to determine target, non-target and new lesions based on RECIST v1.1.

2.

2.1.

2.2.

2.3.

2.3.1. Evaluation of target lesions

Complete Response (CR): Disappearance of all target lesions. Any pathological lymph nodes (whether target or non-target) must have decreased in size to have a short axis of <10 mm.

Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as a reference the baseline sum diameters.

Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as a reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression.)

Stable Disease (SD): Neither sufficient shrinkage to judge as PR nor sufficient increase to judge as PD, taking as a reference the smallest sum diameters while on study.

2.3.2. Special notes on the assessment of target lesions

Lymph nodes: Lymph nodes identified as target lesions should always have the actual short axis measurement recorded (measured in the same anatomical plane as the baseline examination), even if the nodes regress to below 10 mm on study. This means that when lymph nodes are included as target lesions, the 'sum' of lesions may not be zero even if complete response criteria are met, since a normal lymph node is defined as having a short axis of <10 mm. For PR, SD and PD, the actual short axis measurement of the nodes is to be included in the sum of target lesions.

Target lesions that become 'too small to measure': While on study, all lesions (nodal and non-nodal) recorded at baseline should have their actual measurements recorded at each

subsequent evaluation, even when very small (e.g., 2 mm). However, sometimes lesions or lymph nodes which are recorded as target lesions at baseline become so faint on CT scan that the radiologist may not feel comfortable assigning an exact measure and may report them as being 'too small to measure'. When this occurs it is important that a value be recorded on the CRF. If it is the opinion of the radiologist that the lesion has likely disappeared, the measurement should be recorded as 0 mm. If the lesion is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned. (Note: It is less likely that this rule will be used for lymph nodes since they usually have a definable size when normal and are frequently surrounded by fat such as in the retroperitoneum; however, if a lymph node is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned in this circumstance as well). This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). The measurement of these lesions is potentially non-reproducible, therefore providing this default value will prevent false responses or progressions based upon measurement error. To reiterate, however, if the radiologist is able to provide an actual measure, that should be recorded, even if it is below 5 mm.

Lesions that split or coalesce on treatment: When non-nodal lesions "fragment", the longest diameters of the fragmented portions should be added together to calculate the target lesion sum. Similarly, as lesions coalesce, a plane between them may be maintained that would aid in obtaining maximal diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximal longest diameter for the 'coalesced lesion'.

2.3.3. Evaluation of non-target lesions

This section provides the definitions of the criteria used to determine the tumour response for the group of non-target lesions. While some non-target lesions may actually be measurable, they need not be measured and instead should be assessed only qualitatively at the time points specified in the protocol.

Complete Response (CR): Disappearance of all non-target lesions and normalisation of tumour marker level. All lymph nodes must be non-pathological in size (<10 mm short axis).

Non-CR/Non-PD: Persistence of one or more non-target lesion(s) and/or maintenance of tumour marker level above the normal limits.

Progressive Disease (PD): definite progression (see comments below) of existing non-target lesions. (Note: the appearance of one or more new lesions is also considered progression.)

2.3.4. Special notes on assessment of progression of non-target disease

If the subject has additional measurable diseases, the concept of progression of non-target diseases requires additional explanation as follows. In this setting, to achieve 'definite progression' on the basis of the non-target disease, there must be an overall level of substantial worsening in non-target disease such that, even in presence of SD or PR in target disease, the

overall tumour burden has increased sufficiently to merit discontinuation of therapy. A modest 'increase' in the size of one or more non-target lesions is usually not sufficient to qualify for definite progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of SD or PR of target disease will therefore be extremely rare.

2.3.5. New lesions

The appearance of new malignant lesions denotes disease progression; therefore, some comments on detection of new lesions are important. There are no specific criteria for the identification of new radiographic lesions; however, the finding of a new lesion should be definite: i.e. not attributable to differences in scanning technique, change in imaging modality or findings thought to represent something other than tumour (for example, some 'new' bone lesions may be simply healing or flare of pre-existing lesions). This is particularly important when the patient's baseline lesions show partial or complete response. For example, necrosis of a liver lesion may be reported on a CT scan report as a 'new' cystic lesion, which it is not.

A lesion identified on a follow-up study in an anatomical location that was not scanned at baseline is considered a new lesion and will indicate disease progression. An example of this is the patient who has visceral disease at baseline and while on study has a CT or MRI brain ordered which reveals metastases. The patient's brain metastases are considered to be evidence of PD even if he/she did not have brain imaging at baseline.

If a new lesion is equivocal, for example because of its small size, continued therapy and follow-up evaluation will clarify if it represents truly new disease. If repeat scans confirm there is definitely a new lesion, then progression should be declared using the date of the initial scan.

2.4. Evaluation of overall response

The patient's best overall response assignment will depend on the findings of both target and non-target disease and will also take into consideration the appearance of new lesions. It is assumed that at each protocol specified time point, a response assessment occurs. The following table provides a summary of the overall response status calculation at each time point for patients who have measurable disease at baseline.

Table 17-1 Overall response

Time point response: patients with target (+/- non-target) disease			
Target lesions	**Non-target lesions**	**New lesions**	**Overall response**
CR	CR	No	CR
CR	Non-CR/Non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE

PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

Abbreviations: CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease; NE, not evaluable.

2.4.

2.4.1. Missing assessment and inevaluable designation

When no imaging/measurement is done at all at a particular time point, the patient is not evaluable (NE) at that time point. If only a subset of lesion measurements are made at an assessment, usually the case is also considered NE at that time point, unless a convincing argument can be made that the contribution of the individual missing lesion(s) would not change the assigned time point response. This would be most likely to happen in the case of PD. For example, if a patient has a baseline sum of 50 mm with three measured lesions and at follow-up only two lesions are assessed, but those give a sum of 80 mm, the patient will have achieved PD status, regardless of the contribution of the missing lesion.

2.4.2. Special notes on response assessment

When nodal disease is included in the sum of target lesions and the nodes decrease to ‘normal’ size (<10 mm), they may still have a measurement reported on scans. This measurement should be recorded even though the nodes are normal in order not to overstate progression should it be based on increase in size of the nodes. As noted earlier, this means that patients with CR may not have a total sum of 'zero' on the case report form (CRF).

Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as "symptomatic deterioration". Every effort should be made to document objective progression even after discontinuation of treatment. Symptomatic deterioration is not a descriptor of an objective response: it is a reason for stopping study therapy. The objective response status of such patients is to be determined by evaluation of target and non-target disease.

For equivocal findings of progression (e.g. very small and uncertain new lesions; cystic changes or necrosis in existing lesions), treatment may continue until the next scheduled assessment. If at the next scheduled assessment, progression is confirmed, the date of progression should be the earlier date when progression was suspected.

Appendix 2 Immune-related Response Evaluation Criteria in Solid Tumors (irRECIST)

The irRECIST criteria (**Nishino M, 2013**) are an integration of RECIST guideline v1.1 and the initial irRC (**Wolchok JD, 2009**). The latter criterion incorporates important principles used for the assessment of cancer immunotherapy, but is intrinsically different from RECIST in the use of two-dimensional tumor assessments. In contrast, irRECIST is based on one-dimensional tumor measurements. Response assessments should be performed on the basis of irRECIST at each time point based on the following:

- Immune-related complete response (irCR): Complete disappearance of all tumor lesions (target and non-target lesions) with no new measurable or non-measurable diseases for at least 4 weeks from the date of documentation of irCR. All lymph nodes must have a short axis < 10 mm.
- Immune-related partial response (irPR): The sum of diameters of all target lesions is measured and collected as the baseline sum diameters. At each subsequent tumor assessment, the sum of diameters of all target lesions is added to the sum of diameters of new measurable diseases as the overall tumor burden. In the absence of irCR, a decrease of ≥ 30% in overall tumor burden relative to baseline is considered an irPR. It must be determined within no less than 4 weeks after the first irPR assessment.
- Immune-related progressive disease (irPD): At least a 20% increase in tumor burden (ie, sum of diameters of target lesions and any new measurable diseases), taking as an reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. Tumor assessments using immune-related criteria for progressive disease include the contribution of new measurable disease. Each net percentage change in tumor burden per assessment takes into account the size and growth kinetics of new and old lesions at their appearance. Assuming no clinical deterioration, progressive disease will need to be determined by continuous scans (with an interval of at least 4 weeks).
- Immune-related stable disease (irSD): Neither sufficient shrinkage to judge as irPR nor sufficient increase to judge as irPD, taking as reference the smallest sum diameters while on study.

Appendix 3 Listing of Autoimmune Disorders Present Prior to Inclusion

Subjects will be carefully asked if they have any acquired or innate immunodeficiency or autoimmune diseases, and such subjects should be excluded from the study. Unless the likelihood of the subject's history of previous allergic disease and childhood arthralgia being suspected of autoimmune disease is very low. In addition, transient autoimmune manifestations (eg, Lyme arthritis) due to acute infectious disease may be included if cured with antibiotics. Contact the medical monitor if an unconfirmed autoimmune disease requires exclusion.

Diseases associated with autoimmunity, including but not limited to:

Acute sporadic encephalomyelitis
IgA nephropathy
Addison′s disease
Inflammatory bowel disease
Alopecia universalis
Interstitial cystitis
Ankylosing spondylitis
Myasthenia gravis syndrome
Antiphospholipid antibody syndrome
Lupus erythematosus
Aregenerative anemia
Lyme disease-chronic
Asthma
Meniere's syndrome
Autoimmune hemolysis
Anaemia
Corneal ulcer
Autoimmune hepatitis
Localized autoimmune hypophysitis
Multiple sclerosis
Autoimmune hypoparathyroidism
Myasthenia gravis
Autoimmune myocarditis
Neuromyotonia
Autoimmune oophoritis
Myoclonic syndrome
Autoimmune orchitis
Optic neuritis
Autoimmune thrombocytopenic purpura
Ords thyroiditis
Behcet's disease
Pemphigus
Bullous pemphigoid
Pernicious anaemia
Gluten allergy
Multiple arteritis
Chronic fatigue syndrome
Polyarthritis
Chronic inflammatory demyelination
Polyneuropathy
Autoimmune syndrome
Churg-Strauss syndrome
Primary biliary cirrhosis
Crohn's disease
Psoriasis/dermatomyositis
Reiter's syndrome
Diabetes type 1
Rheumatoid arthritis
Autonomic dysfunction
Sarcoidosis
Eczema
Scleroderma
Sjogren's syndrome
Epidermolysis bullosa
Stiff person syndrome
Pemphigoid gestationis
Polyarteritis
Giant cell arteritis
Ulcerative colitis
Pulmonary-renal syndrome
Vitiligo
Graves' disease
Vogt-Kovanagi-Harada disease
Guillain-Barré syndrome
Vulval pain
Hashimoto's disease
Wegeners granulomatosis
Kawasaki's disease

Appendix 4 New York Heart Association Classification

Class	Symptoms
I	No limitation in physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, or dyspnea (shortness of breath).
II	Limitation in physical activity. Comfortable at rest, but ordinary physical activity results in fatigue, palpitation, dyspnea (shortness of breath).
III	Marked limitation in physical activity. Comfortable at rest, but less-than-ordinary physical activity causes fatigue, palpitation, or dyspnea.
IV	Any physical activity causes discomfort. Symptoms of heart failure at rest. If any physical activity is undertaken, discomfort increases.

Adapted from

Dolgin M, Association NYH, Fox AC, Gorlin R, Levin RI, New York Heart Association. CriteriaCommittee.Nomenclature and criteria for diagnosis of diseases of the heart and great vessels.9th ed. Boston, MA:LippincottWilliams and Wilkins; March 1,1994.

Original Source:Criteria Committee, New York Heart Association, Inc. Diseases of the Heart andBloodVessels. Nomenclature and Criteria for diagnosis,6th edition Boston, Little, Brown and Co.1964, p114.

Appendix 5 Guidelines for Diagnostic Evaluation of Immune-related Adverse Events

The recommendations below for the diagnosis of any irAE are intended as a guidance. This document should be used in conjunction with expert clinical judgment (by specialist physicians experienced in the treatment of cancer using immunological agents), and individual institutional guidelines or policies.

Criteria used to diagnose irAEs include blood tests, diagnostic imaging, histopathology, and microbiology assessments to exclude alternative causes such as infection, disease progression, and adverse effects of concomitant drugs. In addition to the results of these tests, the following factors should be considered when making an irAE diagnosis:

- What was the temporal relationship between initiation of anti-PD-1 and CTLA-4 antibodies and the adverse event?
- How did the patient respond after discontinuation of anti-PD-1 and CTLA-4 antibodies?
- Did the event recur after resumption of treatment with anti-PD-1 and CTLA-4 antibodies?
- Was there a clinical response to corticosteroids?
- Is the event an autoimmune endocrinopathy?
- Is disease progression or an alternative diagnosis a more likely explanation?

Immune-related Toxicity	Diagnostic Evaluation Guideline
Thyroid disorders	Scheduled and repeat thyroid function tests (TSH and T4).
Hypophysitis	Check visual fields and consider pituitary endocrine axis blood profile. Perform pituitary and whole brain MRI in patients with headache, visual disturbance, unexplained fatigue, asthenia, weight loss and unexplained constitutional symptoms. Consider consultation with an endocrinologist if an abnormality is detected.
Non-infectious pneumonia	All patients presenting with new or worsened pulmonary symptoms or signs, such as an upper respiratory infection, new cough, shortness of breath or hypoxia should be assessed by high-resolution CT. Consider pulmonary function test including DLCO. Radiographic appearance is often nonspecific. Depending on the location of the abnormality, bronchoscopy and bronchoalveolar lavage or lung biopsy may be considered. Consult with a respiratory medicine physician for cases of uncertain cause.
Neurotoxicity	Perform a comprehensive neurological examination and brain MRI for all CNS symptoms; review alcohol history and other medications. Conduct a diabetic screen, and assess blood B12/folate, HIV status, TFTs, and consider autoimmune serology. Consider the need for brain/spine MRI/MRA and nerve conduction study for peripheral neuropathy. Consult with a neurologist if there are abnormal findings.
Colitis	Review dietary intake and exclude steatorrhea. Consider comprehensive testing, including the following: FBC, UEC, LFTs, CRP, TFTs, stool microscopy and culture, viral PCR, Clostridium difficile toxin, cryptosporidia (drug-resistant organism). In case of abdominal discomfort, consider imaging, eg, X-ray, CT scan. If a patient experiences bleeding, pain or distension, consider colonoscopy with biopsy and surgical intervention, as appropriate.
Eye disorders	If patients experience eye inflammation, blurred vision, or other visual disturbances, refer the patient urgently to an ophthalmologist for evaluation and management.

Immune-related Toxicity	Diagnostic Evaluation Guideline
Hepatitis	Check ALT/AST/total bilirubin, INR/albumin; the frequency will depend on severity of the AE (eg, daily if Grade 3-4; every 2-3 days if Grade 2, until recovering). Review medications (eg, statins, antibiotics) and alcohol history. Perform liver screen including Hepatitis A/B serology, Hepatitis E PCR and assess anti-ANA/SMA/LKM/SLA/LP/LCI, and iron metabolism studies. Consider imaging, eg, ultrasound scan for metastases or thromboembolism. Consult with a hepatologist and consider liver biopsy.
Nephrotoxicity	Check dehydration, water-sodium retention and medication history. Test and culture urine. Consider renal ultrasound scan, protein assessment (dipstick/24-hour urine collection), or phase-contrast microscopy. Refer to nephrology for further management assistance.
Dermatology	Consider other causes by conducting a physical examination, consider dermatology referral for skin biopsy.
Rheumatology	Consult musculoskeletal history and perform complete musculoskeletal examination. Consider joint X-ray and other imaging as required to exclude metastatic disease. Perform autoimmune serology and refer to rheumatology for further management assistance.

Abbreviations: AE=adverse event; ALT=alanine aminotransferase; ANA=antinuclear antibody; AST=aspartate aminotransferase; CNS=central nervous system; CRP=C-reactive protein; CT=computed tomography; DLCO=diffusing capacity for carbon monoxide; FBC=full blood count; HIV=human immunodeficiency virus; INR=international normalized ratio; LCI=liver cystolic antigen; LFT=liver function test; LKM=liver kidney microsomal antibody; LP=liver pancreas antigen; MRA=magnetic resonance angiogram; MRI=magnetic resonance imaging; PCR=polymerase chain reaction; SLA=soluble liver antigen; SMA=smooth muscle antibody; T4=thyroxine; TFT=thyroid function tests; TSH=thyroid stimulating hormone; UEC=urea electrolytes and creatinine.

Sponsor Signature Page

SPONSOR APPROVAL

A PhaseIb/II, Open-label, Multi-center Trial to Evaluate AK104 (an Anti-PD-1 and Anti-CTLA-4 Bispecific Antibody) in the Treatment of Advanced Solid Tumors and as First-line Therapy with Combination Oxaliplatin and Capecitabine in Unresectable Advanced or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

This clinical study protocol has been reviewed and approved by the following representatives of Akeso Biopharma Co., Ltd.

Print Name	**Signature**
Title	**Date (MM/DD/YYYY)**

Investigator's Signature Page

A PhaseIb/II, Open-label, Multi-center Trial to Evaluate AK104 (an Anti-PD-1 and Anti-CTLA-4 Bispecific Antibody) in the Treatment of Advanced Solid Tumors and as First-line Therapy with Combination Oxaliplatin and Capecitabine in Unresectable Advanced or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

I have reviewed the protocol and agree to comply with the ethical principles derived from the Declaration of Helsinki and the ICH GCP - Good Clinical Practice, and the requirements of any regulatory authority and/or Institutional Review Board/Independent Ethics Committee (IRB/IEC) in conducting the protocol.

I understand that no amendments could be made to the protocol without obtaining written approval from the Sponsor. All protocol changes must be submitted to the appropriate regulatory authority and the IRB/IEC and be approved by the IRB/IEC before implementation unless it is necessary to eliminate direct injury to the subject or, in the opinion of the Sponsor, the changes involve only logistics or administration.

I agree to open my subject study records to representatives of the sponsor and relevant regulatory authorities so that they can verify the data I have entered into the case report form. I understand my responsibilities as the Sponsor-designated Principal Investigator.

I understand that the Sponsor may decide to suspend or prematurely terminate the study at any time for any reason; and such decision will be communicated to me in writing. Conversely, if I decide to withdraw from the study, I shall immediately inform the Sponsor in writing.

Print Name	**Signature**
Title	**Date (MM/DD/YYYY)**